



# ROB 2040 SIEMENS LABS



**OAKLAND COMMUNITY COLLEGE®**

ROBOTICS/AUTOMATED SYSTEMS TECHNOLOGY

**Before running a program, check that all the switches for the PLC inputs are in the Down position.**

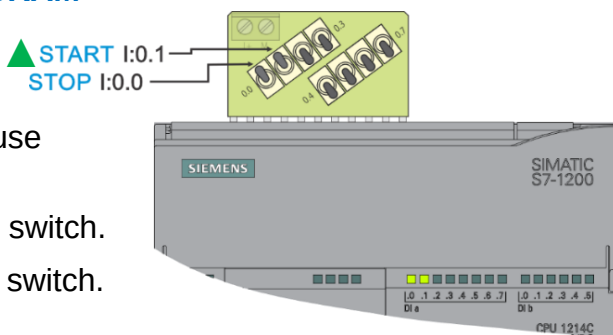
**DO NOT** move switches quickly from open to close or close to open, to allow the HMI screen to react to changes.

## PLC SWITCHES WHEN TESTING A PROGRAM

The steps on when to change the switches use an arrow to show the switch direction:

a GREEN ▲ arrow when to **Push Up** a switch.

a RED ▼ arrow when to **Push Down** a switch.



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**LAB: WASHER STATION****PROCESS FILE:** WASH**OBJECTIVE:** Create ladder structure and assign data tags.**SOFTWARE:** SIEMENS TIA Portal

Create a program that when a momentary normally open pushbutton is pressed and held, two clamp solenoids will be energized. A normally open limit switch for each clamp will be set when that clamp is in position. When the two clamps are in position, a spray wash will clean the part. Releasing the pushbutton will turn off the spray wash and clamps.

**INPUT:**

| Address | PLC Tag       | Comments              |
|---------|---------------|-----------------------|
| I0.0    | PUSHBUTTON    | Pushbutton Wash Cycle |
| I0.1    | SWITCH_CLAMP1 | Limit Switch Clamp 1  |
| I0.2    | SWITCH_CLAMP2 | Limit Switch Clamp 2  |

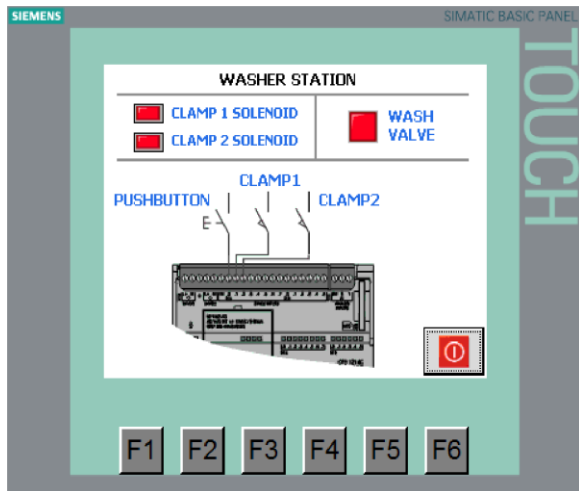
**OUTPUT:**

| Address | PLC Tag         | Comments         |
|---------|-----------------|------------------|
| Q0.0    | SOLENOID_CLAMP1 | Clamp 1 Solenoid |
| Q0.1    | SOLENOID_CLAMP2 | Clamp 2 Solenoid |
| Q0.2    | WASH_VALVE      | Spray Wash Valve |

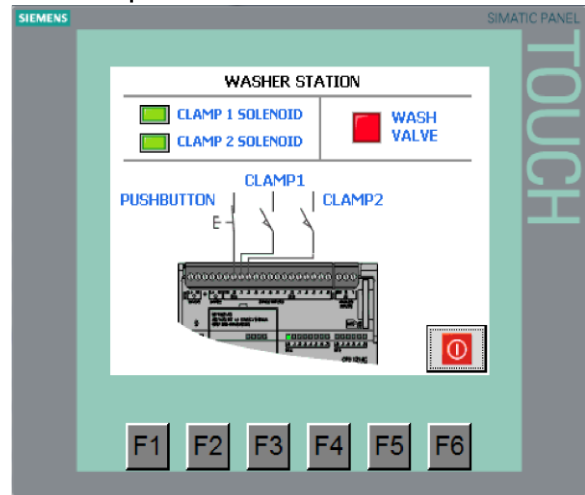
*continued*

## HMI and SWITCHES:

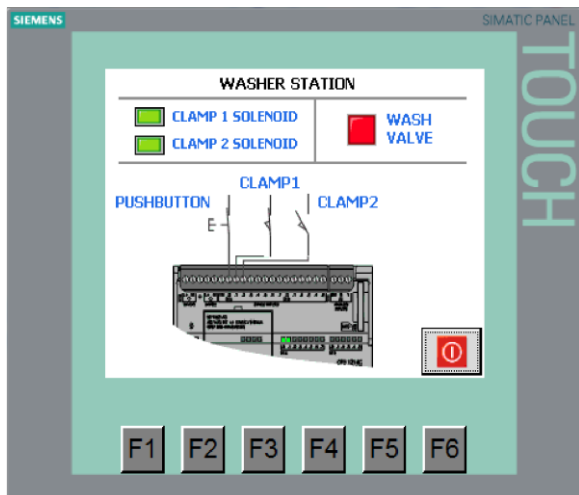
Initial HMI screen.



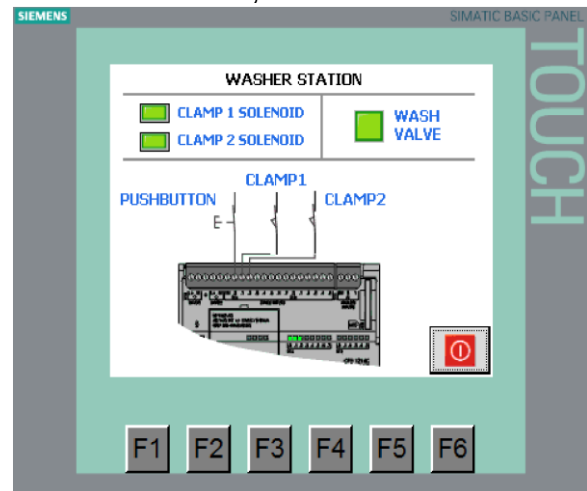
START pressed, SOLENOIDS ON.



CLAMP 1 closed.



CLAMP 2 closed, WASH VALVE ON.

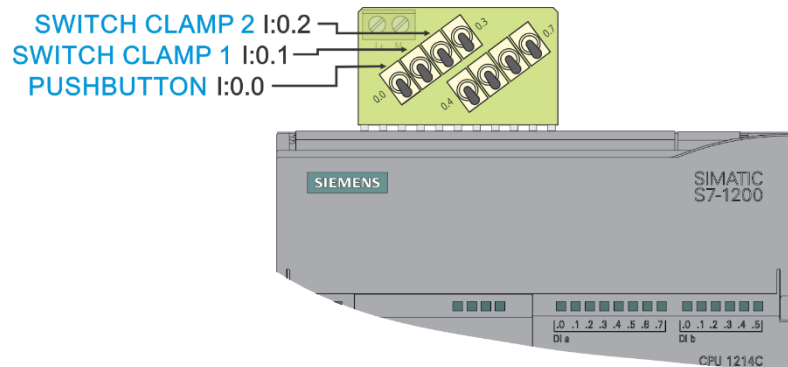


*continued*

The switch will be closed and opened to emulate the operator pressing and releasing the pushbuttons.

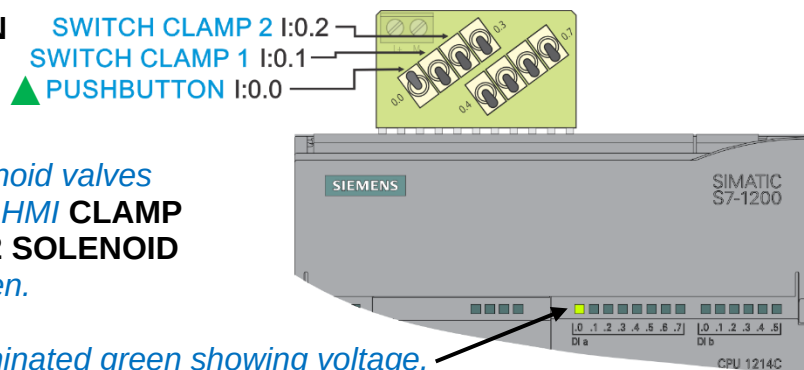
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Initial switch settings, **all switches down.**

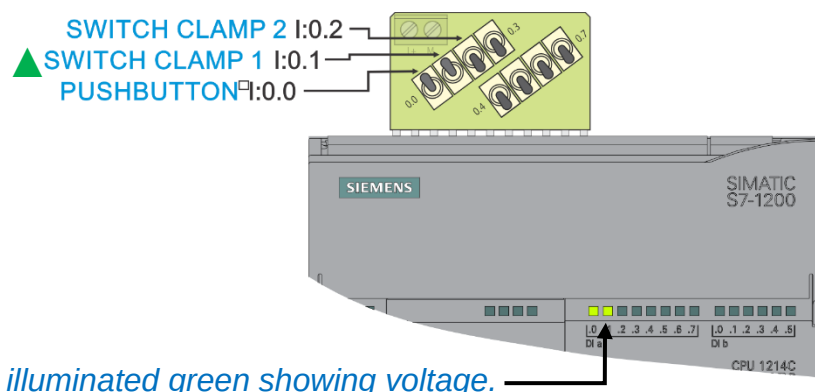


Operator Pressing and Holding the PUSHBUTTON:

**Push up ▲ the PUSHBUTTON** to close.



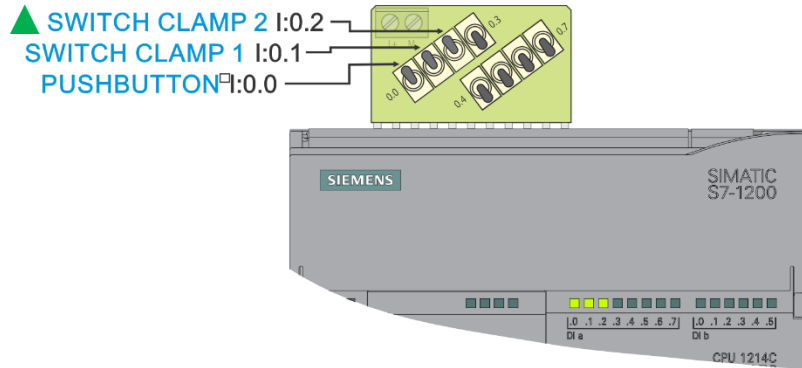
**Push up ▲ the SWITCH CLAMP 1** to close.



---

Push up ▲ the **SWITCH CLAMP 2** to close.

*The spray **WASH** valve turns ON and the HMI indicator will illuminate green on the HMI.*

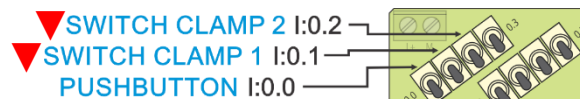
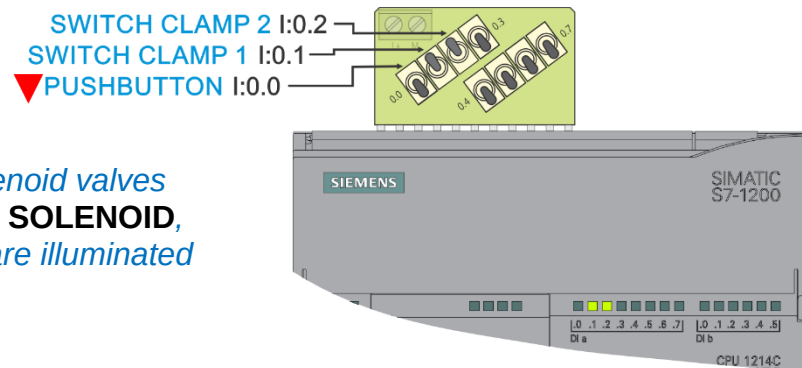


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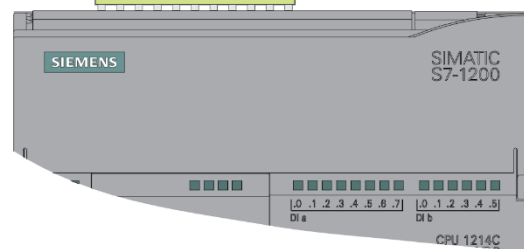
Operator Releasing the PUSHBUTTON:

To turn off the washer, **push down ▼** the **PUSHBUTTON** switch to open.

*The spray wash and the solenoid valves turn OFF. **WASH**, **CLAMP 1 SOLENOID**, and **CLAMP 2 SOLENOID** are illuminated red on the HMI.*

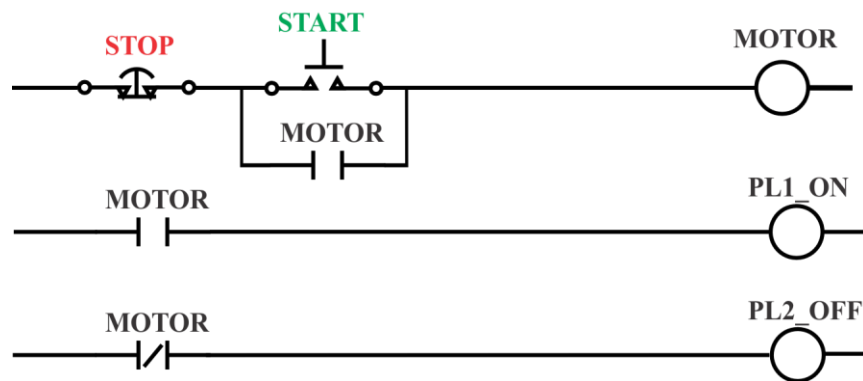


**Push down ▼** switches **SWITCH CLAMP 1** and **SWITCH CLAMP 2** to emulate the clamps releasing.



**LAB: SS1****PROCESS FILE:** SS1**OBJECTIVE:** Using start-stop button with a holding circuit. Use of Normally Closed pushbutton.**SOFTWARE:** SIEMENS TIA Portal

Create the control of a START - STOP circuit for a single motor control with ON - OFF indicators.

**STOP-START CIRCUIT WITH INDICATOR****INPUT:**

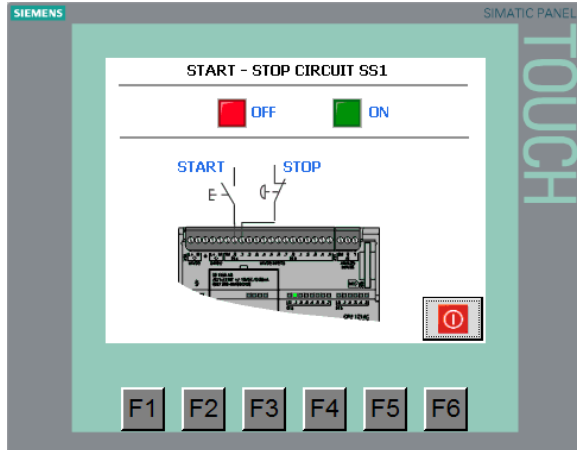
| Address | PLC Tag | Comments                      |
|---------|---------|-------------------------------|
| I0.0    | STOP    | Momentary NC Pushbutton Stop  |
| I0.1    | START   | Momentary NO Pushbutton Start |

**OUTPUT:**

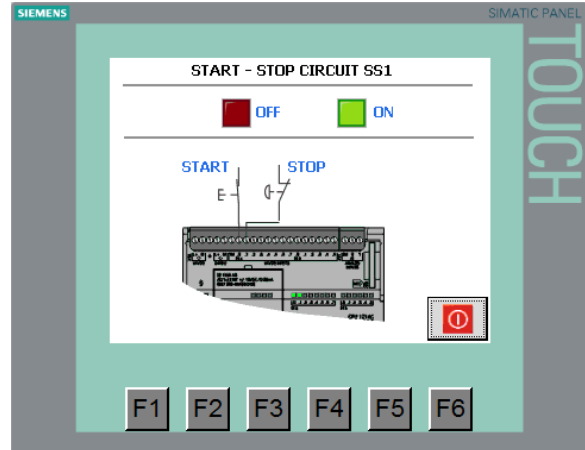
| Address | PLC Tag | Comments            |
|---------|---------|---------------------|
| Q0.0    | MOTOR   | Motor Control Relay |
| Q0.1    | PL1_ON  | Indicator Light ON  |
| Q0.2    | PL2_OFF | Indicator Light OFF |

## HMI and SWITCHES:

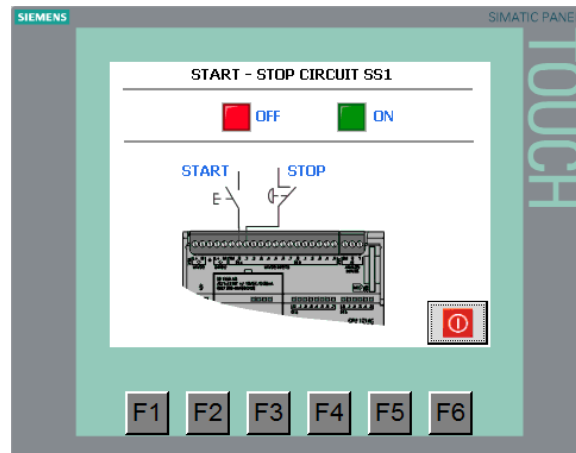
Initial HMI screen.



HMI screen when MOTOR started.



HMI screen when MOTOR stopped.



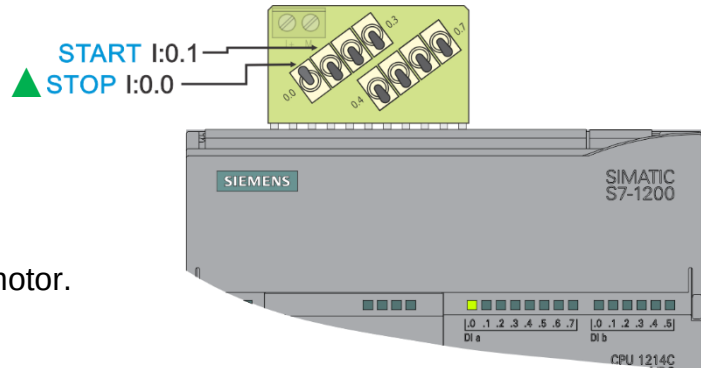
*continued*

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Initial switch settings, the STOP is wired as a **Normally Closed** field device, **push up ▲** the **STOP** to close the switch.

All the other switches are down.

Proceed to the next step to cycle the motor.



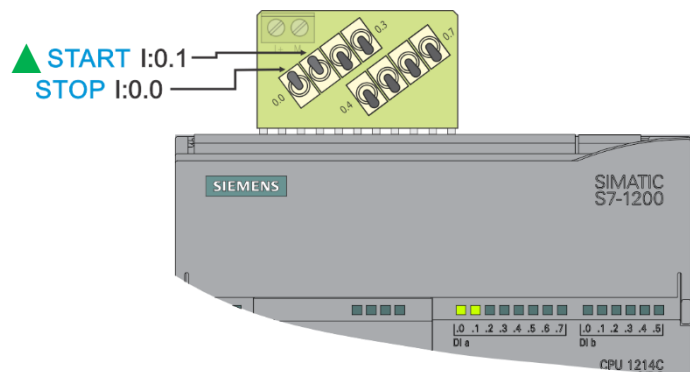
---

### Operator Pressing the FORWARD:

**Push up ▲** the **START** switch to close.

*The HMI **On** indicator is illuminated green and the **MOTOR** is running.*

Push down the **START** switch to open.



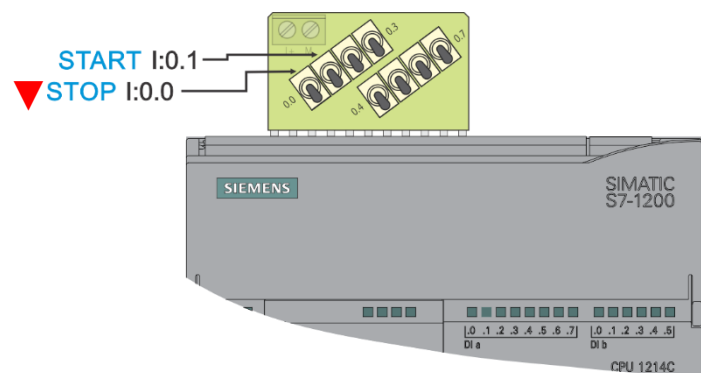
---

### Operator Pressing the STOP:

**Push down ▼** the **STOP** switch to open.

*The HMI **Off** indicator is illuminated red and the **Motor** is stopped.*

**Push up ▲** on the **STOP** switch to close.



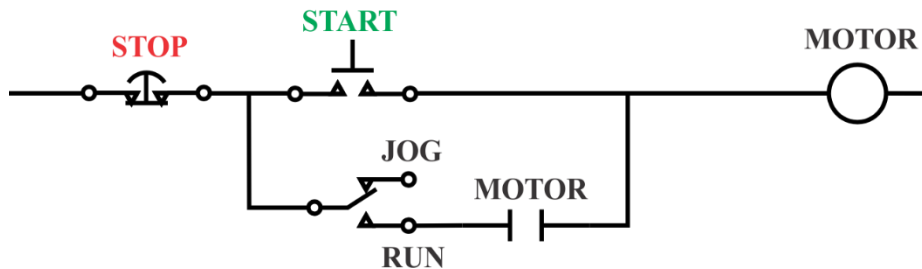


**LAB: SS2****PROCESS FILE:** SS2**OBJECTIVE:** Using selector switch with a holding circuit for a JOG/RUN circuit.**SOFTWARE:** SIEMENS TIA Portal

Create the control of a START - STOP circuit that will allow for the continuous running of a motor control, or by selecting the momentary (jogging) motion of the controlled motor when the START pushbutton is pressed.

**NOTE:** The selector switch for JOG - RUN is a two position maintained contact selector switch. Only the RUN contact of the switch is wired to the terminal on the PLC module.

### JOG/RUN SELECTOR SWITCH

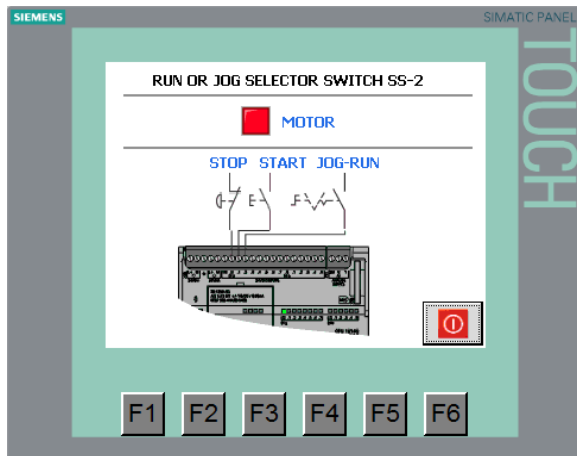
**INPUT:**

| Address | PLC Tag | Comments                       |
|---------|---------|--------------------------------|
| I0.0    | STOP    | Momentary NC Pushbutton Stop   |
| I0.1    | START   | Momentary NO Pushbutton Start  |
| I0.2    | SEL_SW1 | Selector Switch for JOG or RUN |

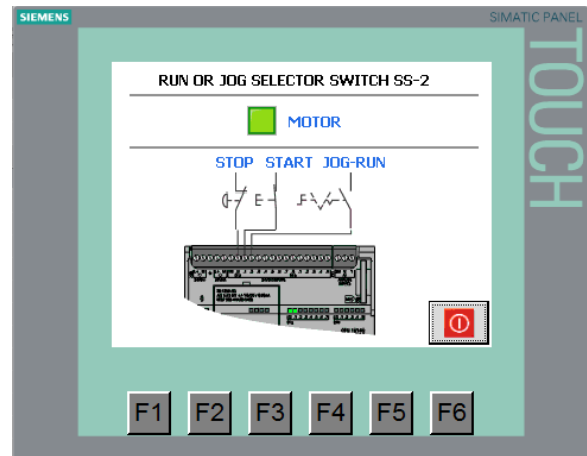
**OUTPUT:**

| Address | PLC Tag | Comments            |
|---------|---------|---------------------|
| Q0.0    | MOTOR   | Motor Control Relay |

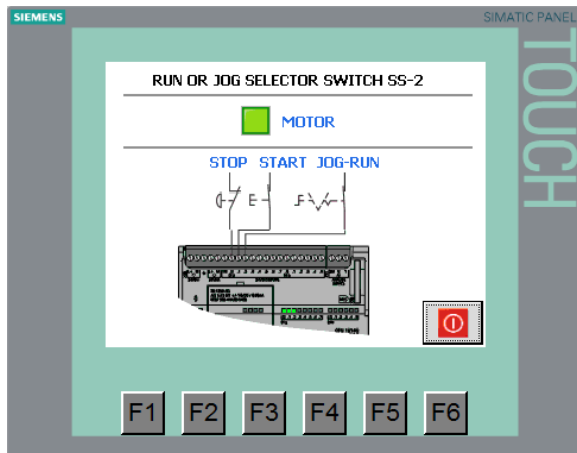
Initial HMI screen.



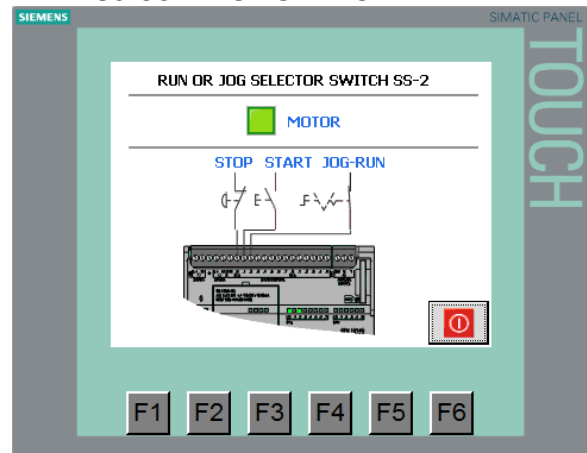
HMI screen for START- MOTOR JOG.



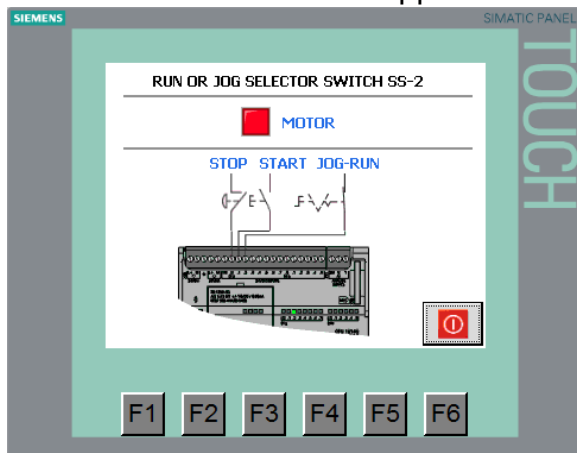
HMI screen for START-MOTOR RUN.



HMI screen MOTOR RUN

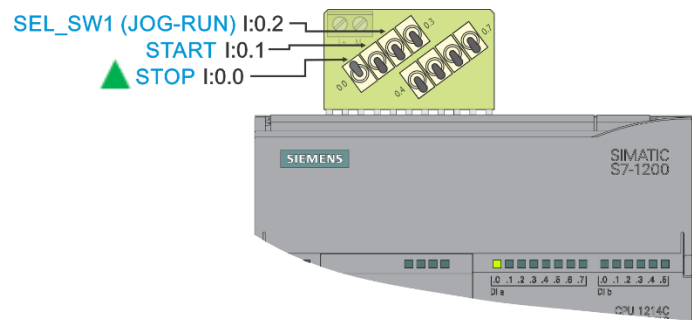


HMI screen for MOTOR stopped.



---

Initial switch settings, the STOP is wired as a **Normally Closed** field device, **push up ▲** the **STOP** to close the switch.

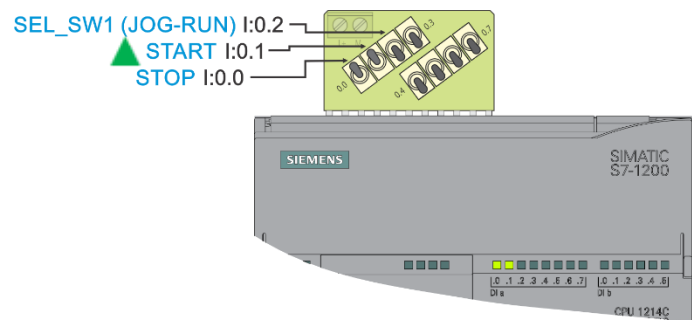


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Operator Pressing the START to JOG:

**Push up ▲** the **START** switch to close.

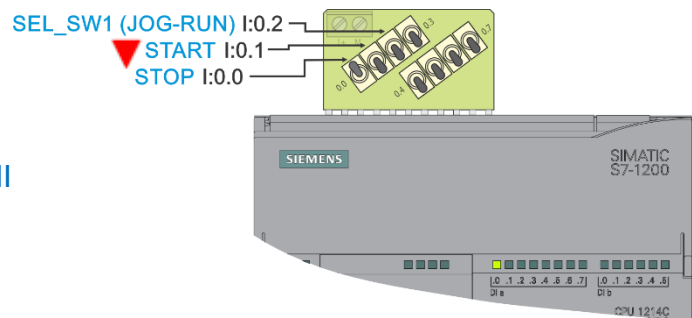
*The MOTOR is running, and the HMI indicator is illuminated green.*



-- THEN --

**Push down ▼** the **START** switch to open.

*The MOTOR is stopped, and the HMI indicator is illuminated red.*

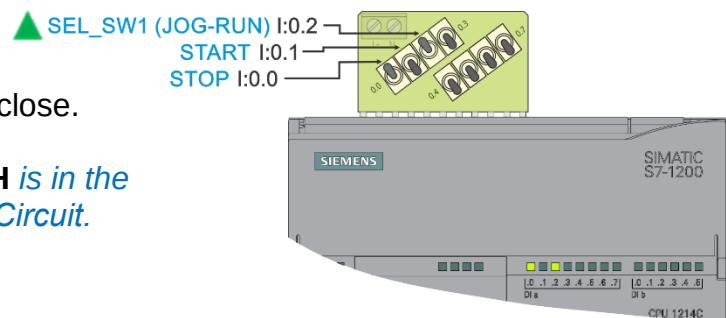


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Operator Switching to RUN mode:

**Push up ▲** the **SEL\_SW1** switch to close.

*The JOG-RUN SELECTOR SWITCH is in the Jog position to “enable” the Holding Circuit.*



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Operator Pressing the START to RUN:

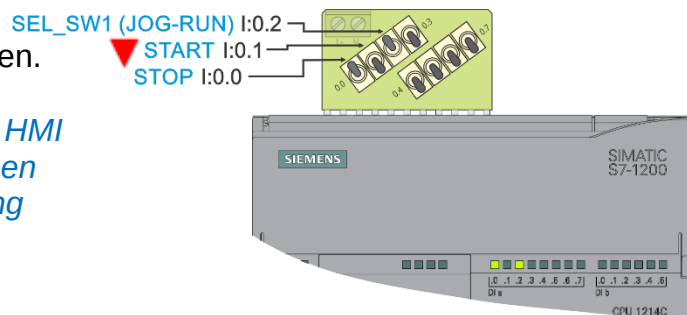
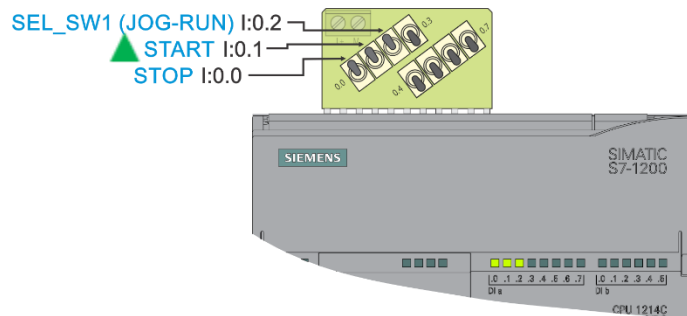
Push up ▲ the **START** switch to close.

*The **MOTOR** is running, and the HMI indicator is illuminated green.*

-- THEN --

Push down ▼ the **START** switch to open.

*The **MOTOR** is running, and the HMI indicator remains illuminated green because of the alternative Holding Circuit logic continuity path.*



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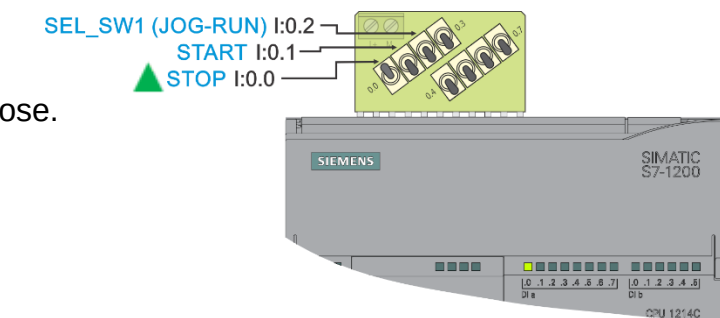
Operator Pressing the STOP:

Push down ▼ on the **STOP** switch to open.

*The HMI indicator is illuminated red, and the **MOTOR** is stopped.*

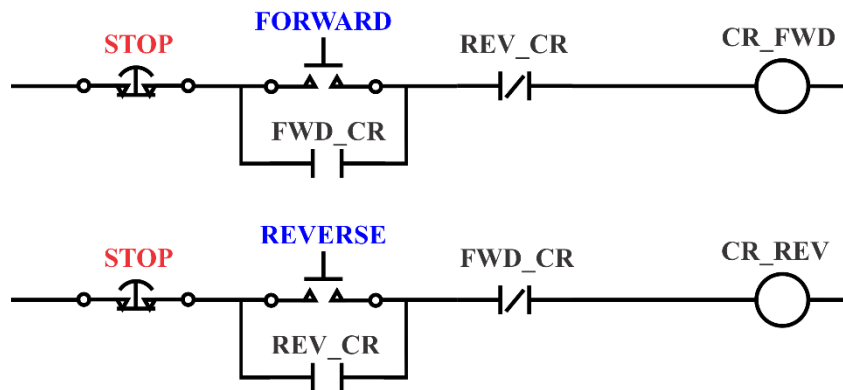
-- THEN --

Push up ▲ on the **STOP** switch to close.



**LAB: SS3****PROCESS FILE:** SS3**OBJECTIVE:** Using simple start-stop circuits to illustrate holding circuits and the programming of normally closed field devices.**SOFTWARE:** SIEMENS TIA Portal

Create the control of a single motor with FORWARD and REVERSE control relays where the motor must be stopped before changing directions.

**FORWARD/REVERSE WITH STOPPING****INPUT:**

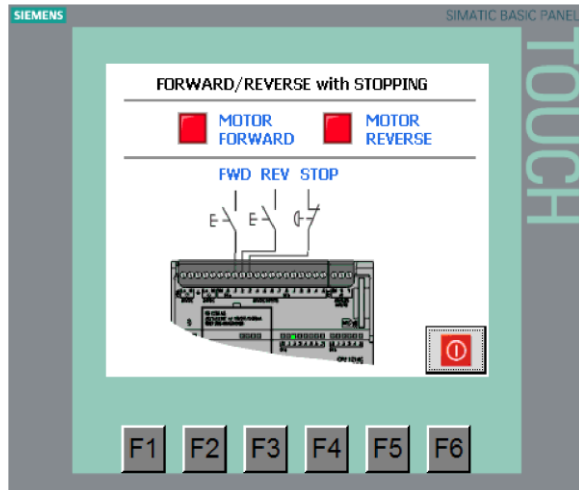
| Address | PLC Tag | Comments                        |
|---------|---------|---------------------------------|
| I0.0    | FWD     | Momentary NO Pushbutton Forward |
| I0.1    | REV     | Momentary NO Pushbutton Reverse |
| I0.2    | STOP    | Momentary NC Pushbutton Stop    |

**OUTPUT:**

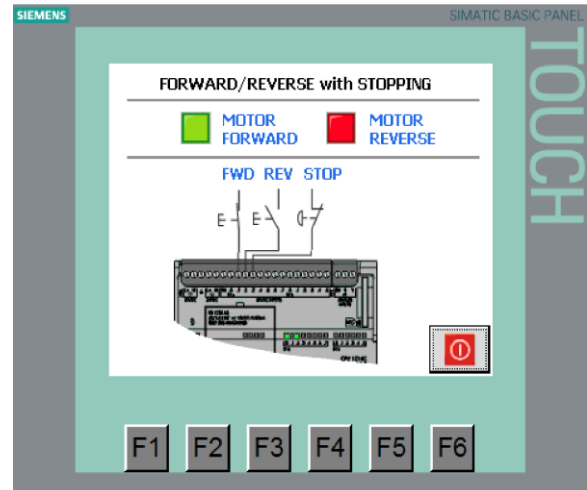
| Address | PLC Tag | Comments                    |
|---------|---------|-----------------------------|
| Q0.0    | CR_FWD  | Motor Control Relay Forward |
| Q0.1    | CR_REV  | Motor Control Relay Reverse |

## HMI and SWITCHES:

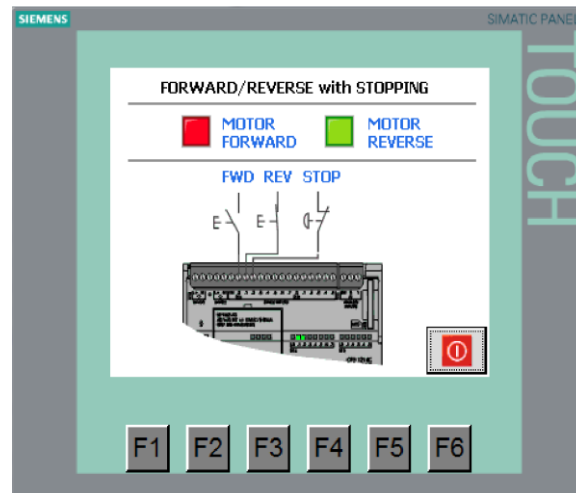
Initial HMI screen.



HMI screen when motor is Forward.



HMI screen when motor is Reversed.

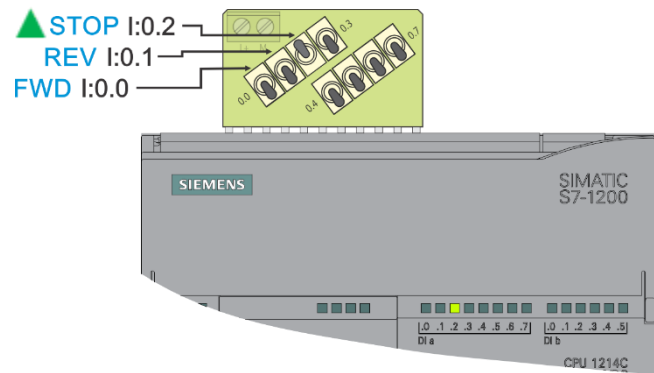


*continued*

---

Initial switch settings, the STOP is wired as a Normally Closed field device, **push up ▲ STOP** to close the switch.

Proceed to the next step to cycle the motor FWD and REV.



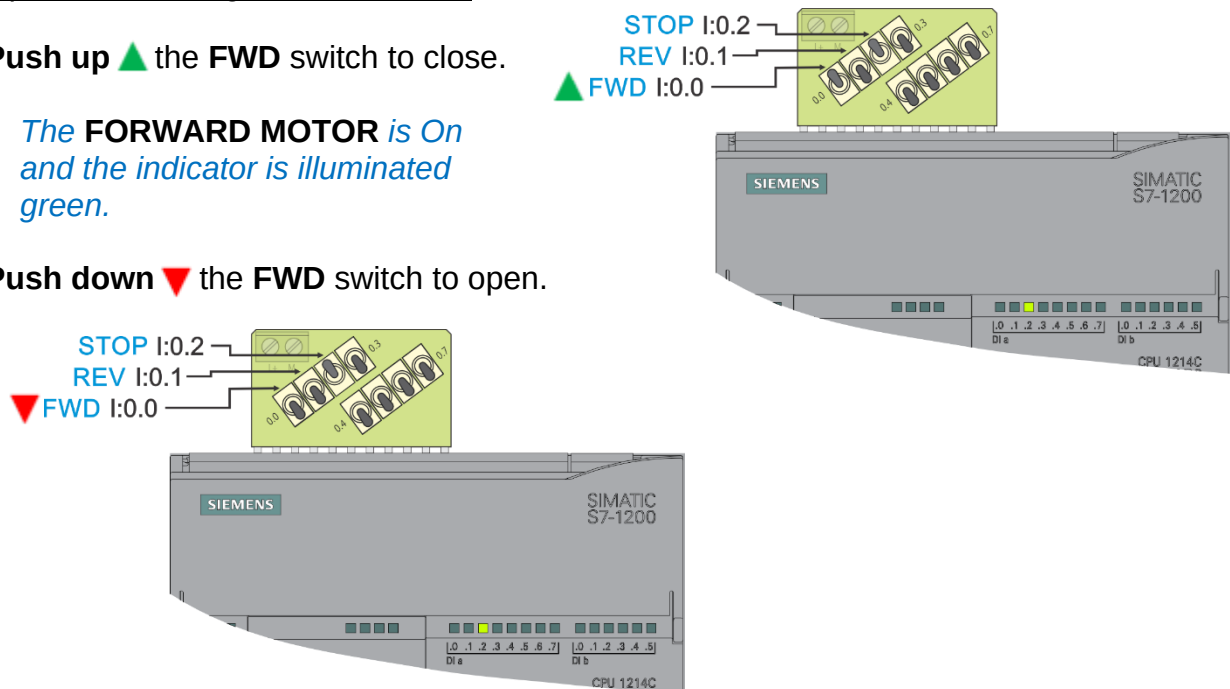
---

Operator Pressing the FORWARD:

**Push up ▲ the FWD switch to close.**

*The FORWARD MOTOR is On and the indicator is illuminated green.*

**Push down ▼ the FWD switch to open.**



*continued*

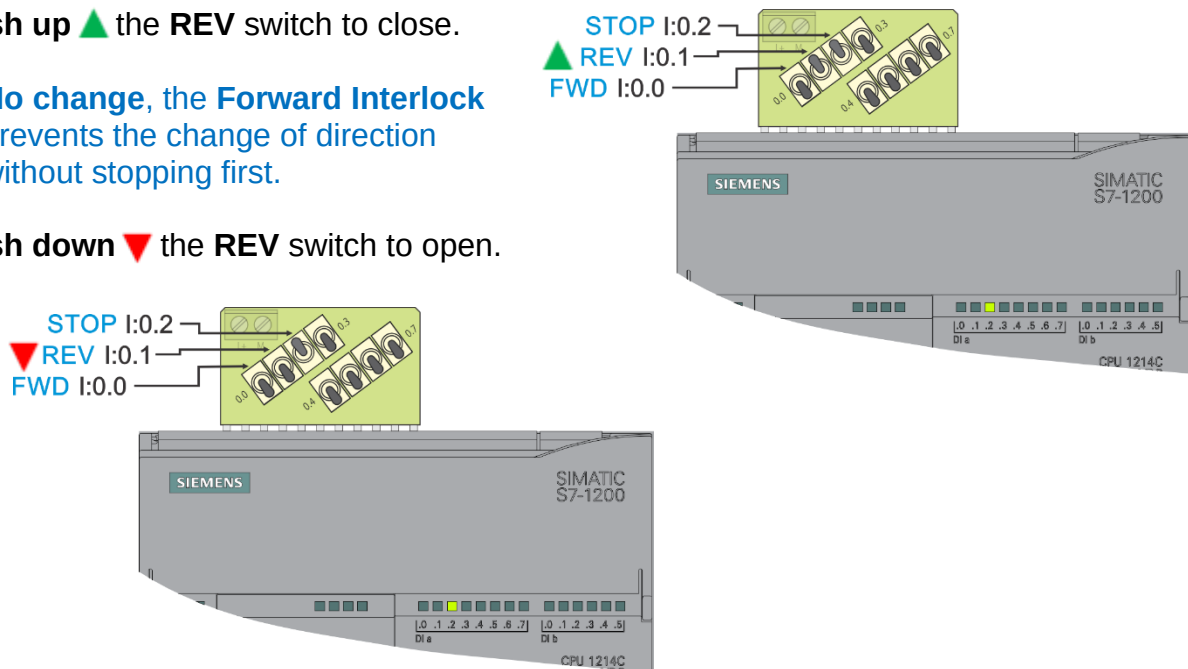
---

### Operator Pressing the REVERSE:

Push up ▲ the **REV** switch to close.

**No change**, the **Forward Interlock** prevents the change of direction without stopping first.

Push down ▼ the **REV** switch to open.



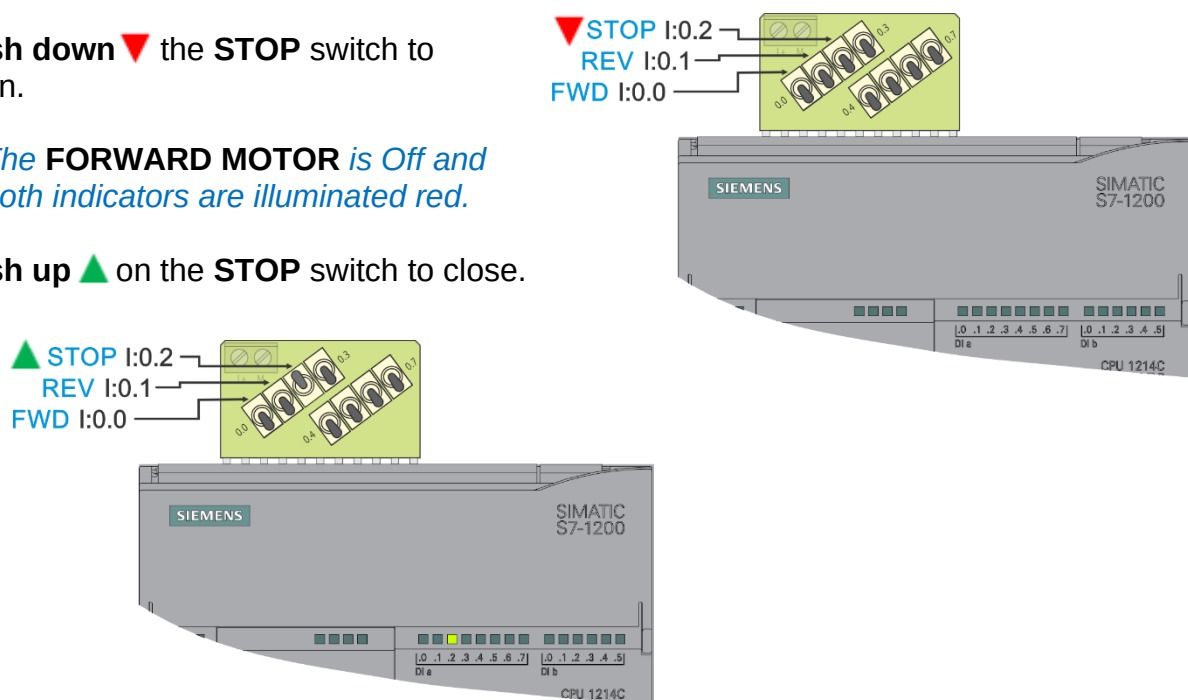
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### Operator Pressing the STOP:

Push down ▼ the **STOP** switch to open.

*The **FORWARD MOTOR** is Off and both indicators are illuminated red.*

Push up ▲ on the **STOP** switch to close.





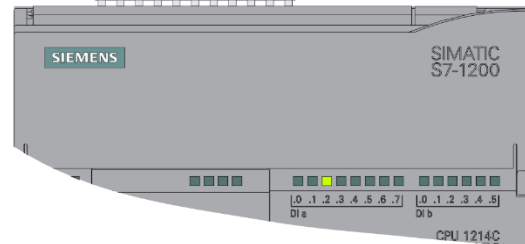
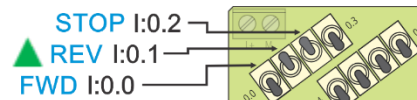
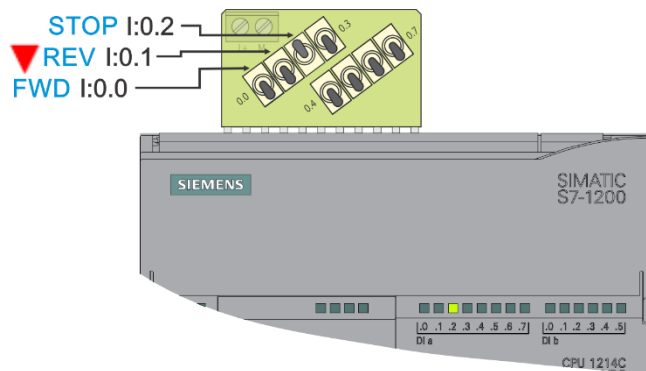
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### Operator Pressing the REVERSE:

Push up ▲ the **REV** switch to close.

*The **REVERSE MOTOR** is On and the indicator is illuminated green*

Push down ▼ the **REV** switch to open.



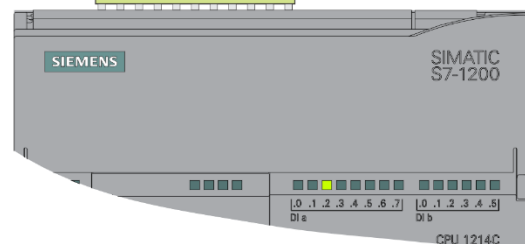
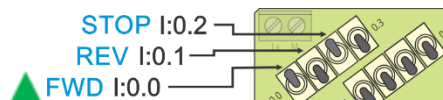
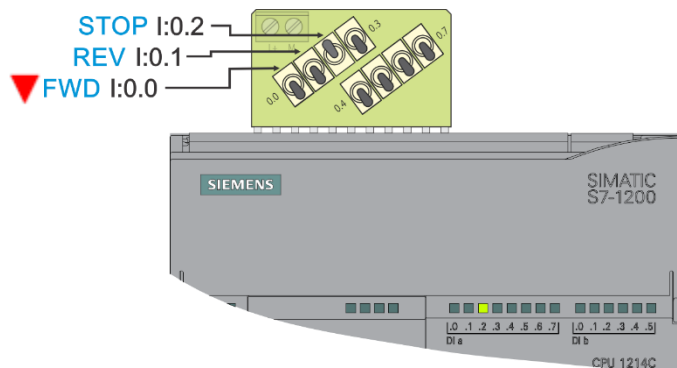
---

### Operator Pressing the FORWARD:

Push up ▲ the **FWD** switch to close.

*No change, the **Reverse Interlock** prevents the change of direction without stopping first.*

Push down ▼ the **FWD** switch to open.



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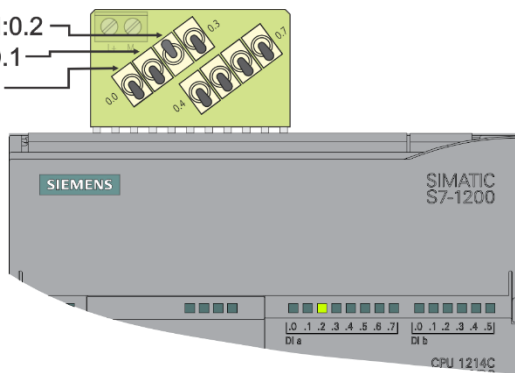
### Operator Pressing the STOP:

Push down ▼ the **STOP** switch to open.

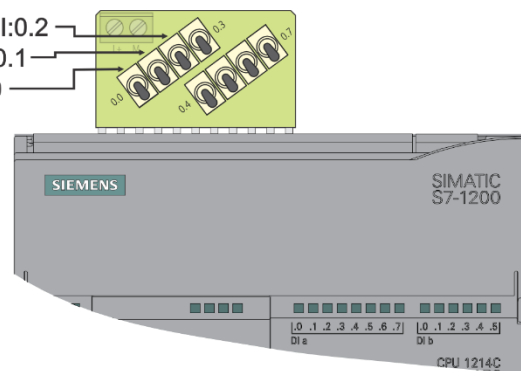
*The REVERSE MOTOR is Off and both indicators are illuminated red.*

Push up ▲ the **STOP** switch to close.

▲ **STOP** I:0.2  
REV I:0.1  
FWD I:0.0



▼ **STOP** I:0.2  
REV I:0.1  
FWD I:0.0



**LAB: CONVEYOR****PROCESS FILE:** CONVEYOR**OBJECTIVE:** Programming normally closed field devices.**SOFTWARE:** SIEMENS TIA Portal

Create a program for a conveyor system to transport a single parts container. When a bin is placed on the line, the operator will energize the conveyor via a Start/Stop pushbutton panel. When the bin reaches the end of the line, a photoelectric switch de-energizes the conveyor.

The photoelectric switch is wired normally closed and any absence of the signal, that is the beam is interrupted, will stop the conveyor.

When creating the tag, the description MUST BE entered.

**INPUT:**

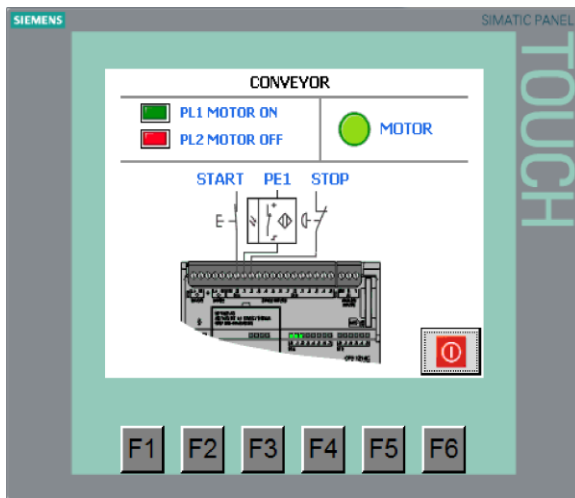
| Address | PLC Tag | Description                          |
|---------|---------|--------------------------------------|
| I0.0    | START   | Momentary Normally Open Pushbutton   |
| I0.1    | PE1     | Photoelectric Switch Normally Closed |
| I0.2    | STOP    | Momentary Normally Closed Pushbutton |

**OUTPUT:**

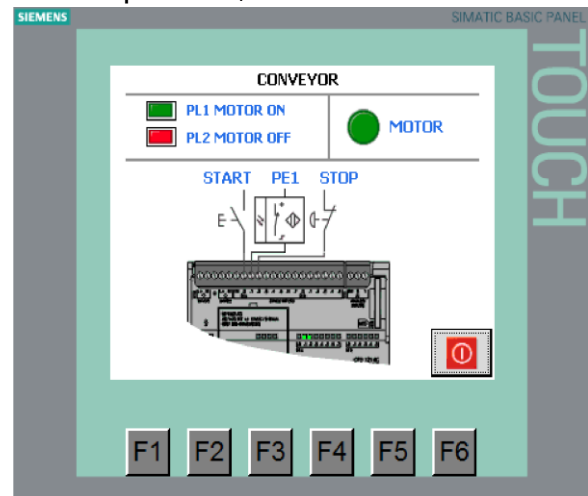
| Address | PLC Tag | Description                            |
|---------|---------|----------------------------------------|
| Q0.0    | MCR1    | Motor Control Relay for Conveyor Motor |
| Q0.1    | PL1     | Pilot Light (GREEN) Conveyor ON        |
| Q0.2    | PL2     | Pilot Light (RED) Conveyor OFF         |

## HMI and SWITCHES:

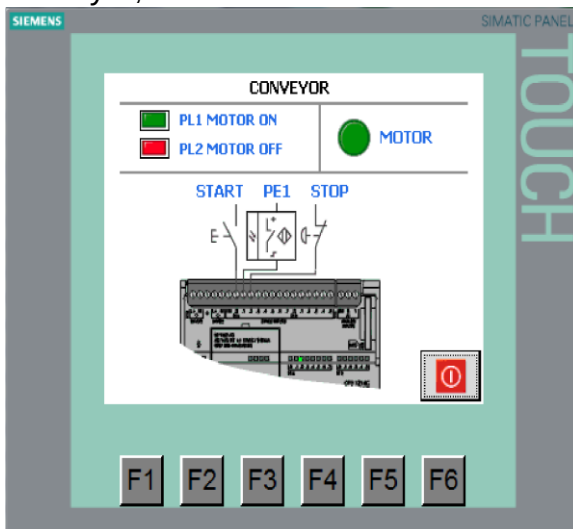
Initial HMI screen.



START pressed, Motor ON.



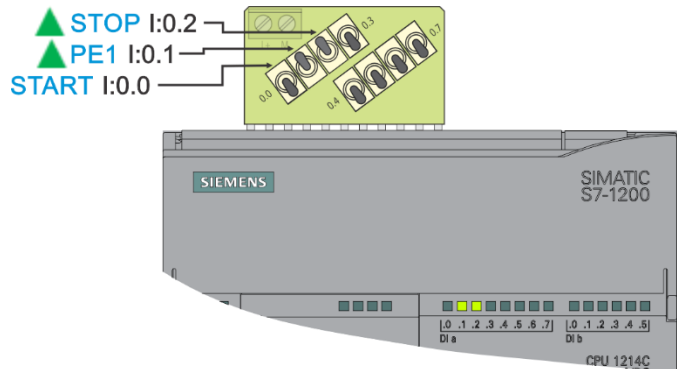
Proximity switch open, box at end of conveyor, Motor OFF.



*continued*

---

Initial switch settings, push up ▲ PE1 and STOP switches to close.

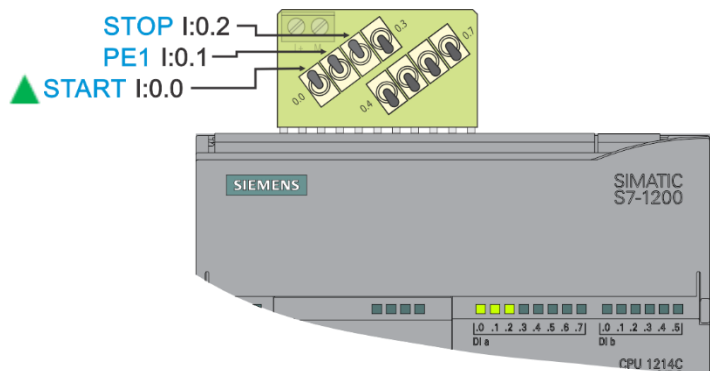


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Operator Pressing the START pushbutton:

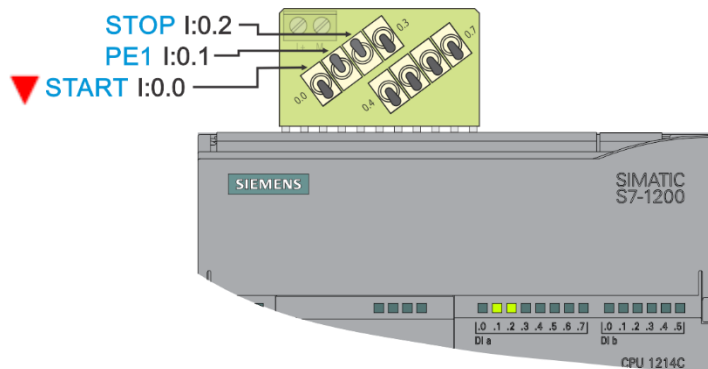
Push up ▲ the START switch to close.

*The PL1 Motor ON indicator is illuminated green.*



-- THEN --

Push down ▼ the START switch to open.



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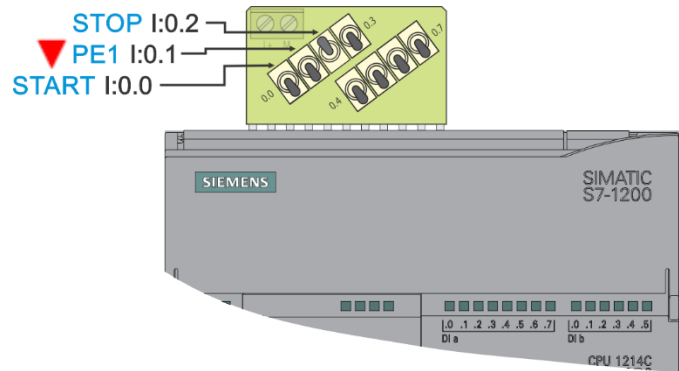
*continue*

---

Interrupting of the Photoelectric at the end of the conveyor:

**Push down ▼ the PE1 switch to open.**

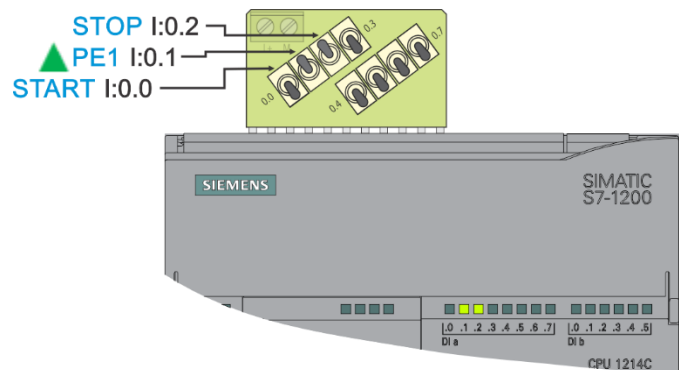
*The PL2 Motor OFF indicator is illuminated red.*



---

Return Photoelectric to normal state:

**Push up ▲ the PE1 switch to close.**



---

**Push up ▲ and push down ▼ START.**



*The PL1 Motor ON indicator is illuminated green*

---

**Push down ▼ and push up ▲ STOP.**



*The PL2 Motor OFF indicator is illuminated red*

**LAB: MIXING VAT****PROCESS FILE:** VAT**OBJECTIVE:** Use of internal to control sequence of operations and scope of tags.**SOFTWARE:** SIEMENS TIA Portal

When a START pushbutton is pressed/released the filler valve on the vat will open and the drain valve will close. A normally closed float switch will open when the tank is filled, and the filler valve will be shutoff. The motor will then start to mix the contents of the tank.

The pressing/releasing of the STOP pushbutton will turn off the motor and open the drain valve.

NOTES: The pressing of START pushbutton closes the de-energizes the drain valve.

When de-energized by the logic, the Filler valve is closed.

*Emulation\_Data and Emulation\_Prog are used by the HMI Tank animation.*

When creating the tag, the description MUST BE entered.

**INPUT:**

| Address | PLC Tag      | Comments                         |
|---------|--------------|----------------------------------|
| I0.0    | START        | START Pushbutton Momentary N. O. |
| I0.1    | FLOAT_SWITCH | Vat Filled Float Switch N. C.    |
| I0.2    | STOP         | STOP Pushbutton Momentary N. C.  |

**OUTPUT:**

| Address | PLC Tag      | Comments          |
|---------|--------------|-------------------|
| Q0.0    | DRAIN_VALVE  | Vat Drain Valve   |
| Q0.1    | FILLER_VALVE | Vat Filler Valve  |
| Q0.2    | MIX_MOTOR    | Vat Motor Control |

*continued*

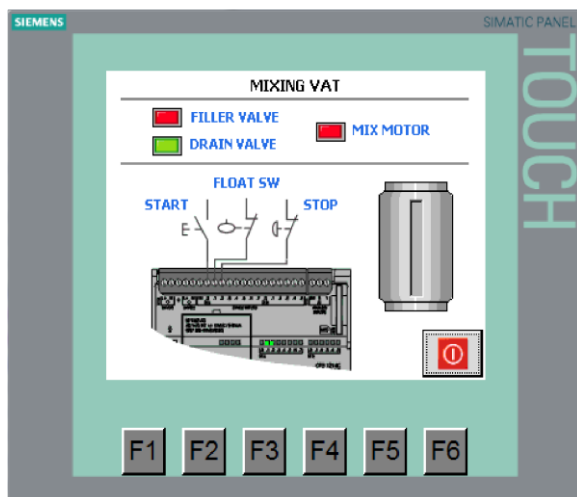
## INTERNAL: Data\_block\_1

[DB1] the Data Block and Tag has been created.

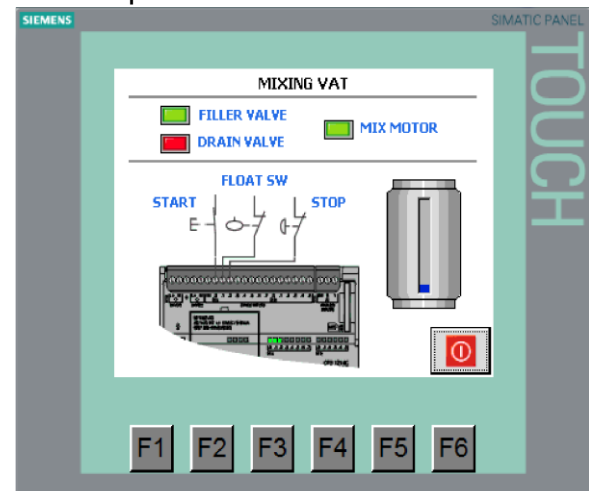
| Static Tag | Data Type | Comments                                       |
|------------|-----------|------------------------------------------------|
| INTERNAL   | Boolean   | Enable<br>1- Fill/Mix Cycle<br>0 - Drain Cycle |

## HMI and SWITCHES:

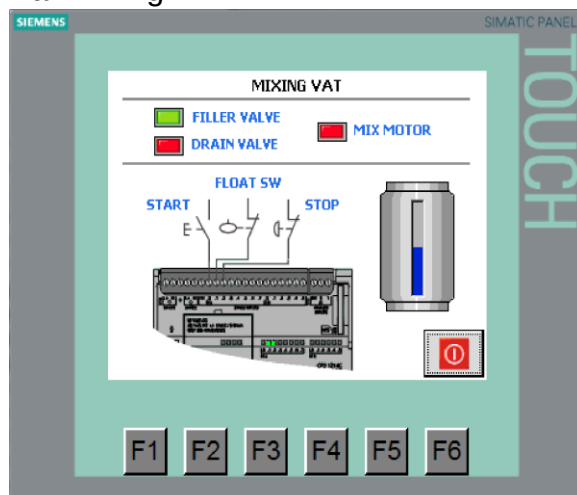
Initial HMI screen.



START pressed.

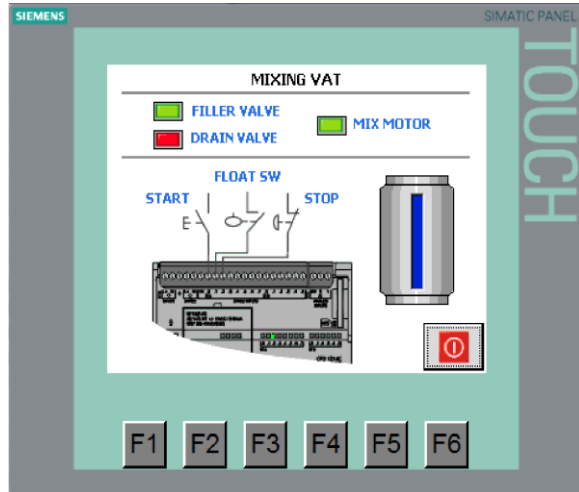


Tank filling

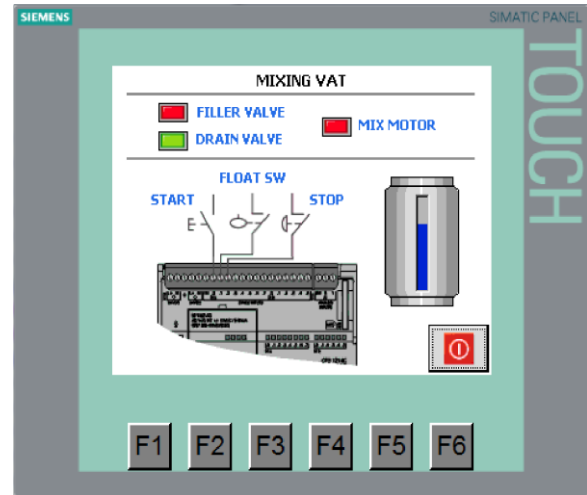




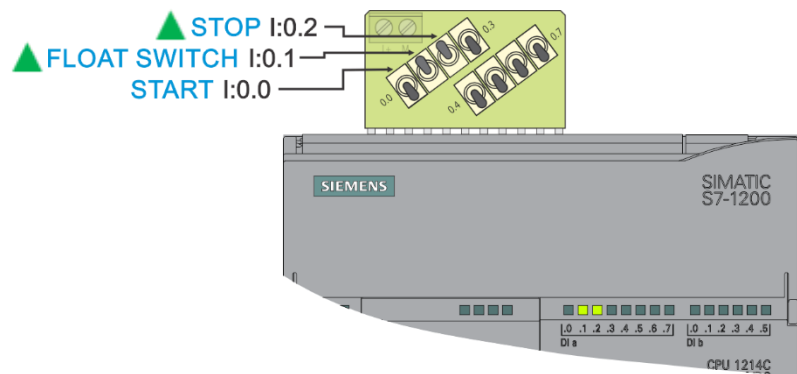
Tank filled, Mixing Motor ON.



STOP pressed, tank draining.



Initial switch settings, **push up ▲** the **FLOAT SWITCH** and **STOP** switches to close.



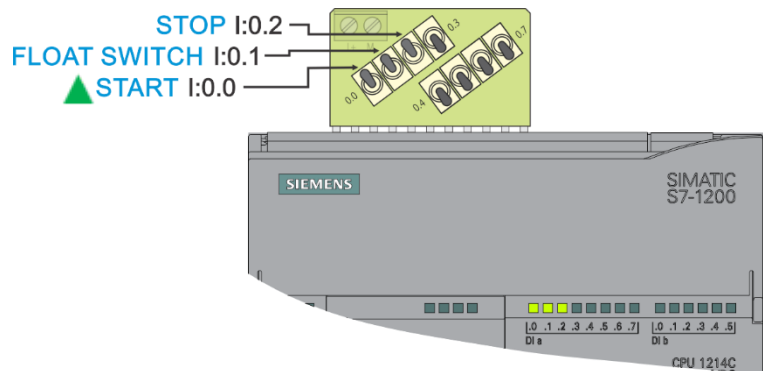
*continued*

---

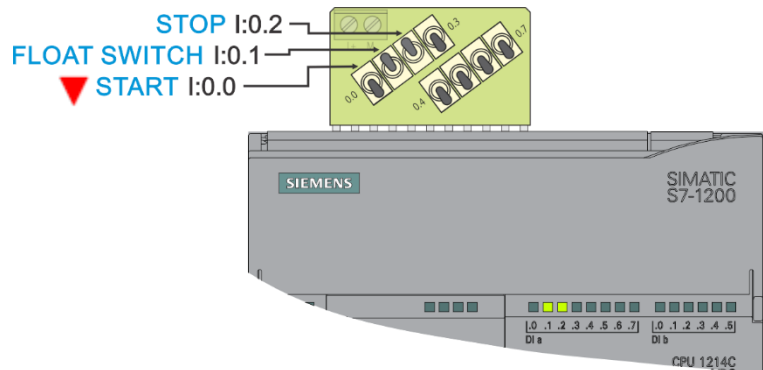
Operator Pressing the START pushbutton:

Push up ▲ the **START** switch to close.

*The DRAIN VALVE turns OFF and the FILLER VALVE will turn ON.*



Push down ▼ the **START** switch to open.

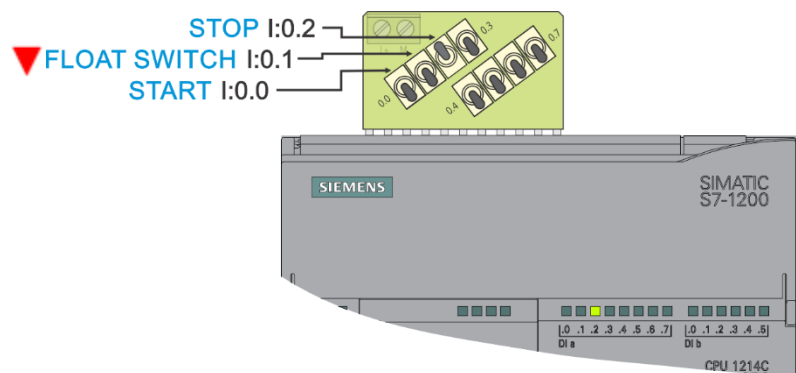


---

Full tank opens FLOAT switch:

Push down ▼ the **FLOAT SWITCH** to open.

*The FILLER VALVE will turn OFF and the MIXER MOTOR will turn ON.*

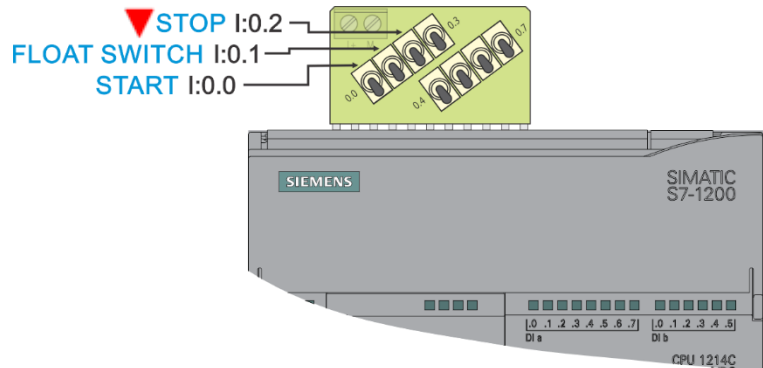


---

Operator Pressing the STOP pushbutton:

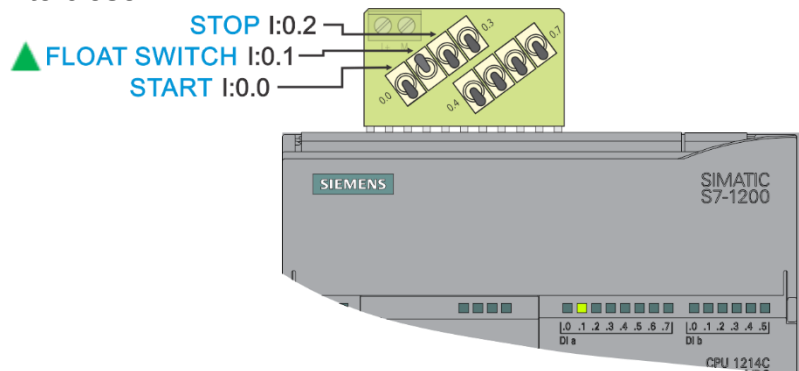
Push down ▼ the **STOP** switch to open.

The **MIXER MOTOR** will turn OFF and the **DRAIN VALVE** turns ON.

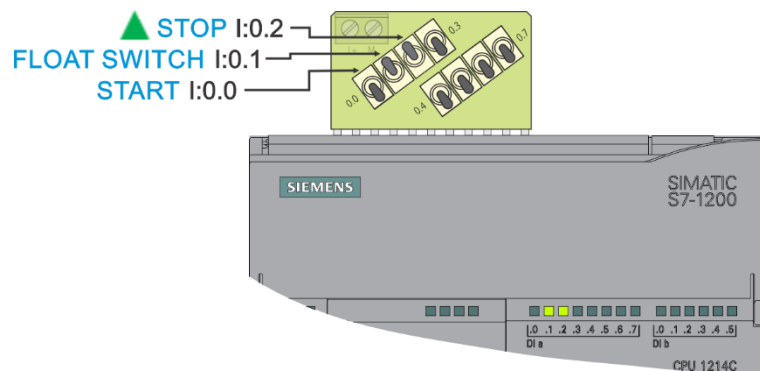


Return FLOAT to normal state:

Push up ▲ the **FLOAT SWITCH** to close.



Push up ▲ the **STOP** switch to close.



## LAB: TIMER ON STATUS BITS

**PROCESS FILE:** TIMER ON

**OBJECTIVE:** Program a TIMER ON instruction and the status bits.

**SOFTWARE:** SIEMENS TIA Portal

Create the program and complete the following questions. All inputs are XIC instructions.

When creating the tag, the description MUST BE entered.

### INPUT:

| Address | PLC Tag | Comments                   |
|---------|---------|----------------------------|
| I0.0    | SWITCH  | Timer Control N. O. Switch |

### OUTPUT:

| Address | PLC Tag | Comments              |
|---------|---------|-----------------------|
| Q0.0    | IN      | IN Status Indicator   |
| Q0.1    | DONE    | DONE Status Indicator |

### TIMER:

*(Do not comment)*

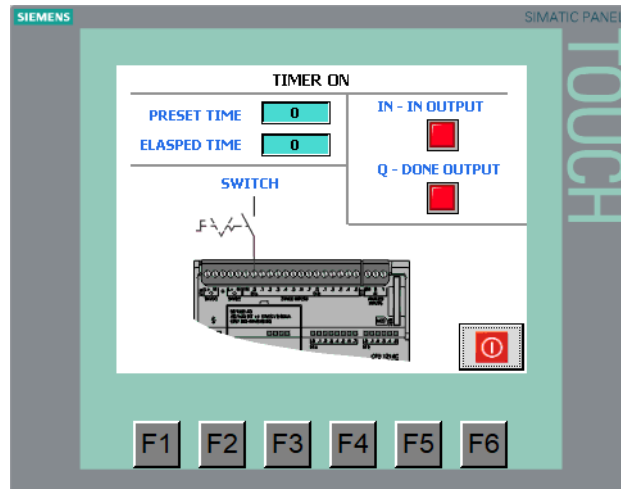
| Timer Data Block Tag       | Comments        |
|----------------------------|-----------------|
| DWELL_ON                   | Timer On        |
| DWELL_ON.IN <sup>(1)</sup> | IN Status Bit   |
| DWELL_ON.Q <sup>(1)</sup>  | DONE Status Bit |

<sup>(1)</sup> Timer status bits created automatically

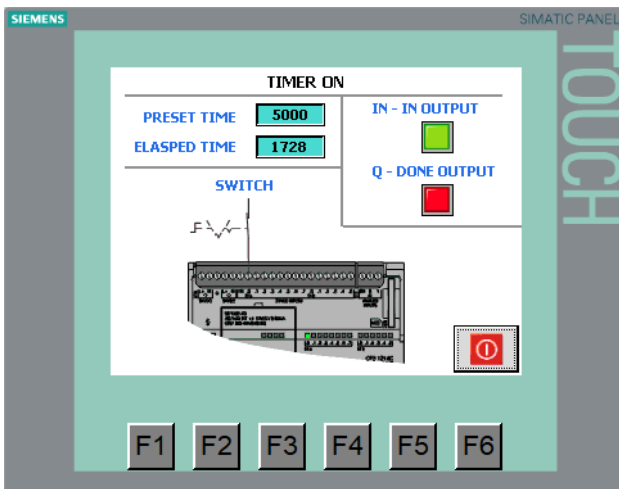
*continued*

## HMI and SWITCH:

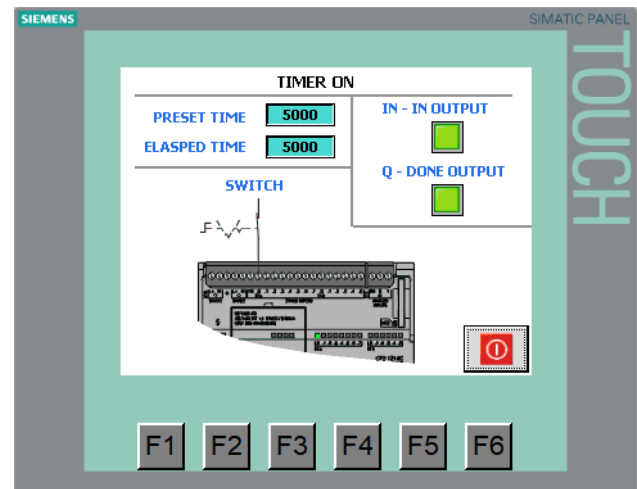
SWITCH opened, initial HMI screen.



SWITCH closed, Timer elapsing.



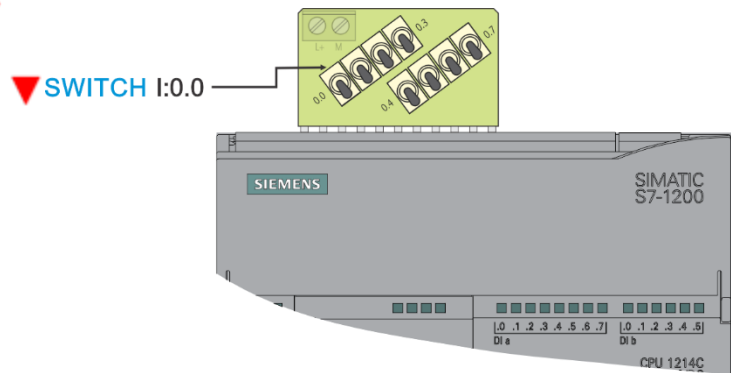
SWITCH closed, Timing complete.



*continued*

---

Initial switch settings, **push down ▼** the **SWITCH**.



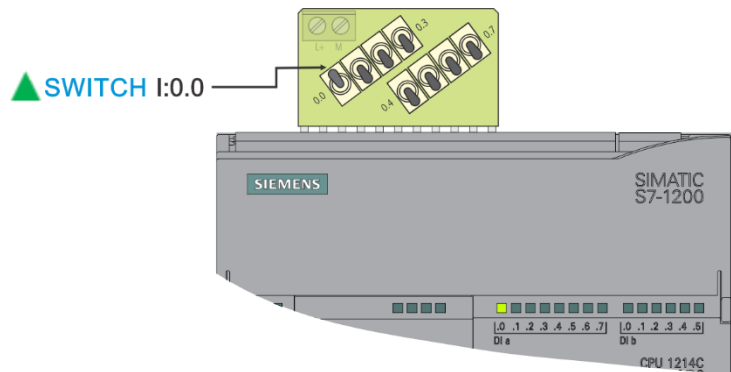
---

**Push up ▲** the **SWITCH** to close.

*IN bit turns On.*

*The PRESET VALUE PV begins elapsing.*

*When PRESET VALUE PV and ELAPSED VALUE EY are equal, Q DONE turns to ON.*

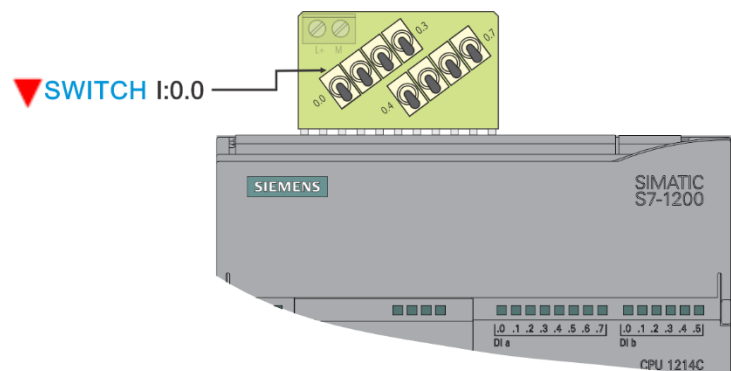


---

When Q DONE is **ON**, **push down ▼** the **SWITCH** to open.

*IN bit turns Off.*

*Q DONE turns Off and ELAPSED VALUE EV returns to zero.*



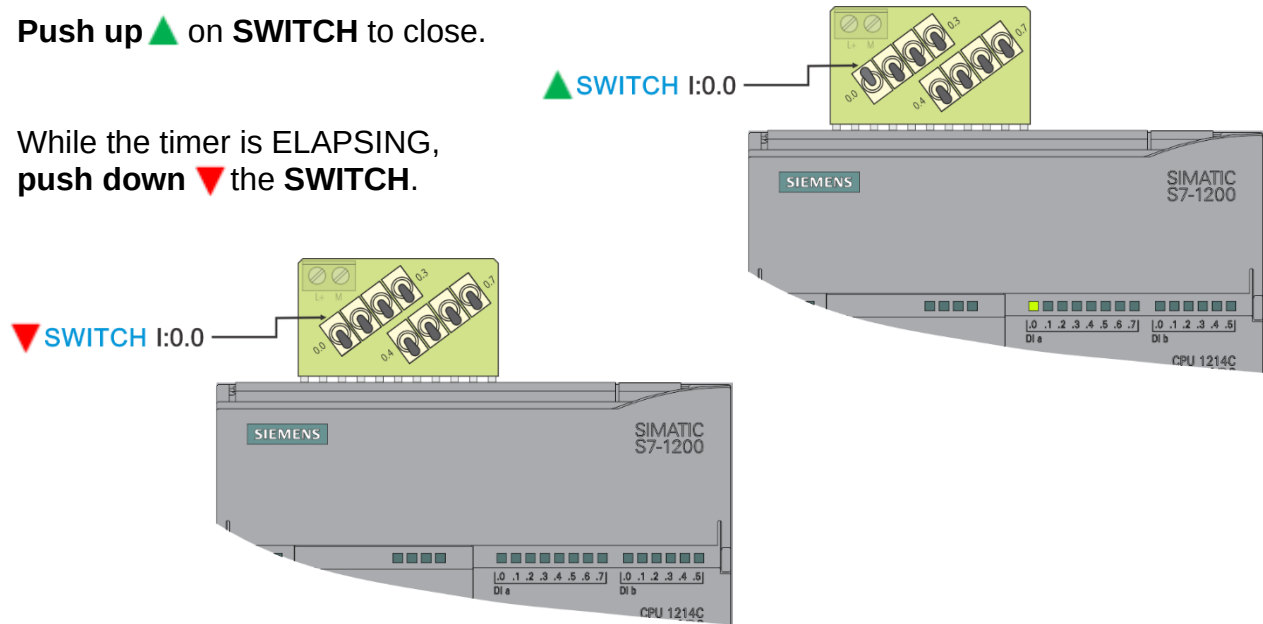
---

*continued*

---

Push up ▲ on **SWITCH** to close.

While the timer is ELAPSING,  
push down ▼ the **SWITCH**.



*ELAPSED VALUE EV returns to zero*

---

## LAB: TIMER OFF STATUS BITS

**PROCESS FILE:** TIMER OFF

**OBJECTIVE:** Program a TIMER OFF instruction and the status bits.

**SOFTWARE:** SIEMENS TIA Portal

Create the program and complete the following questions. All inputs are XIC instructions.

When creating the tag, the description MUST BE entered.

### INPUT:

| Address | PLC Tag | Comments                   |
|---------|---------|----------------------------|
| I0.0    | SWITCH  | Timer Control N. O. Switch |

### OUTPUT:

| Address | PLC Tag | Comments              |
|---------|---------|-----------------------|
| Q0.0    | IN      | IN Status Bit         |
| Q0.1    | DONE    | DONE Status Indicator |

### TIMER:

*(Do not comment)*

| Timer Data Block Tag        | Comments        |
|-----------------------------|-----------------|
| DWELL_OFF                   | Timer Off       |
| DWELL_OFF.IN <sup>(1)</sup> | IN Status Bit   |
| DWELL_OFF.Q <sup>(1)</sup>  | DONE Status Bit |

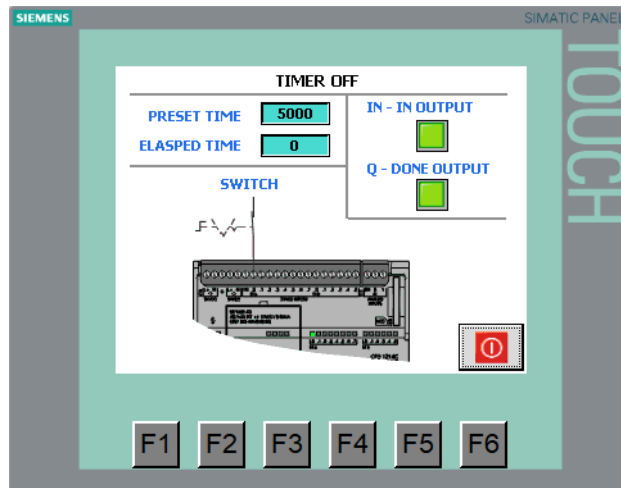
<sup>(1)</sup> Timer status bits created automatically

*continued*

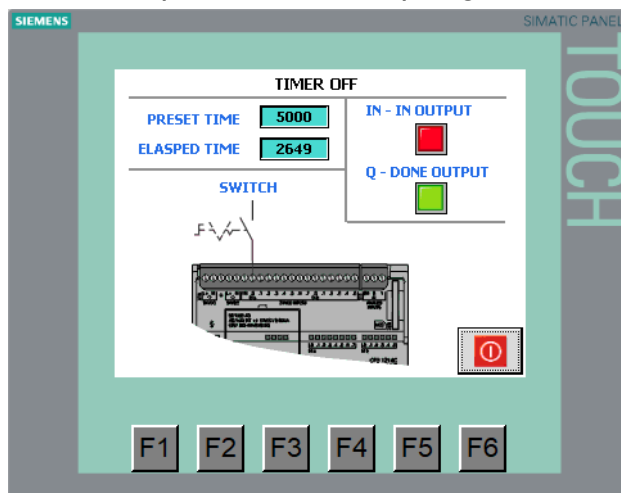


## HMI and SWITCH:

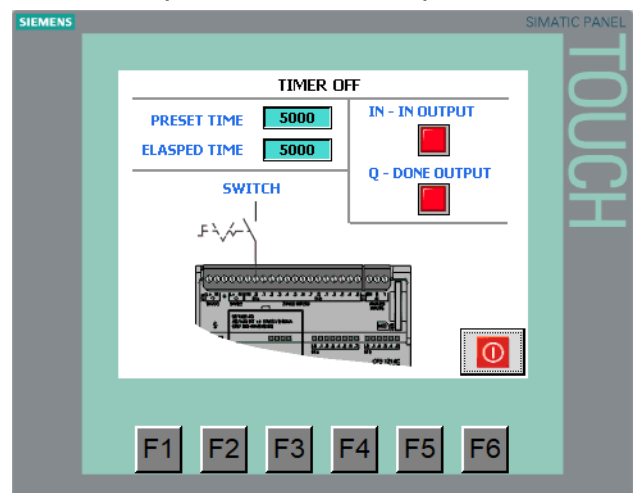
SWITCH closed, initial HMI screen.



SWITCH opened, Timer elapsing.

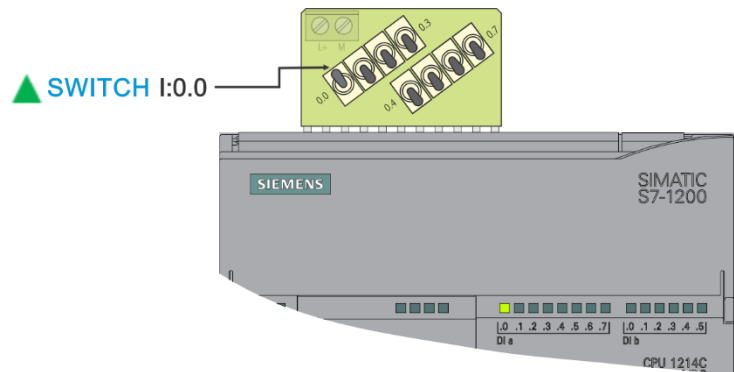


SWITCH opened, Timer complete.



---

Initial switch settings,  
**push up ▲** the **SWITCH**  
to close.



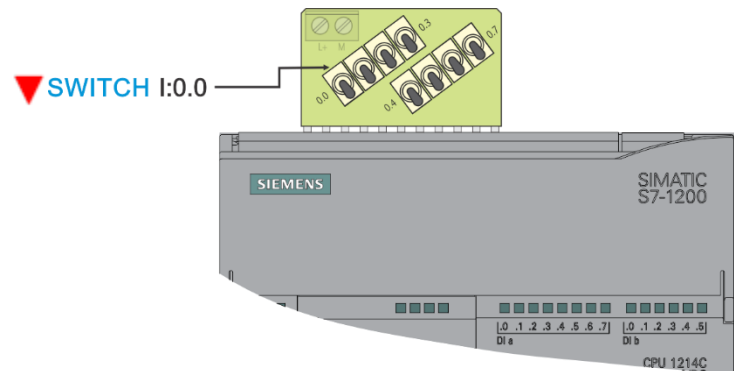
---

**Push down ▼** the **SWITCH** to  
open.

*IN bit turns OFF.*

*The PRESET VALUE PV begins  
elapsing*

*When PRESET VALUE PV and  
ELAPSED VALUE EV are equal,  
Q DONE turns to OFF.*

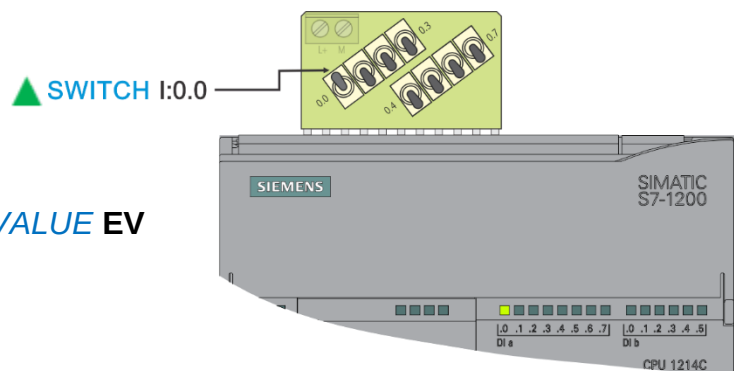


---

When **Q DONE** is **Off**, push up ▲  
the **SWITCH** to close.

*IN bit turns ON.*

*Q DONE turns On and ELAPSED VALUE EV  
returns to zero.*



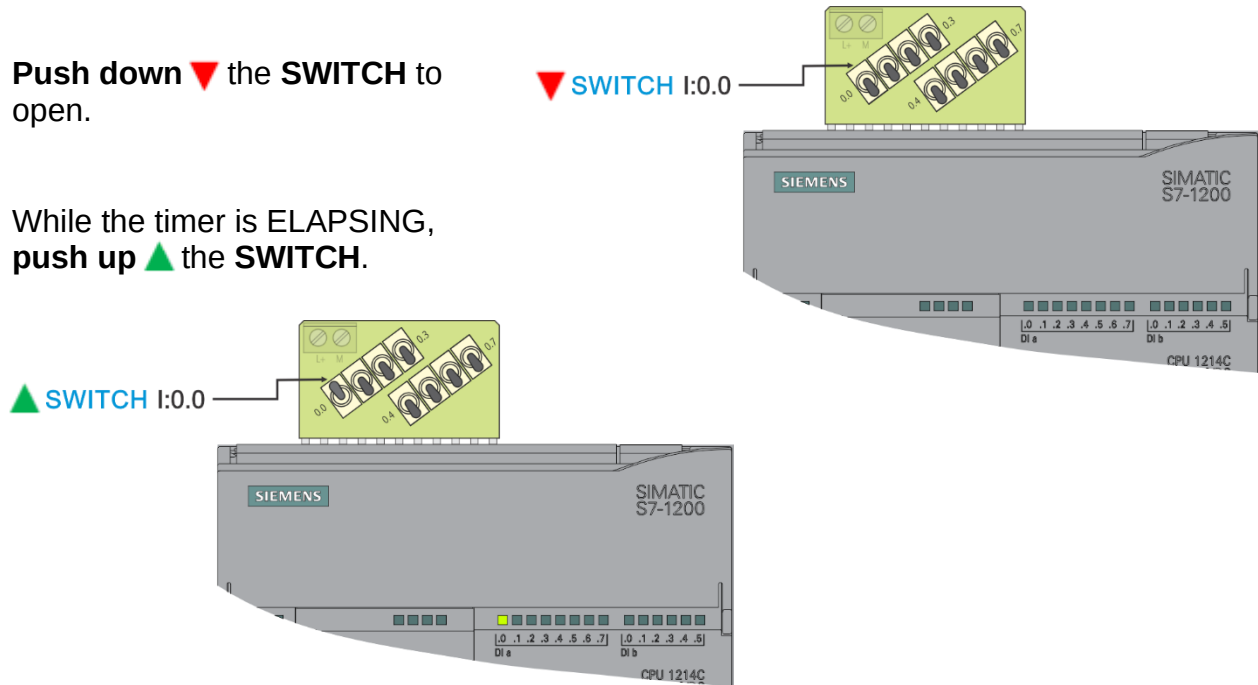
---

*continued*

---

Push down ▼ the **SWITCH** to open.

While the timer is ELAPSING,  
push up ▲ the **SWITCH**.



*ELAPSED VALUE EV returns to zero.*

---

**LAB: STARTUP TIMER****PROCESS FILE:** STARTUP**OBJECTIVE:** Determine the type of timer required for the required logic and program the appropriate relay instruction controlled by the DONE bit.**SOFTWARE:** SIEMENS TIA Portal

When a START pushbutton is pressed/released, a warning horn will sound for five (5) seconds. When the horn turns off, an output will then be turned on to run a conveyor. The conveyor can be turned off by the STOP pushbutton.

When creating the tag, the description MUST BE entered.

**INPUT:**

| Address | PLC Tag | Comments                        |
|---------|---------|---------------------------------|
| I0.0    | START   | Momentary N.O. START Pushbutton |
| I0.3    | STOP    | Momentary N.C. STOP Pushbutton  |

**OUTPUT:**

| Address | PLC Tag | Comments       |
|---------|---------|----------------|
| Q0.0    | HORN    | Warning Horn   |
| Q0.1    | MOTOR   | Conveyor Motor |

**TIMER:**

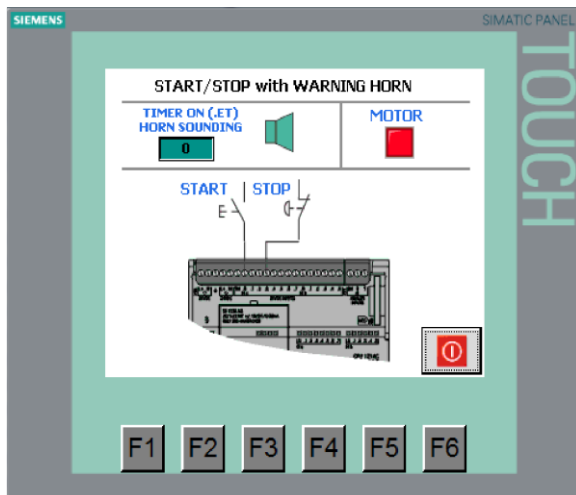
*(Do not comment)*

| Timer Data Block Tag         | Comments              |
|------------------------------|-----------------------|
| WARN_DWELL                   | Start-up Timer        |
| WARN_DWELL.Q <sup>(1)</sup>  | Timer Done Status Bit |
| WARN_DWELL.IN <sup>(1)</sup> | Timer Enable Bit      |

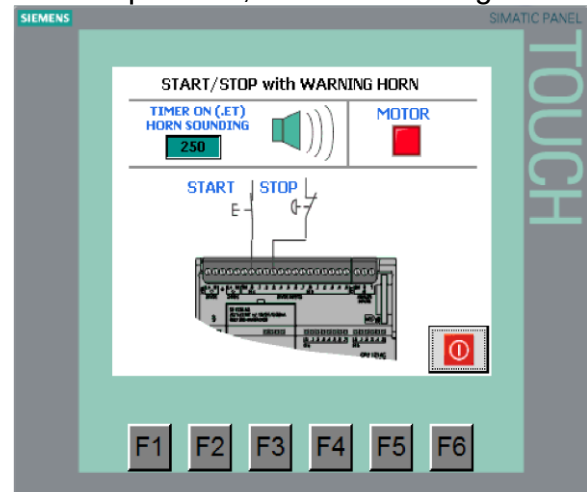
<sup>(1)</sup> Timer status bits created automatically

## HMI and SWITCHES:

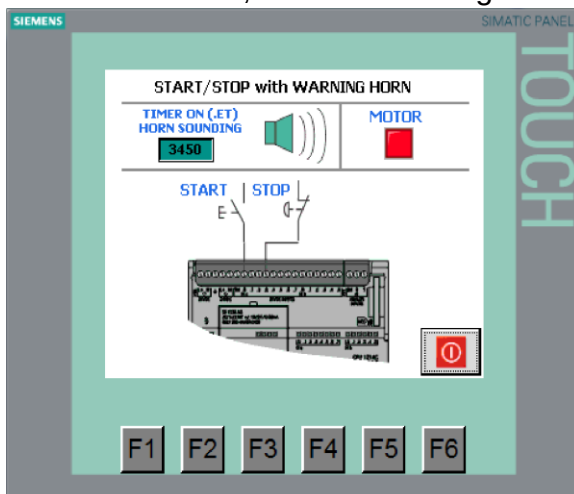
Initial HMI screen.



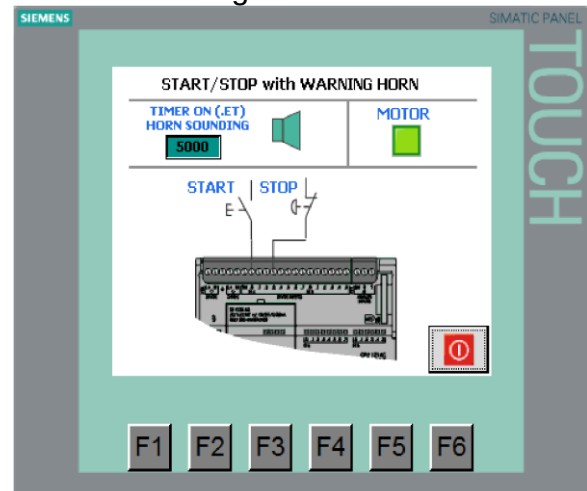
START pressed, HORN sounding.



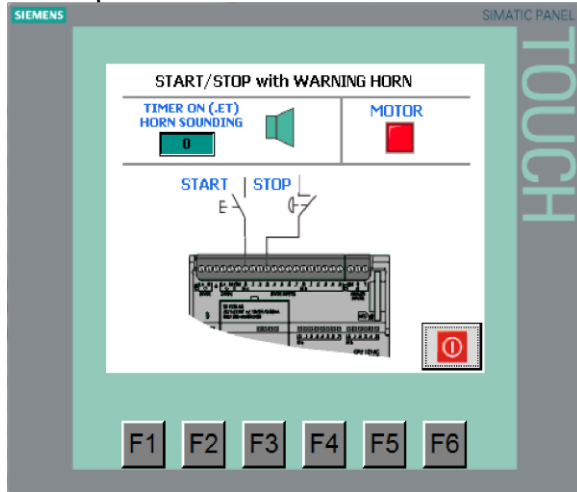
START released, HORN sounding.



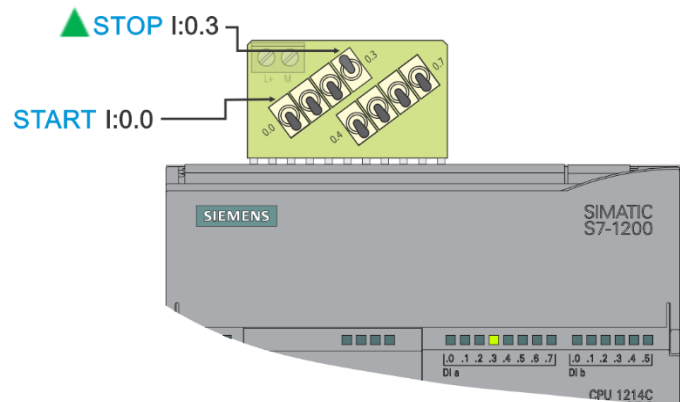
MOTOR running.



STOP pressed, MOTOR OFF



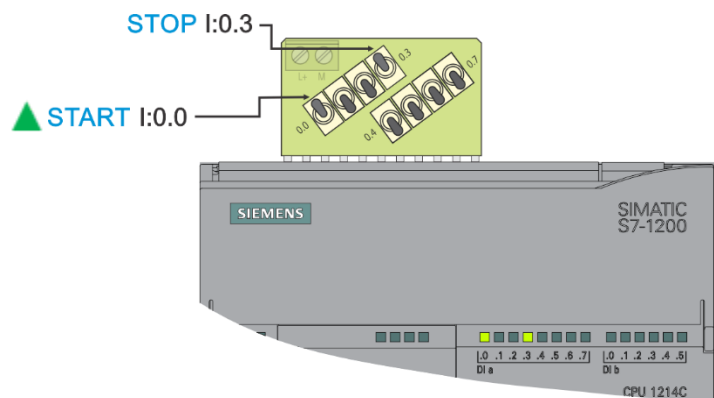
Initial switch settings, **push up ▲** the **STOP** switch to close.



Operator Pressing the START pushbutton:

**Push up ▲** the **Start** switch to close.

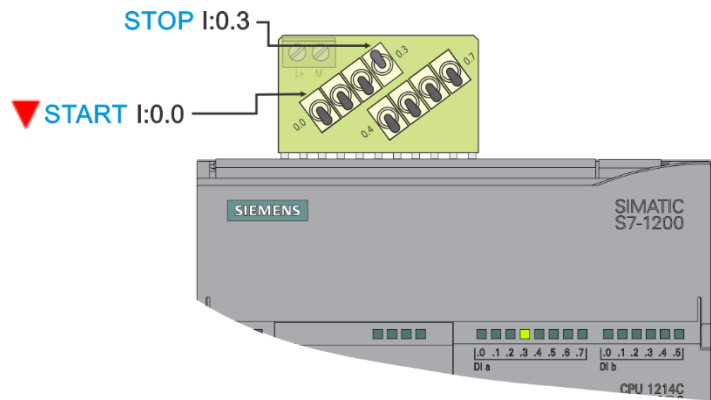
*The HORN will sound.*



---

Push down ▼ the **Start** switch to open.

*After five seconds the **HORN** turns OFF and the conveyor **MOTOR** turns ON with the indicator illuminated green.*

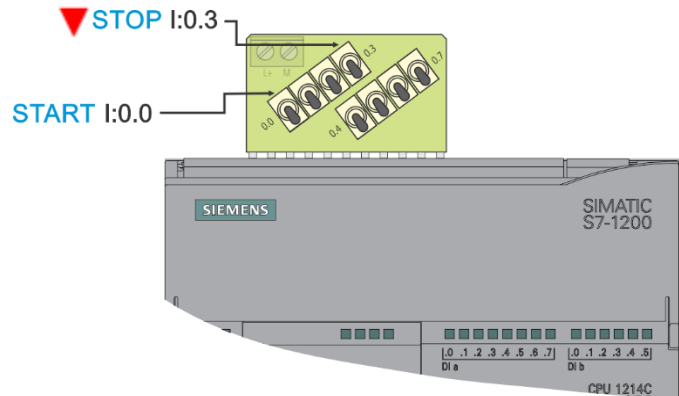


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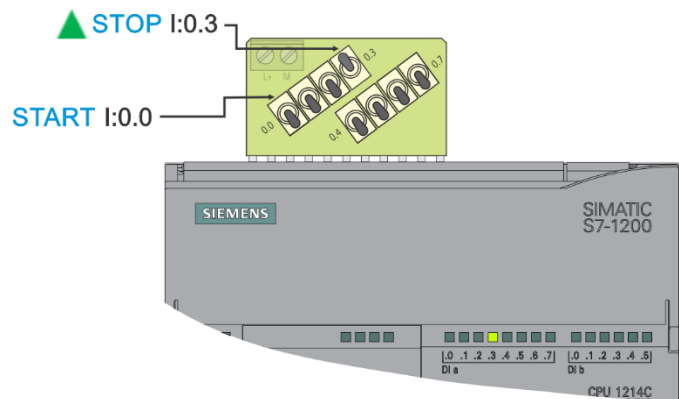
Operator Pressing the STOP pushbutton:

Push down ▼ the **STOP** switch to open.

*The **MOTOR** indicator is illuminated red.*



Push up ▲ the **STOP** switch to close.



**LAB: TIMED WASHER STATION****PROCESS FILE:** WASH.TON**OBJECTIVE:** Identify the type of timer instruction and the controlling logic.**HMI:** WASH\_HMI**SOFTWARE:** SIEMENS TIA Portal

Modify the WASHER STATION lab so that a timer will allow the spray wash to be on for 10 sec. while the pushbutton is being held.

When creating the tag, the description MUST BE entered.

**INPUT:**

| Address | PLC Tag       | Comments              |
|---------|---------------|-----------------------|
| I0.0    | PUSHBUTTON    | Pushbutton Wash Cycle |
| I0.1    | SWITCH_CLAMP1 | Limit Switch Clamp 1  |
| I0.2    | SWITCH_CLAMP2 | Limit Switch Clamp 2  |

**OUTPUT:**

| Address | PLC Tag         | Comments         |
|---------|-----------------|------------------|
| Q0.0    | SOLENOID_CLAMP1 | Clamp 1 Solenoid |
| Q0.1    | SOLENOID_CLAMP2 | Clamp 2 Solenoid |
| Q0.2    | WASH_VALVE      | Spray Wash Valve |

**TIMER:**

*(Do not comment)*

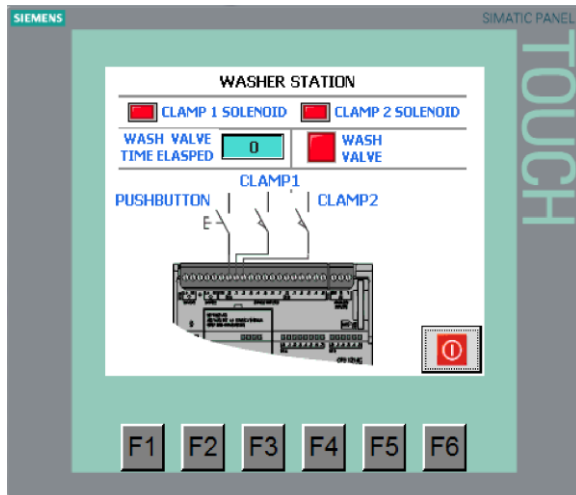
| Timer Data Block Tag        | Comments                  |
|-----------------------------|---------------------------|
| WASH_DELAY                  | Spray Wash Timer          |
| WASH_DELAY.Q <sup>(1)</sup> | Spray Wash Delay Complete |

<sup>(1)</sup> Timer status bits created automatically

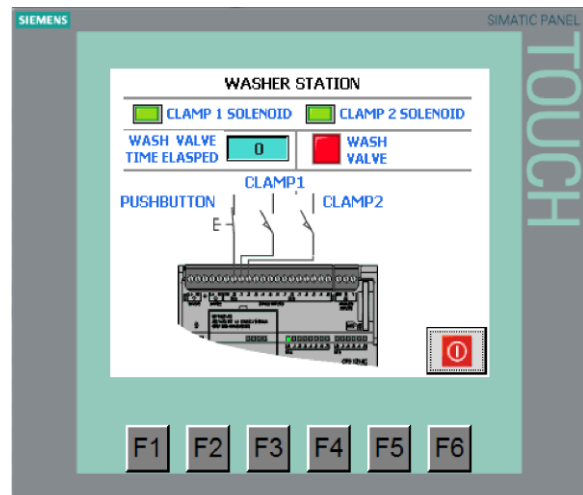


## HMI and SWITCHES:

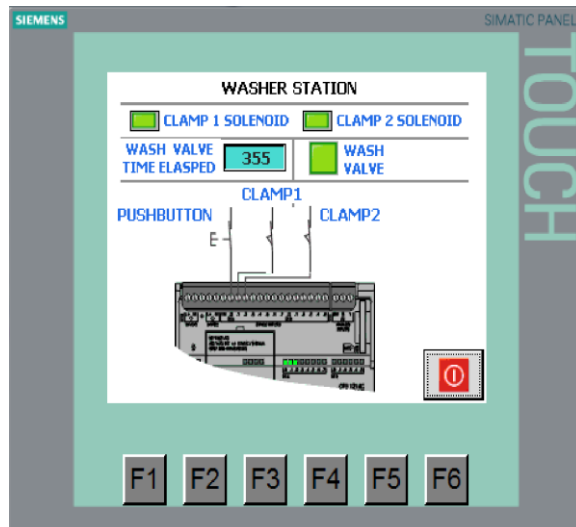
Initial HMI screen.



START pressed, SOLENOIDS ON.



CLAMPS closed, WASH ON, Timer



SIEMENS SIMATIC PANEL TOUCH

### WASHER STATION

|                                                                                                    |                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  CLAMP 1 SOLENOID |  CLAMP 2 SOLENOID                                                                             |
| WASH VALVE<br>TIME ELAPSED                                                                         |   WASH VALVE |

CLAMP1  
PUSHBUTTON

CLAMP2



F1 F2 F3 F4 F5 F6

**WASHER STATION**

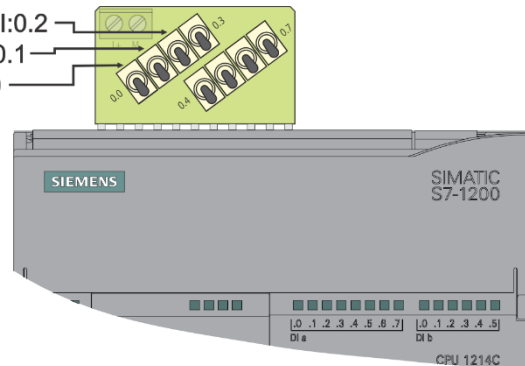
|                                                                                                    |                                                                                                      |
|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
|  CLAMP 1 SOLENOID |  CLAMP 2 SOLENOID |
| WASH VALVE<br>TIME ELAPSED                                                                         |  WASH VALVE       |

CLAMP1  
PUSHBUTTON



F1 F2 F3 F4 F5 F6

SWITCH CLAMP 2 I:0.2  
SWITCH CLAMP 1 I:0.1  
PUSHBUTTON I:0.0



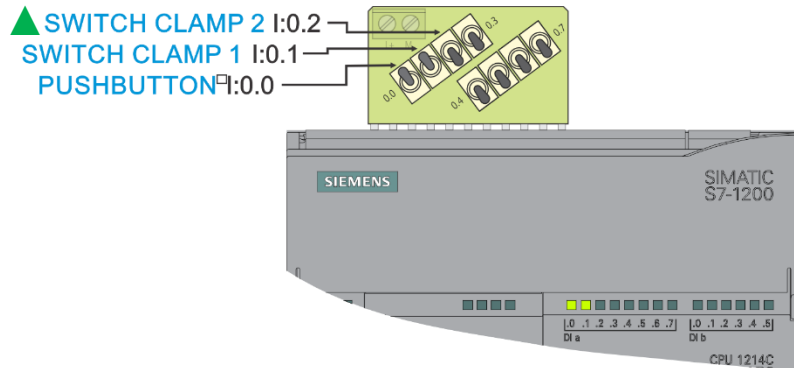
Push up ▲ the **PUSHBUTTON** to close.

SWITCH CLAMP 2 I:0.2  
SWITCH CLAMP 1 I:0.1  
PUSHBUTTON I:0.0

Diagram illustrating the connection of a 12VDC power supply to the CPU 1214C terminal block. The power supply is connected to terminals 0.0 (GND) and 0.1 (+12VDC). The terminal block is labeled with SIEMENS, SIMATIC S7-1200, and CPU 1214C.

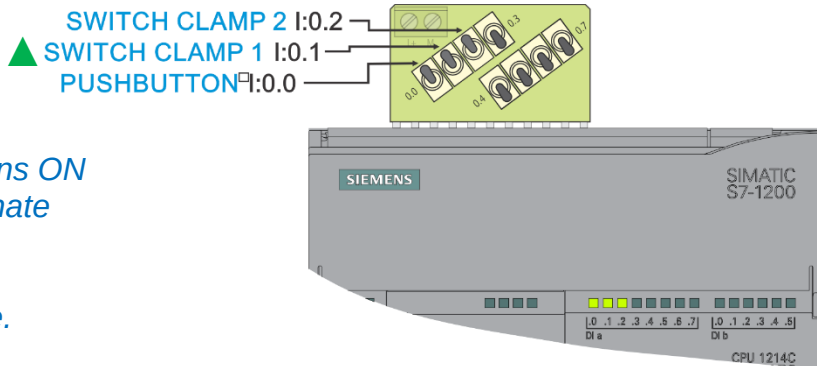
---

Push up ▲ the **SWITCH CLAMP 1** to close.



---

Push up ▲ the **SWITCH CLAMP 2** to close.



*The spray **WASH** valve turns ON and the indicator will illuminate green.*

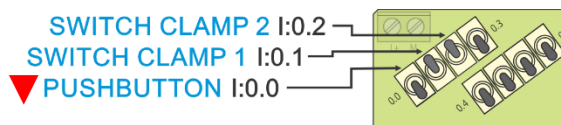
*The Timer begins to elapse.*

*After ten seconds the spray **WASH** turns OFF.*

---

#### Operator Releasing the PUSHBUTTON:

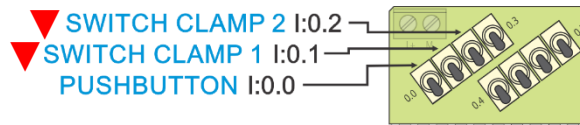
To turn off the washer, push down ▼ the **PUSHBUTTON** switch to open.



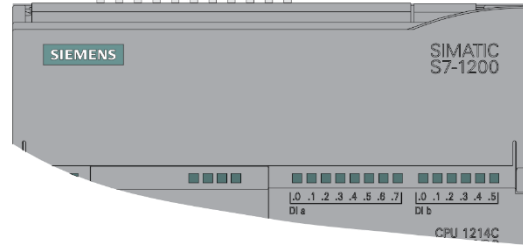
*The solenoid valves turn OFF. **CLAMP 1 SOLENOID**, and **CLAMP 2 SOLENOID** are illuminated red on the HMI.*

---

*continued*



**Push down ▼ switches SWITCH CLAMP 1 and SWITCH CLAMP 2 to emulate the clamps releasing.**



## LAB: BUILDING LIGHTS

**PROCESS FILE:** BUILDING

**OBJECTIVE:** Use of TOF instruction and cascading timer logic.

**HMI:** BUILDING\_HMI

**SOFTWARE:** SIEMENS TIA Portal

Create a program that will use cascading timers to control when building and parking lot lights are turned off. When the light switch is switched to the off position, a Normally Open switch returns to the normal position, the building light will turn off immediately. The lights for the Exit Door will remain on for 5 seconds. The Parking Lot lights will remain on for an additional 10 seconds.

When creating the tag, the description MUST BE entered.

### INPUT:

| Address | PLC Tag      | Comments        |
|---------|--------------|-----------------|
| I0.0    | LIGHT_SWITCH | Building Switch |

### OUTPUT:

| Address | PLC Tag         | Comments           |
|---------|-----------------|--------------------|
| Q0.0    | BUILDING_LIGHTS | Building Lights    |
| Q0.1    | EXIT_LIGHTS     | Exit Door Lights   |
| Q0.2    | LOT_LIGHTS      | Parking Lot Lights |

### TIMERS:

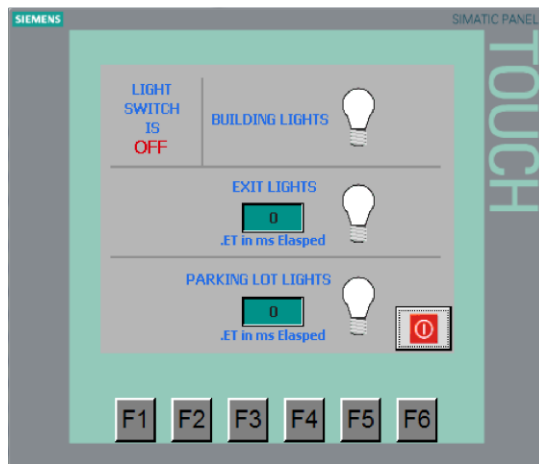
*(Do not comment)*

| Timer Data Block Tag        | Comments              |
|-----------------------------|-----------------------|
| EXIT_DELAY                  | Timer for Exit        |
| EXIT_DELAY.Q <sup>(1)</sup> | Exit Light Off        |
| LOT_DELAY                   | Timer for Parking Lot |
| LOT_DELAY.Q <sup>(1)</sup>  | Parking Lights Off    |

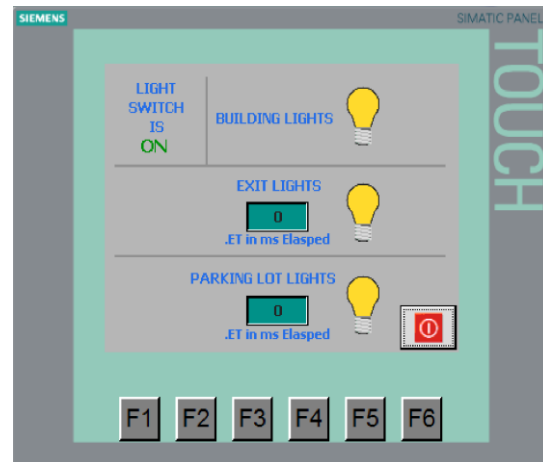
<sup>(1)</sup> Timer status bits created automatically

## HMI and SWITCH:

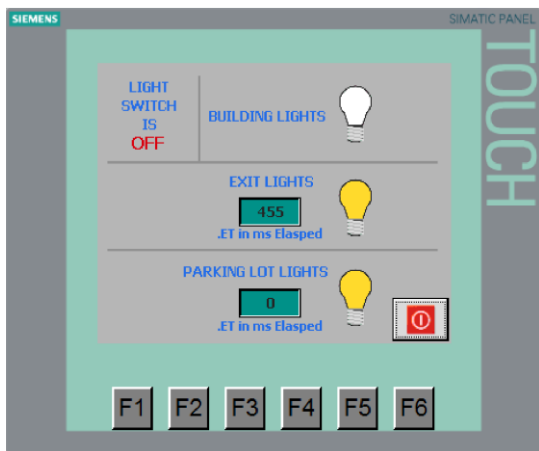
Initial HMI screen.



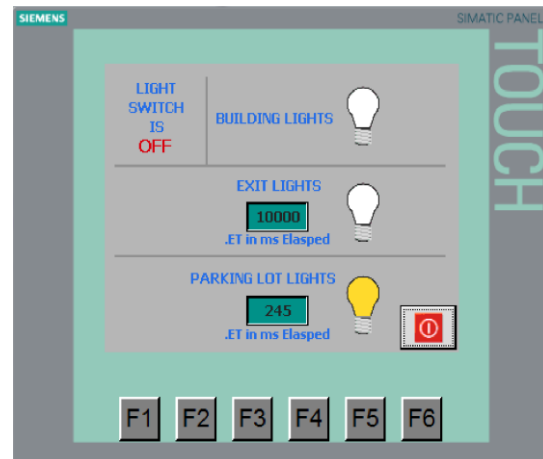
SWITCH ON, all lights ON.



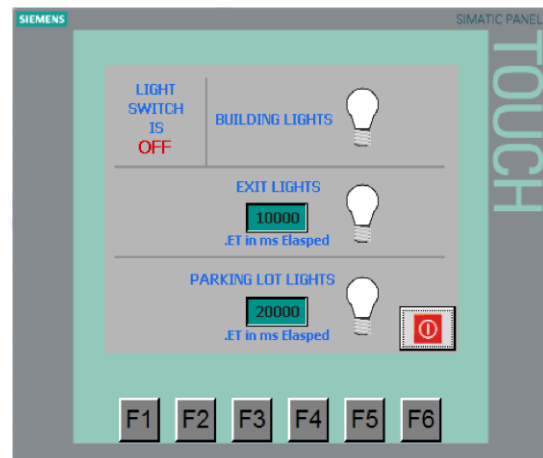
SWITCH OFF, BUILDING light OFF, EXIT Timer elapsing.



EXIT Timer complete, EXIT light OFF, LOT Timer elapsing.



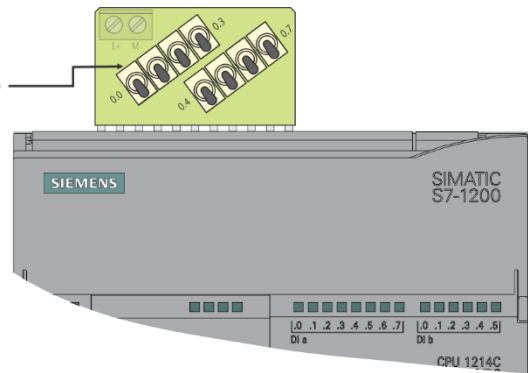
LOT Timer complete, LOT light OFF.



---

Initial switch setting, push down ▼ the **LIGHT SWITCH**.

▼ **LIGHT SWITCH** I:0.0

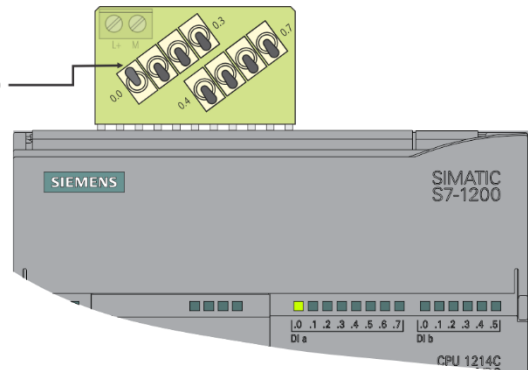


---

Push up ▲ the **LIGHT SWITCH** to close.

▲ **LIGHT SWITCH** I:0.0

*All the lights should turn ON.*



---

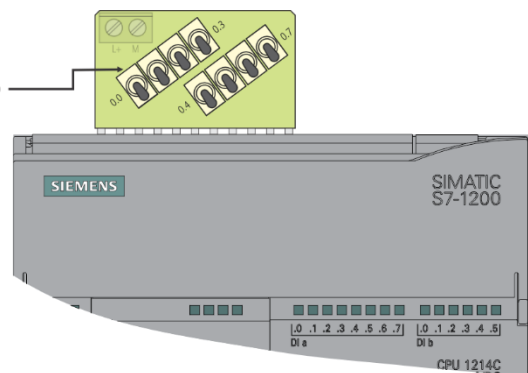
Push down ▼ the **LIGHT SWITCH** to open.

*The **BUILDING LIGHTS** will turn OFF immediately.*

▼ **LIGHT SWITCH** I:0.0

*The **EXIT LIGHTS** will turn OFF after 5 seconds.*

*The parking **LOT LIGHTS** will turn OFF after an additional 10 seconds.*



## LAB: TIMED WASHER STATION - RETENTIVE TIMER

**PROCESS FILE:** WASH TONR

**OBJECTIVE:** Use of retentive timer during wash cycle and identifying logic to reset timer.

**SOFTWARE:** SIEMENS TIA Portal

Use a Retentive Timer On (TONR) instruction for controlling the washer valve.

If the START pushbutton is released before the timing cycle is complete, the elapsed value of the timer *will not be reset*. Pressing the pushbutton will start the timing cycle from the current accumulated value.

When creating the tag, the description MUST BE entered.

### INPUT:

| Address | PLC Tag       | Comments              |
|---------|---------------|-----------------------|
| I0.0    | PUSHBUTTON    | Pushbutton Wash Cycle |
| I0.1    | SWITCH_CLAMP1 | Limit Switch Clamp 1  |
| I0.2    | SWITCH_CLAMP2 | Limit Switch Clamp 2  |

### OUTPUT:

| Address | PLC Tag         | Comments         |
|---------|-----------------|------------------|
| Q0.0    | SOLENOID_CLAMP1 | Clamp 1 Solenoid |
| Q0.1    | SOLENOID_CLAMP2 | Clamp 2 Solenoid |
| Q0.2    | WASH_VALVE      | Spray Wash Valve |

### TIMER:

(Do not comment)

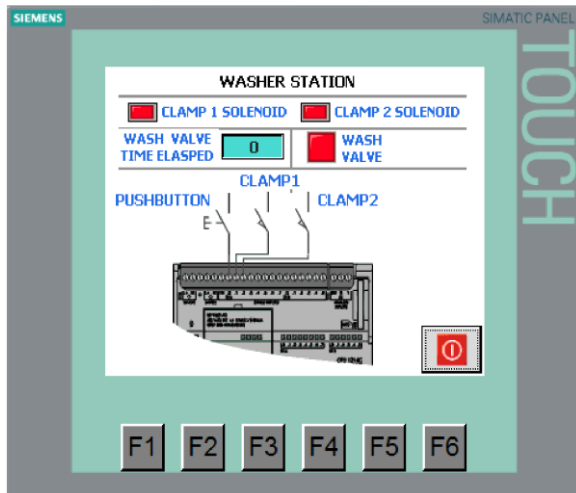
| Timer Data Block Tag        | Comments                  |
|-----------------------------|---------------------------|
| WASH_DELAY                  | Spray Wash Timer          |
| WASH_DELAY.Q <sup>(1)</sup> | Spray Wash Delay Complete |

<sup>(1)</sup> Timer status bits created automatically

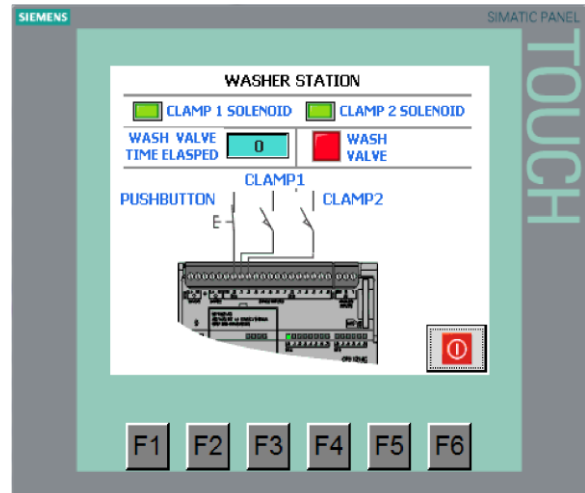


## HMI and SWITCHES:

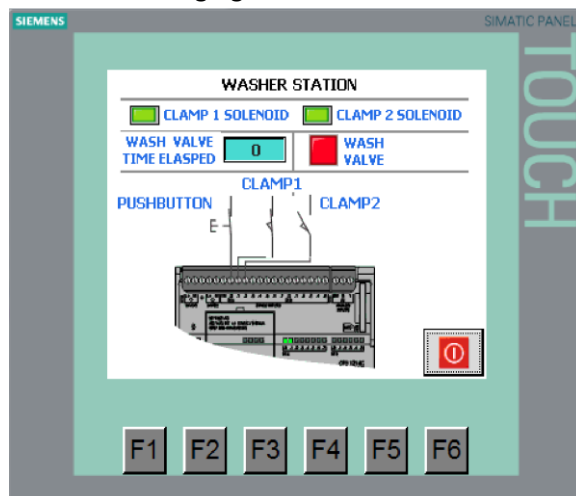
Initial HMI screen.



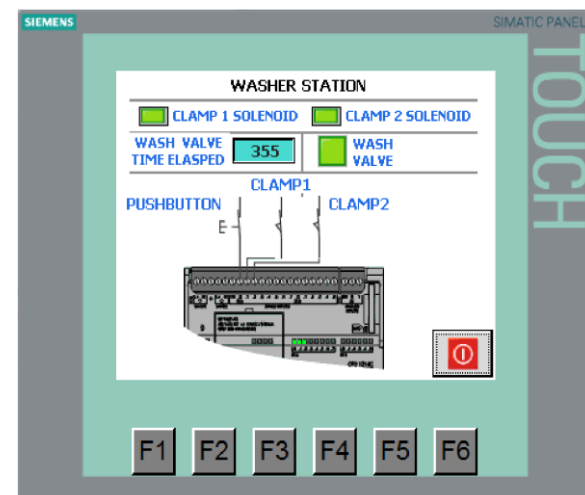
START presses, SOLENOIDS ON.



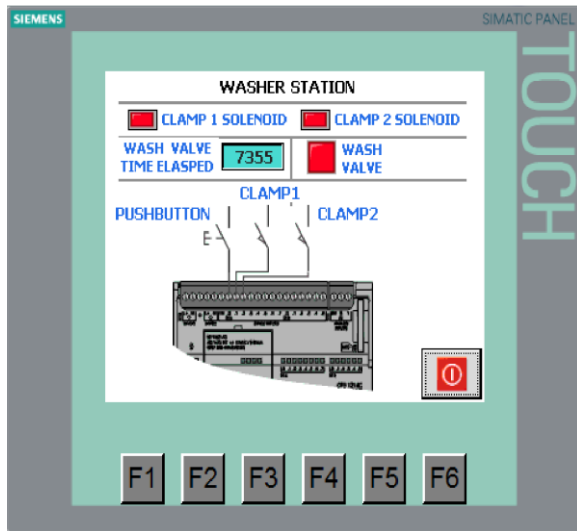
CLAMP 1 engaged.



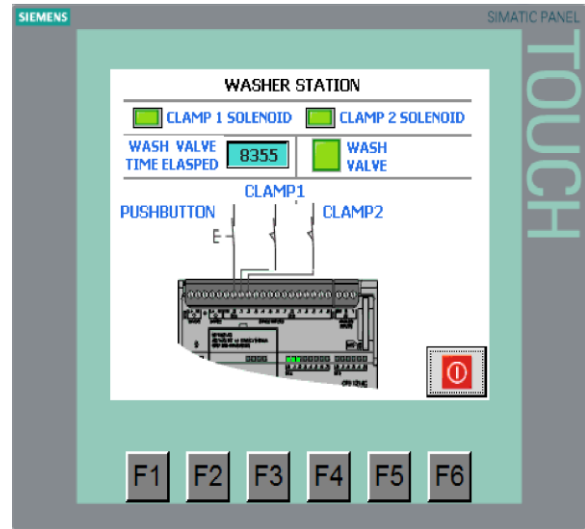
CLAMP 2 engaged, WASH valve ON.



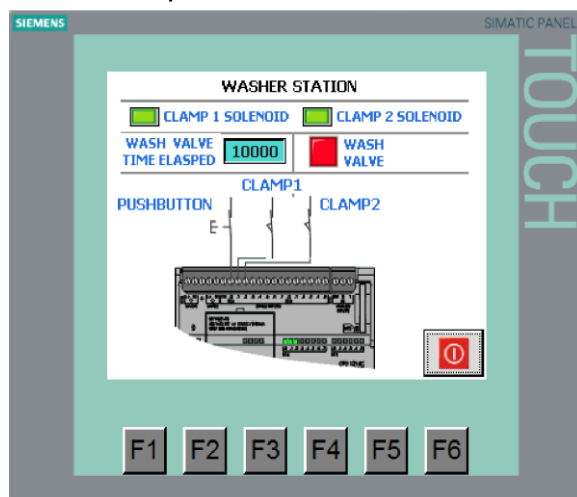
START released, SOLENOIDS and WASH VALVE OFF, Timer retained.



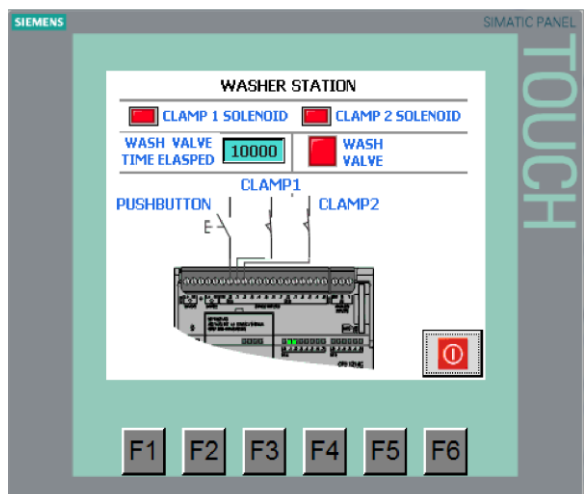
START pressed, SOLENOIDS and WASH VALVE ON, Timer elapsing.



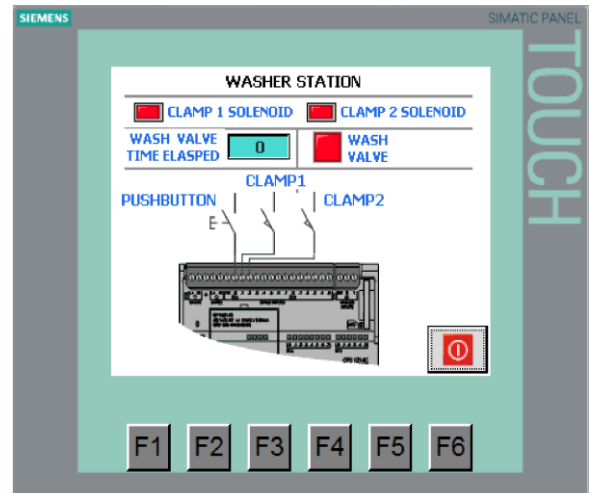
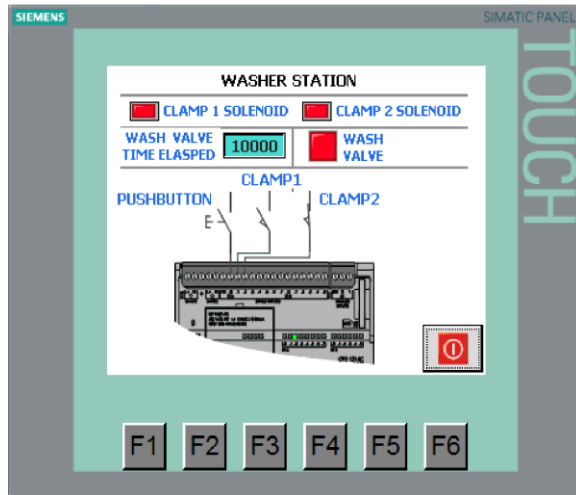
Timer Complete and WASH VALVE OFF.



START released, SOLENOIDS OFF.

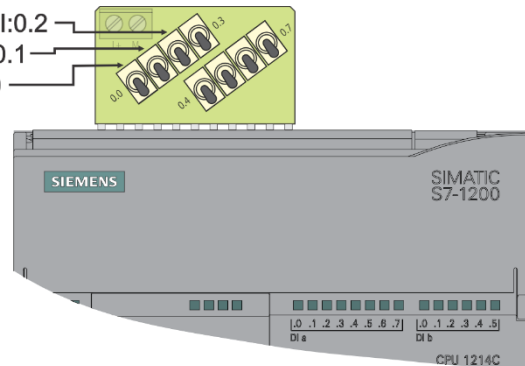


CLAMP 1 opened.



Initial switch settings, **all switches down.**

SWITCH CLAMP 2 I:0.2  
SWITCH CLAMP 1 I:0.1  
PUSHBUTTON I:0.0

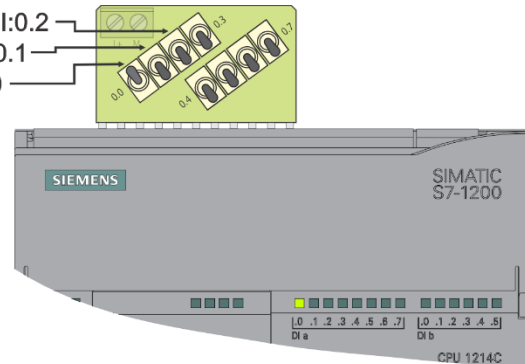


Operator Pressing and Holding the PUSHBUTTON:

**Push up ▲ the PUSHBUTTON**  
to close.

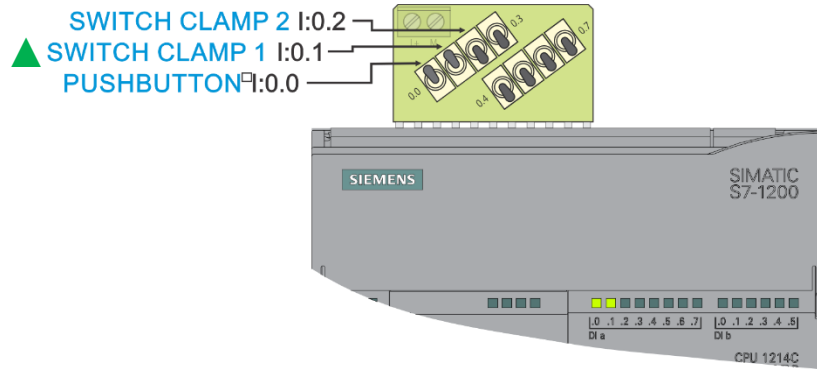
SWITCH CLAMP 2 I:0.2  
SWITCH CLAMP 1 I:0.1  
▲ PUSHBUTTON I:0.0

*The two outputs for the solenoid valves for the clamps turn ON. The HMI **CLAMP 1 SOLENOID** and **CLAMP 2 SOLENOID** indicators will illuminate green.*



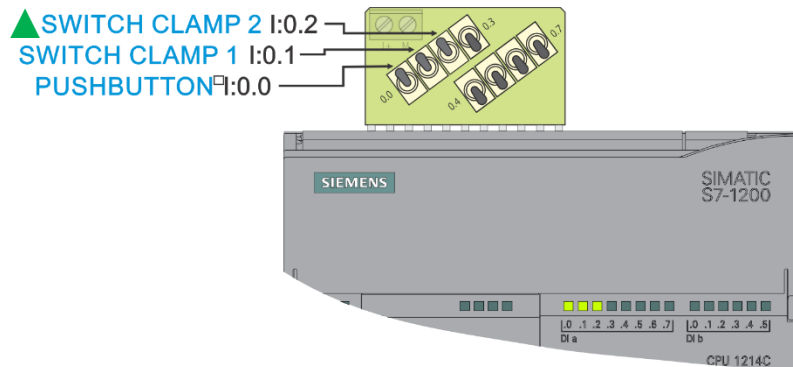
---

Push up ▲ the **SWITCH CLAMP 1** to close.



✓ *PUSHBUTTON should be pushed down before the washer times out. Before doing this step, read this STEP and the next STEP.*

Push up ▲ the **SWITCH CLAMP 2** to close.



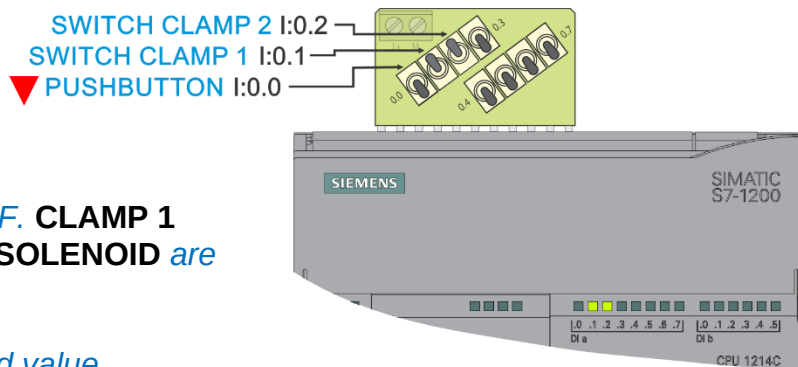
*The spray WASH valve turns ON and the indicator will illuminate green.*

*The Timer begins to elapse.*

---

#### Operator Releasing the PUSHBUTTON:

To turn off the washer, push down ▼ the **PUSHBUTTON** switch to open.



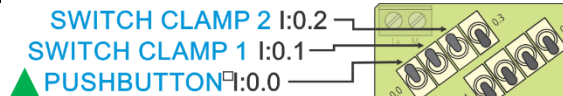
*The solenoid valves turn OFF. CLAMP 1 SOLENOID, and CLAMP 2 SOLENOID are illuminated red on the HMI.*

*Timer retains current elapsed value.*

*continued*

---

Operator Pressing and Holding  
the PUSHBUTTON:



Push up ▲ the **PUSHBUTTON** to close.

*Timer resumes from retained value.*

*The spray **WASH** valve turns ON and the indicator will illuminate green.*

*The two outputs for the solenoid valves for the clamps turn ON. The HMI **CLAMP 1 SOLENOID** and **CLAMP 2 SOLENOID** indicators will illuminate green*

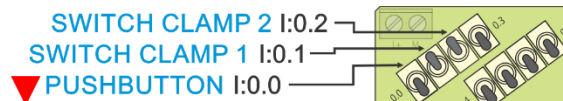
*When the ten seconds have completed the spray **WASH** turns OFF.*

*If the **WASH** valve remains ON and the Elapsed Time **ET** begins to increment again there is a problem with the logic.*

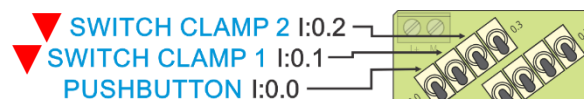
---

Operator Releasing the PUSHBUTTON:

To turn off the washer,  
**push down ▼** the  
**PUSHBUTTON** switch  
to open.



*The solenoid valves turn OFF. **CLAMP 1 SOLENOID**, and **CLAMP 2 SOLENOID** are illuminated red on the HMI.*



**Push down ▼** switches **SWITCH CLAMP 1** and **SWITCH CLAMP 2** to emulate the clamps releasing.

## LAB: CASCADING CONVEYOR TIMER

**PROCESS FILE:** CONVEYOR TIMER

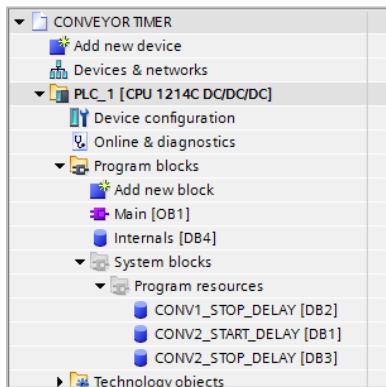
**OBJECTIVE:** Use of TIMER ON and OFF instructions with cascading logic.

**SOFTWARE:** SIEMENS TIA Portal

Use delay to energize and delay to de-energize timers to create a delay Start/Stop Circuit.

When the Start pushbutton is depressed, the first conveyor output will be energized immediately, a second conveyor output will be energized after five (5) seconds.

The depressing of the Stop pushbutton will de-energize the first conveyor output after a five (5) second delay, and the second conveyor output will be de-energized after an additional five (5) second delay.



The program will use a Global Data Block.

**Data Block Name:** Internals

See [Program Assist](#) on details to create the Data Block.

When creating the tag, the description MUST BE entered.

### INPUT:

| Address | PLC Tag | Comments                  |
|---------|---------|---------------------------|
| I0.0    | START   | Momentary N.O. Pushbutton |
| I0.3    | STOP    | Momentary N.C. Pushbutton |

### OUTPUT:

| Address | PLC Tag    | Comments             |
|---------|------------|----------------------|
| Q0.0    | CONVEYOR_1 | Conveyor Motor No. 1 |
| Q0.1    | CONVEYOR_2 | Conveyor Motor No. 2 |

## TIMERS:

*(Do not comment)*

| Timer Data Block Tag               | Comments                                   |
|------------------------------------|--------------------------------------------|
| CONV2_START_DELAY                  | Delay Energize Timer for Conveyor No. 2    |
| CONV2_START_DELAY.Q <sup>(1)</sup> | Conveyor No. 2 Start                       |
| CONV1_STOP_DELAY                   | Delay De-energize Timer for Conveyor No. 1 |
| CONV1_STOP_DELAY.Q <sup>(1)</sup>  | Delay for Halting/Stop Conveyor No. 1      |
| CONV2_STOP_DELAY                   | Delay De-energize Timer for Conveyor No. 2 |
| CONV2_STOP_DELAY.Q <sup>(1)</sup>  | Delay for Halting/Stop Conveyor No. 2      |

<sup>(1)</sup> Timer status bits created automatically

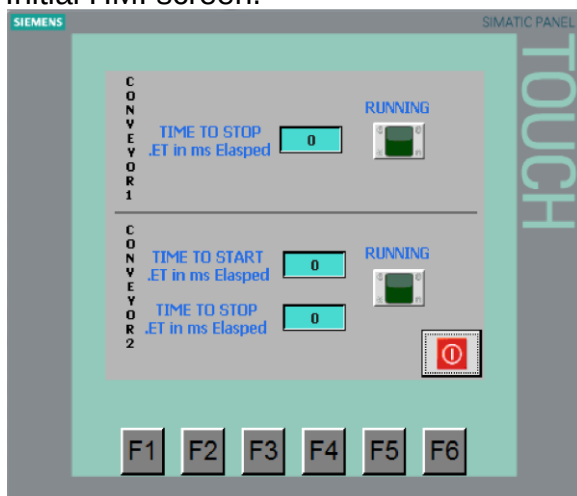
Name of Data Block: **INTERNALS**

## DATA BLOCK INTERNALS:

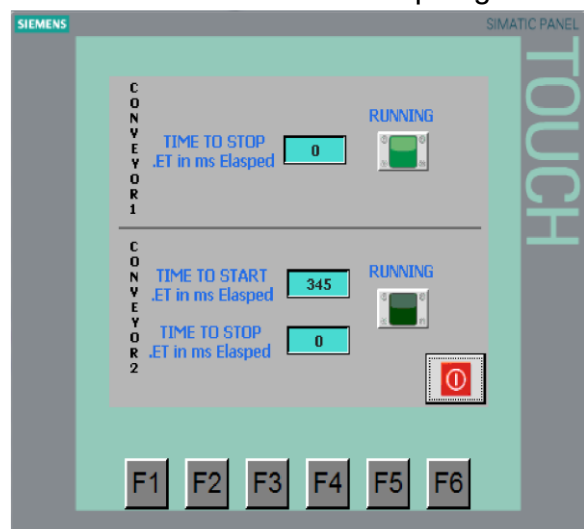
| Static Tag  | Data Type | Comments                    |
|-------------|-----------|-----------------------------|
| RUN_CONTACT | Boolean   | Start Motors/Motors Running |

## HMI and SWITCHES:

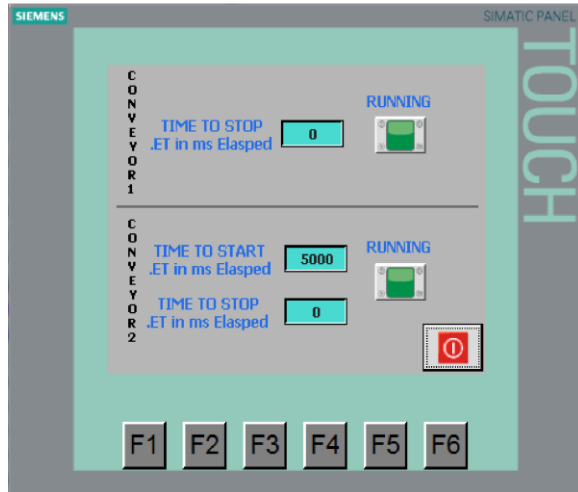
Initial HMI screen.



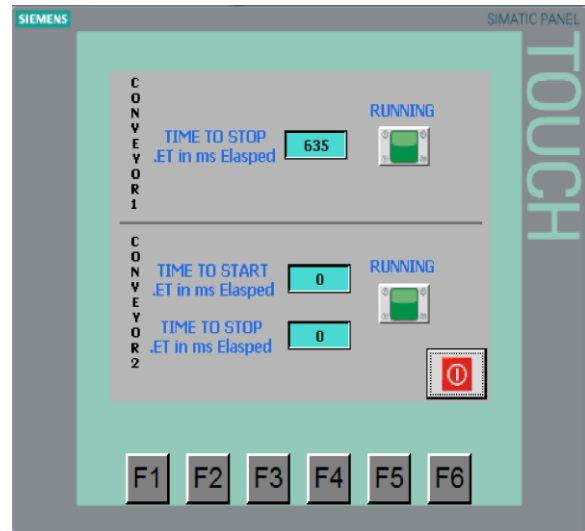
START pressed, CONVEYOR 1 ON, CONVEYOR 2 Timer On elapsing.



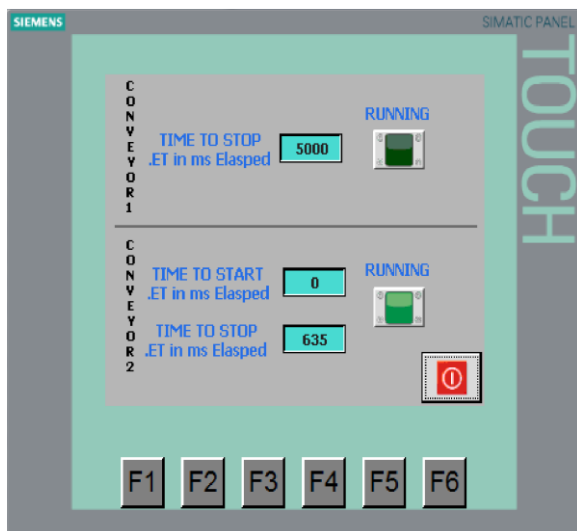
CONVEYOR 2 ON.



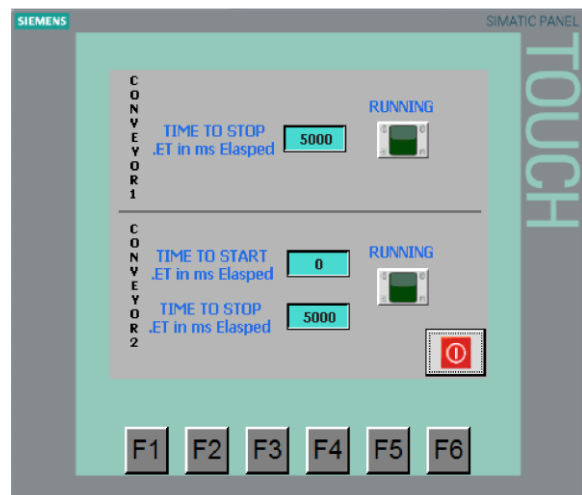
STOP pressed, CONVEYOR 1 Timer Off elapsing.



CONVEYOR 1 OFF, CONVEYOR 2 Timer Off elapsing.



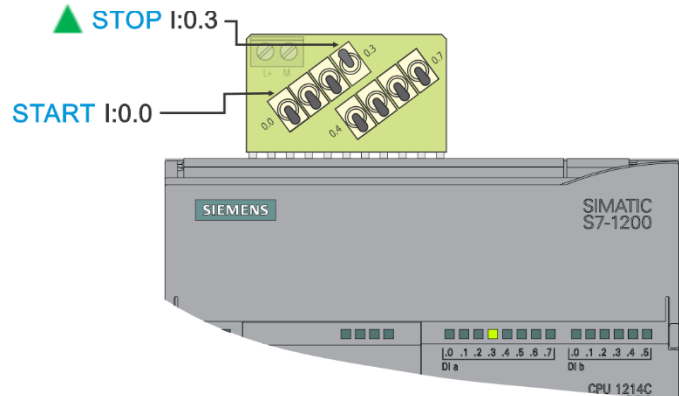
CONVEYOR 2 OFF



*continued*



Initial switch settings, **push up ▲** the **STOP** switch to close.



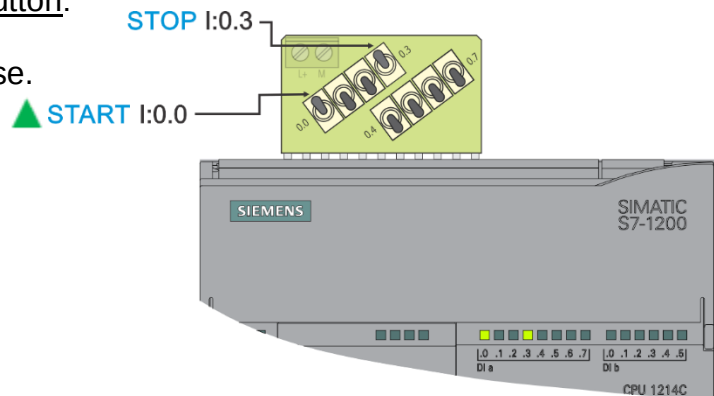
Operator Pressing the START pushbutton:

**Push up ▲** the **START** switch to close.

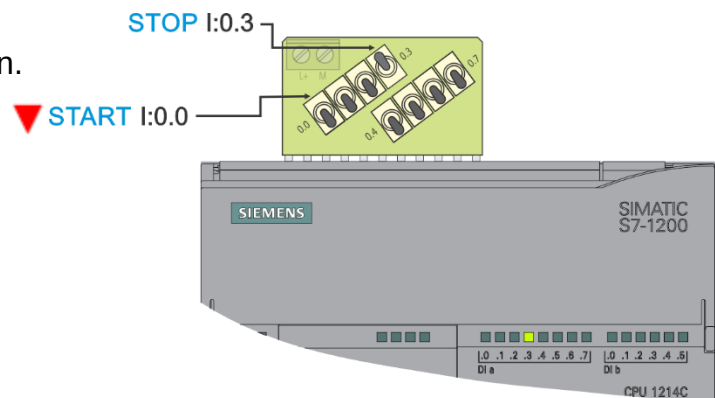
**CONVEYOR 1 turns ON**  
*immediately.*

**Conveyor 2 Timer On begins**  
*elapsing.*

-- THEN --



**Push down ▼** the **Start** switch to open.



*After the time delay, CONVEYOR 2 turns ON.*

*continued*

---

Operator Pressing the STOP pushbutton:

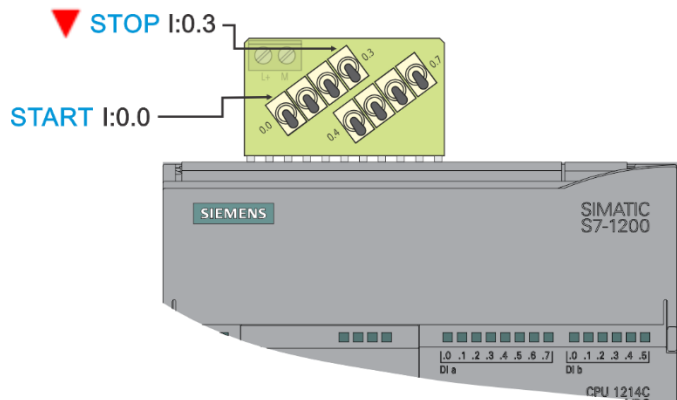
Push down ▼ the **STOP** switch to open.

**Conveyor 1 Timer Off begins elapsing.**

**CONVEYOR 1 turns OFF after the time delay.**

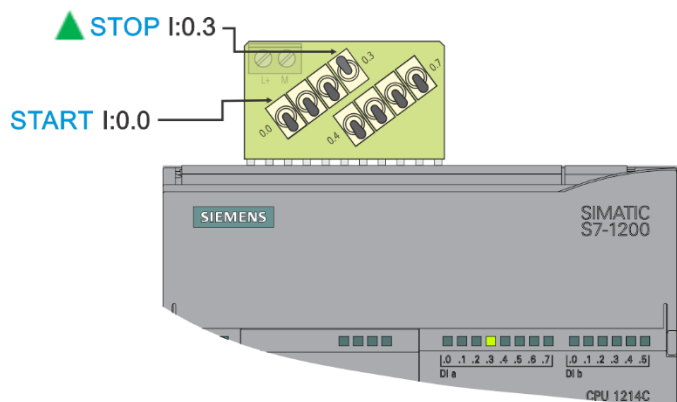
**Conveyor 2 Timer Off begins elapsing.**

**CONVEYOR 2 turns OFF after the time delay.**



---

Push up ▲ the **STOP** switch to close.



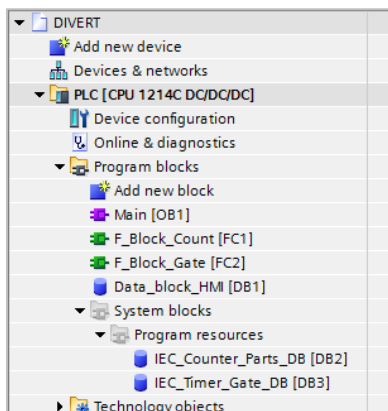
## LAB: PARTS DIVERTER

**PROCESS FILE:** DIVERT

**OBJECTIVE:** Use of counter to initiate a timed output.

**SOFTWARE:** SIEMENS TIA Portal

Create a program that diverts one in ten parts for a quality inspection. A START/STOP pushbutton station controls the parts conveyor. Parts are detected by a normally open proximity switch. When a count of ten has been reached, a solenoid for a diverter gate is opened. The gate stays open for two seconds. The parts counter will reset to zero automatically after the tenth part. A manual reset pushbutton is provided to also return the counter's accumulated value to zero.



The program will use two Function programming blocks.

**Function Name:** F\_Block\_Count  
F\_Block\_Gate

See [Program Assist](#) on details to create the Function block.

When creating the tag, the description MUST BE entered.

### INPUT:

| Address | PLC Tag | Comments                        |
|---------|---------|---------------------------------|
| I0.0    | START   | Momentary N.O. Pushbutton       |
| I0.1    | STOP    | Momentary N.O. Pushbutton       |
| I0.3    | PX1     | N.O. Proximity Switch for Count |
| I0.7    | RESET   | Momentary N.C. Pushbutton       |

### OUTPUTS:

| Address | PLC Tag | Comments             |
|---------|---------|----------------------|
| Q0.0    | GATE    | Diverter Gate        |
| Q0.1    | MOTOR   | Parts Conveyor Motor |

## TIMER:

*(Do not comment)*

| Timer Data Block Tag        | Comments              |
|-----------------------------|-----------------------|
| GATE_TIMER                  | Diverter Open Delay   |
| GATE_TIMER.Q <sup>(1)</sup> | Close Inspection Gate |

<sup>(1)</sup> Timer status bits created automatically

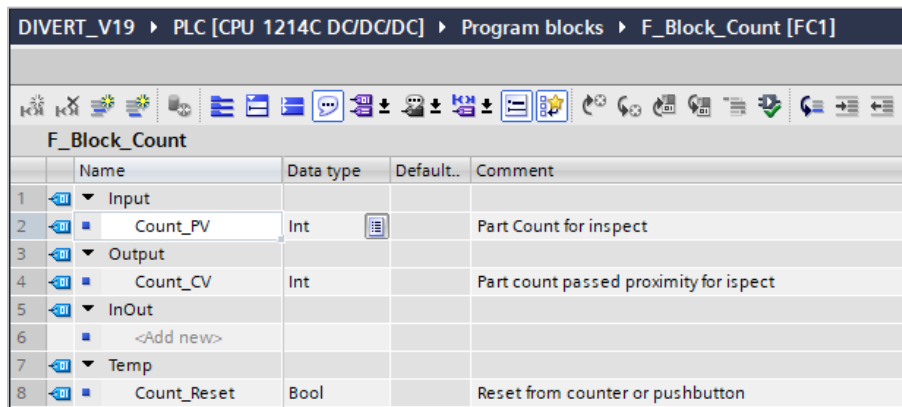
## COUNTER:

*(Do not comment)*

| Counter Data Block Tag       | Comments                          |
|------------------------------|-----------------------------------|
| PART_COUNT                   | Parts Counter                     |
| PART_COUNT.QU <sup>(1)</sup> | Part Count Setpoint for Gate Open |

<sup>(1)</sup> Counter status bits created automatically

## Name of Function Block: **F\_Block\_Count**



| Name          | Data type | Default... | Comment                                 |
|---------------|-----------|------------|-----------------------------------------|
| 1 Input       |           |            |                                         |
| 2 Count_PV    | Int       |            | Part Count for inspect                  |
| 3 Output      |           |            |                                         |
| 4 Count_CV    | Int       |            | Part count passed proximity for inspect |
| 5 InOut       |           |            |                                         |
| 6 <Add new>   |           |            |                                         |
| 7 Temp        |           |            |                                         |
| 8 Count_Reset | Bool      |            | Reset from counter or pushbutton        |

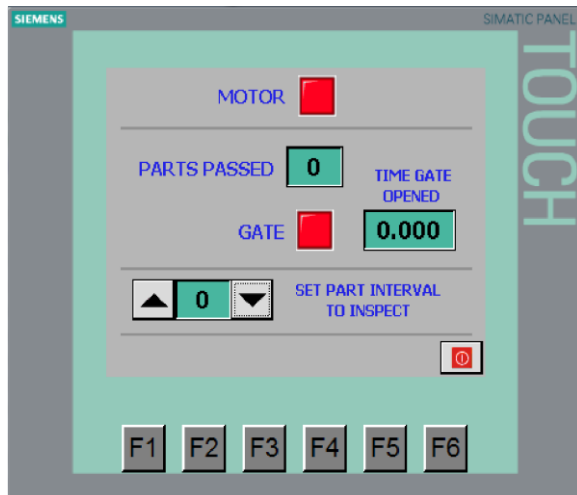
See **Program Assist** on details to enter the Function block Interface data.

## FUNCTION BLOCK INTERFACE:

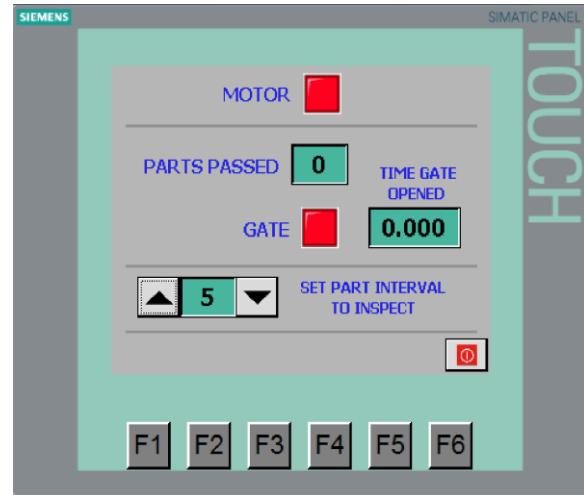
| Static Tag  | Data Type | Comments                              |
|-------------|-----------|---------------------------------------|
| Count_PV    | Integer   | Part Count for Inspect                |
| Count_CV    | Integer   | Part Count Past Proximity for Inspect |
| Count_Reset | Boolean   | Reset from Counter or Pushbutton      |

## HMI and SWITCHES:

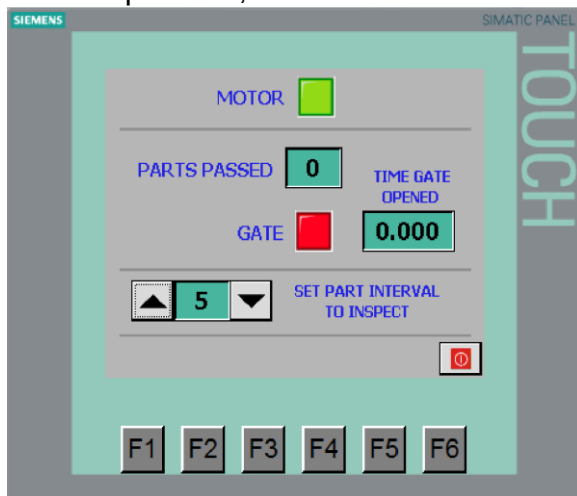
Initial HMI screen.



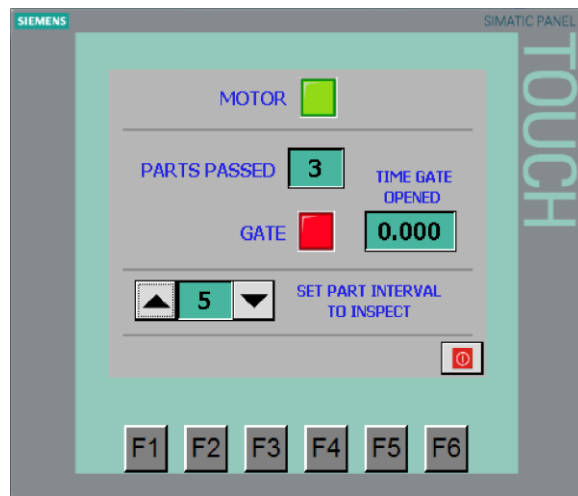
Part interval set.



START pressed, MOTOR ON.

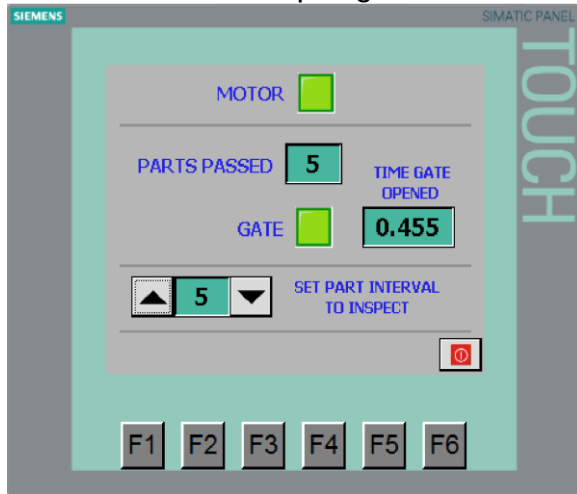


Three Parts detected by Proximity switch.

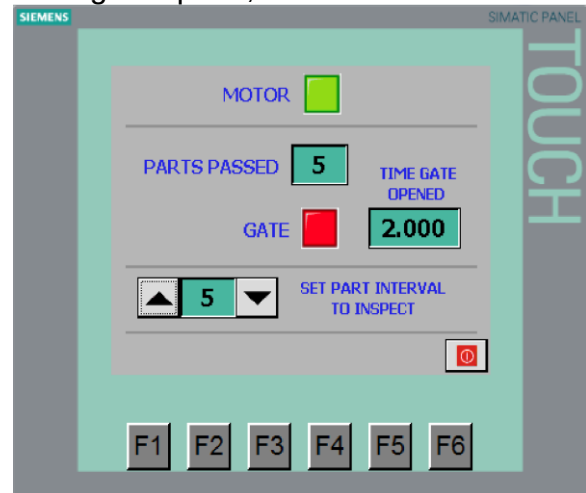


*continue*

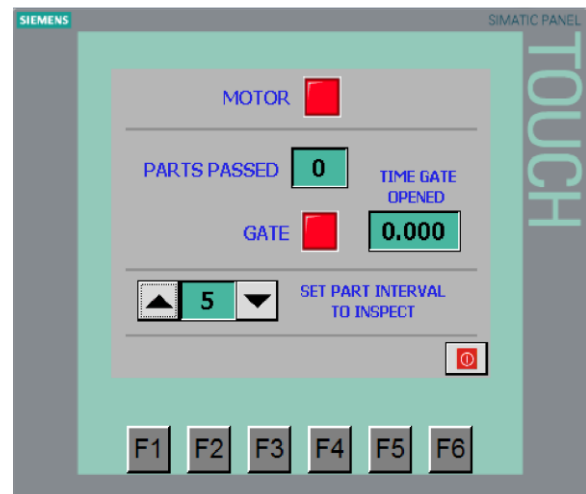
GATE ON, Timer elapsing.



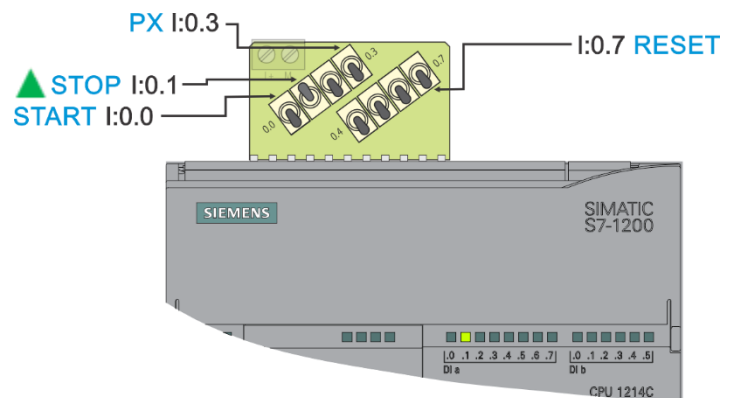
Timing complete, GATE OFF.



Part count and Timer reset.



Initial switch settings,  
**push up ▲ the STOP**  
switch to close.



---

Use the **UP** and **DOWN ARROWHEAD** buttons to change the **SET PART INTERVAL TO INSPECT** value.

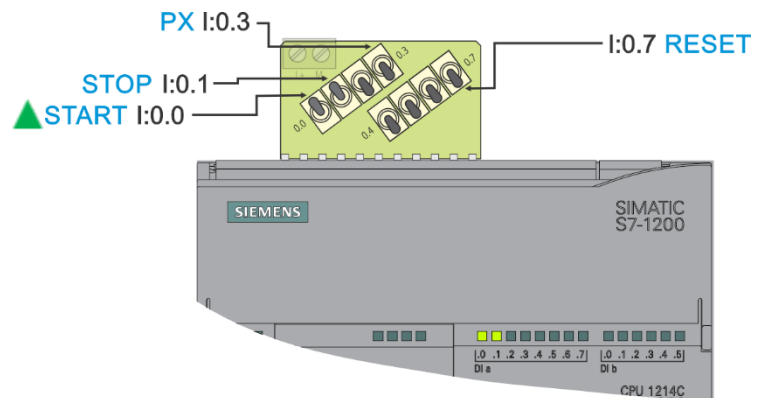


---

Operator Pressing the START pushbutton:

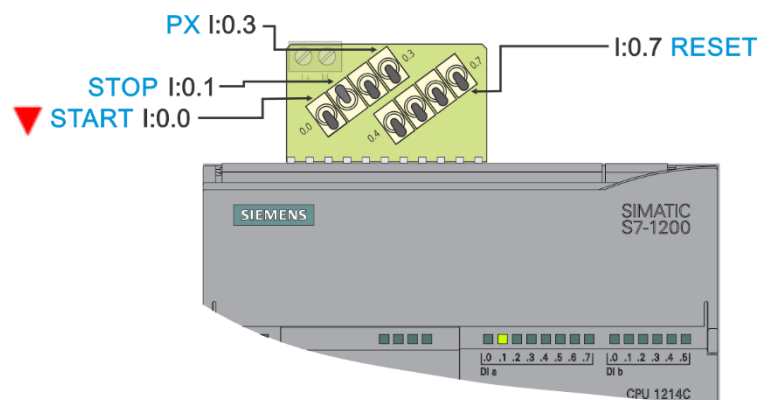
Push up ▲ the **START** switch to close.

*The conveyor MOTOR will turn ON.*



---

Push down ▼ the **START** switch to open.



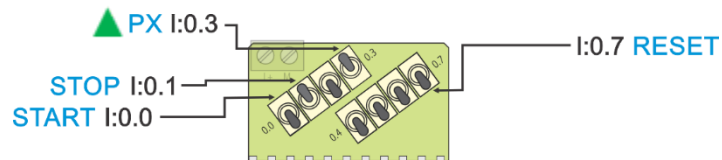
---

*continue*

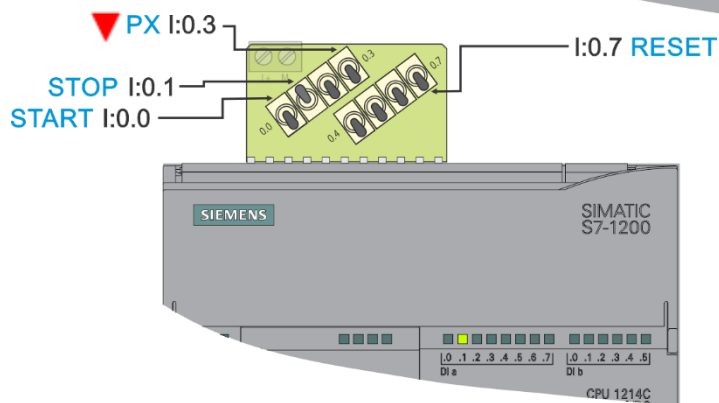
---

As the Conveyor runs, **toggle up** and **down** the **Proximity Switch** until the Part count is reached.

Push up ▲ the PX switch.



Push down ▼ the PX switch.



When the **PART Count Value CV** is at the part interval, the **DIVERTER** turns ON.

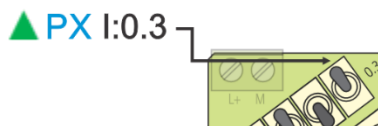
After the time delay, the **DIVERTER** turns OFF.

The **PART Count Value CV** resets to zero.

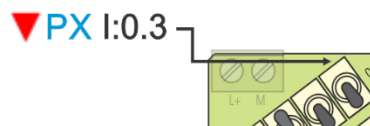
---

Toggle up and down **Proximity Switch** for a few counts.

Push up ▲ the PX switch.



Push down ▼ the PX switch



---

continue

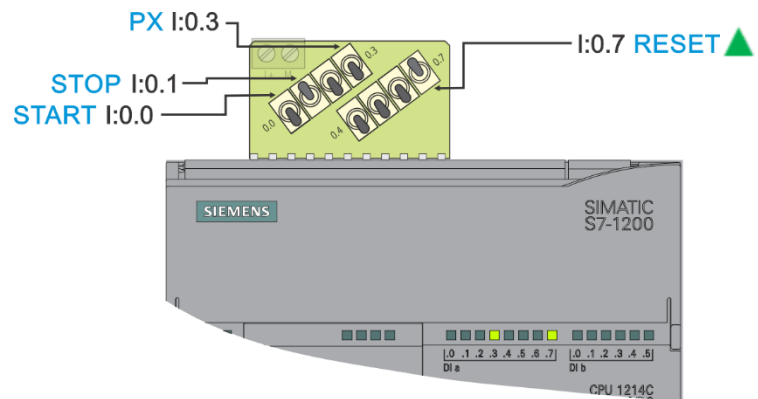


---

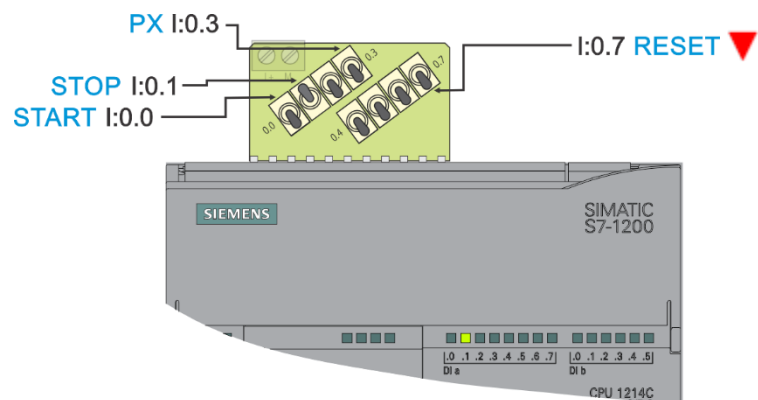
Operator Pressing the RESET pushbutton:

Push up ▲ the **RESET** to close.

*The **PART Count Value CV** is zeroed.*



Push down ▼ the **RESET** to open.

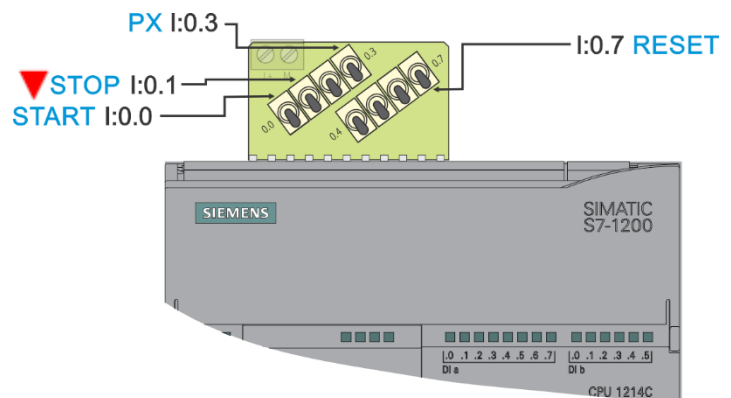


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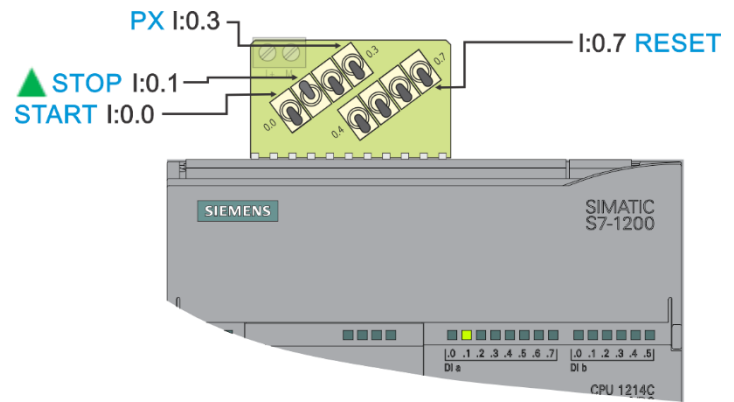
Operator Pressing the STOP pushbutton:

Push down ▼ the **STOP** switch to open.

*The **Motor** turns OFF.*



Push up ▲ the **STOP** switch to close.



## LAB: SPOOL

**PROCESS FILE:** SPOOL

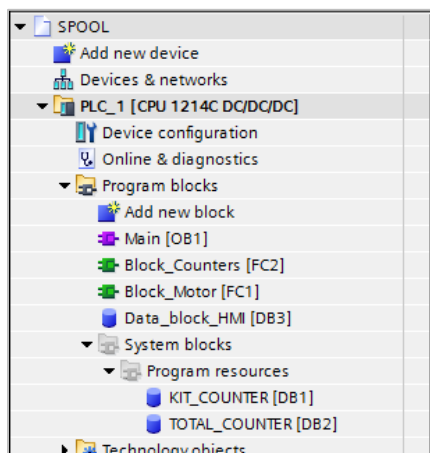
**OBJECTIVE:** Input logic to control two counters and to have one counter control the spool motor output.

**SOFTWARE:** SIEMENS TIA Portal

Write a program to implement the process outlined. A company which assembles electronic kits needs a counter to count and control the number of resistors placed into each kit. The controller must stop the take-up spool at a predetermined amount of resistors (100). An operator will then cut the resistor strip and place it into the kit. Pressing and releasing the start pushbutton repeats the spooling/counting cycle. The circuit operates as follows:

**FOR TESTING:** Set Kit Counter to 4.

1. A START/STOP pushbutton station is used to turn the spool motor drive on and off manually. Pressing and releasing the START pushbutton begins the spooling/counting cycle.
2. A thru-beam sensor counts the resistors as they pass by.
3. A Kit counter preset for 100 (the amount of resistors in each kit) will automatically stop the take-up spool when the accumulated count reaches 100.
4. The operator cuts and removes the resistors from the take-up spool and then presses the Kit Counter RESET pushbutton.
5. A second counter is provided to count the grand total used.
6. A Manual reset button is provided for the Grand Total Counter.



The program will use two Function programming blocks.

**Function Name:** Block\_Counters  
Block\_Motor

See [Program Assist](#) on details to create the Function block

When creating the tag, the description MUST BE entered.

**INPUT:**

| Address | PLC Tag             | Comments                                      |
|---------|---------------------|-----------------------------------------------|
| I0.0    | START               | Momentary N.O. Pushbutton                     |
| I0.3    | STOP                | Momentary N. C. Pushbutton                    |
| I0.4    | SENSOR_PE1          | Thru-beam Sensor Detect Resistors             |
| I0.6    | RESET_KIT_COUNTER   | Kit Counter RESET, Momentary N. O. Pushbutton |
| I0.7    | RESET_TOTAL_COUNTER | Grand Total RESET, Momentary N.O. Pushbutton  |

**OUTPUT:**

| Address | PLC Tag | Comments      |
|---------|---------|---------------|
| Q0.0    | SPOOL   | Take-up Spool |

**COUNTERS:**

*(Do not comment)*

| Counter Data Block Tag        | Comments                      |
|-------------------------------|-------------------------------|
| KIT_COUNTER                   | Kit Parts Counter             |
| KIT_COUNTER.QU <sup>(1)</sup> | Kit Count Complete Stop Spool |
| TOTAL_COUNTER                 | Grand Total Counter Per Spool |

<sup>(1)</sup> Counter status bits created automatically

The tags below are in series or paralleled with the corresponding physical pushbuttons in the logic to provide HMI equivalent of the wired pushbuttons.

The Data Block and Tags has been created and does not need to be modified.

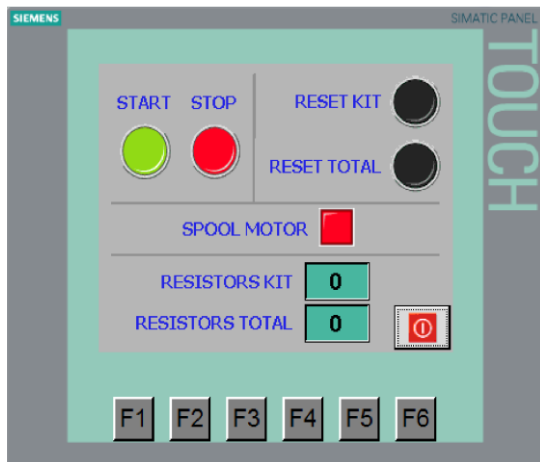
**Name of Data Block: Data\_block\_HMI**

## DATA BLOCK Data\_block\_HMI:

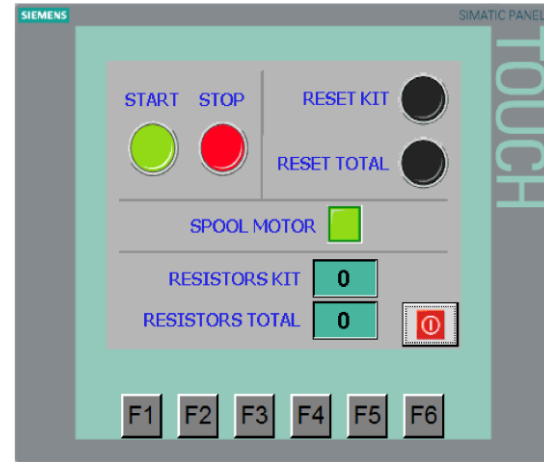
| Static Tag              | Data Type | Comments                     |
|-------------------------|-----------|------------------------------|
| HMI_START               | Boolean   | Oper. Panel START Pushbutton |
| HMI_STOP                | Boolean   | Oper. Panel STOP Pushbuttons |
| HMI_RESET_KIT_COUNTER   | Boolean   | Oper. Panel RESET KIT PB     |
| HMI_RESET_TOTAL_COUNTER | Boolean   | Oper. Panel RESET TOTAL PB   |

## HMI and SWITCHES:

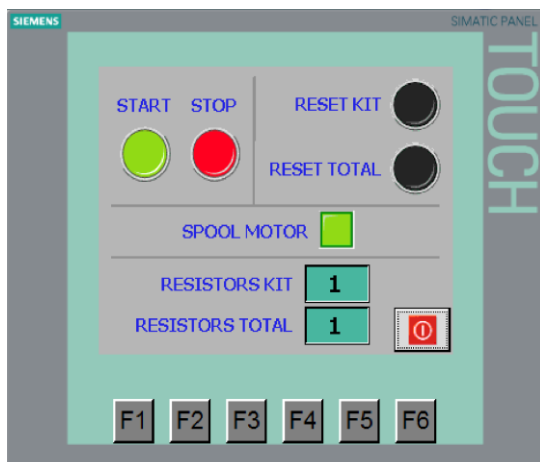
Initial HMI screen.



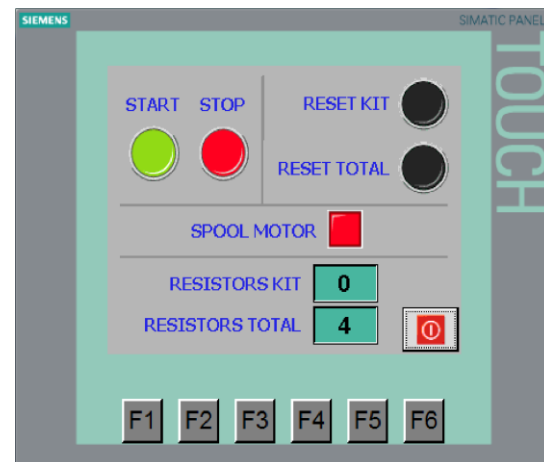
START pressed, MOTOR ON.



PHOTOELECTRIC counting.

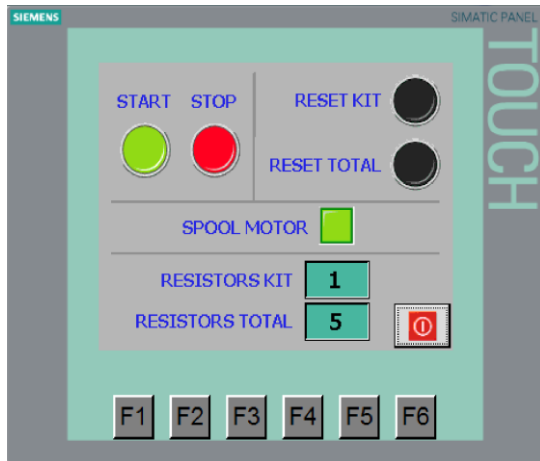


Count complete, MOTOR OFF.

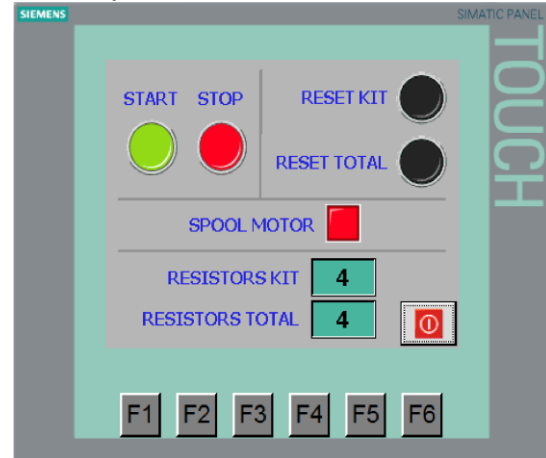


*continued*

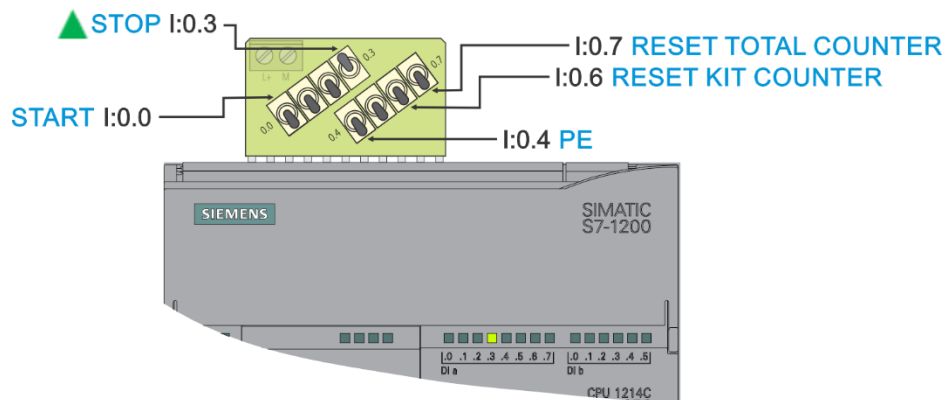
RESET pressed, Count zeroed.



START pressed, MOTOR ON, new count.



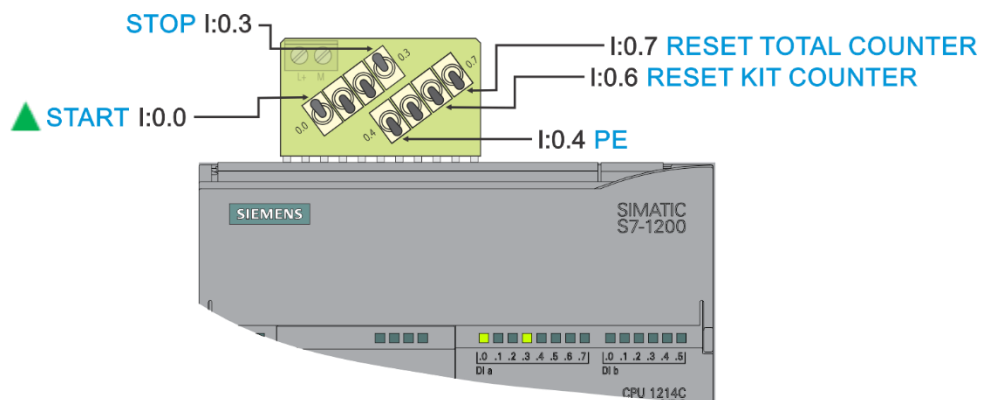
Initial switch settings, **push up ▲** the **STOP** switch to close.



THIS SECTION WILL START A CYCLE TO TEST THE **RESET KIT COUNTER**.

Operator Pressing the START pushbutton:

**Push up ▲** the **START** pushbutton. *The take-up SPOOL turns ON.*

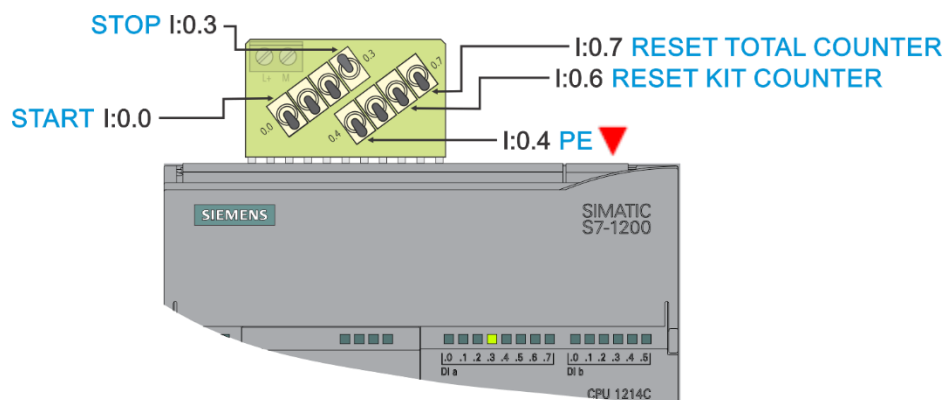
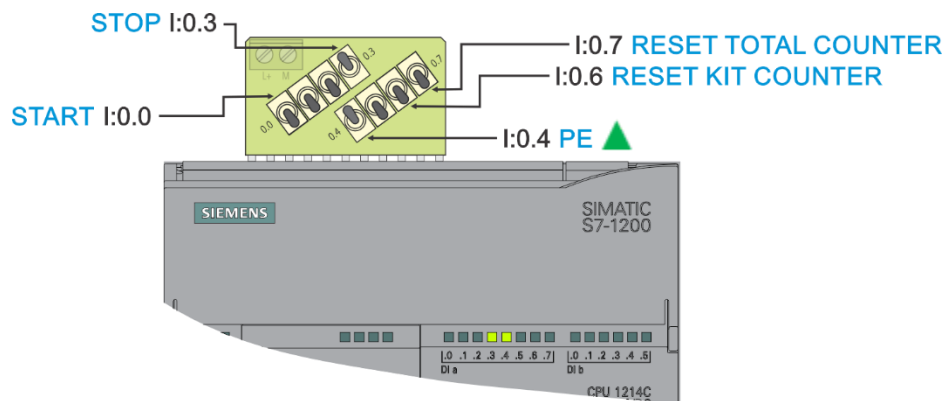


Push down ▼ the **START** pushbutton.

▼ **START** I:0.0



As the Motor runs, toggle up and down PHOTOELECTRIC switch until the part count is reached.



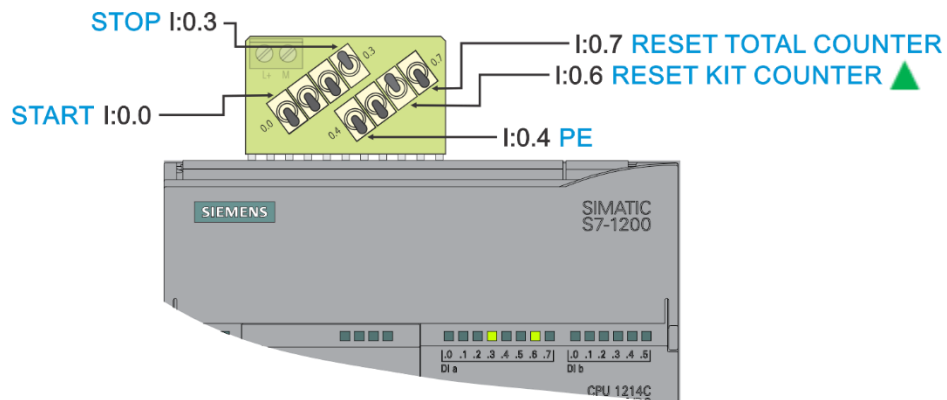
When the count reaches the number of resistors in a kit, the **MOTOR** turns OFF.

*continued*

---

Operator Pressing the RESET KIT pushbutton:

Push up ▲ the **RESET KIT** switch to close. **KIT COUNT** *returns to zero*..



Push down ▼ the **RESET KIT** switch

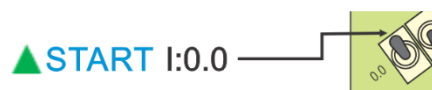


---

**THIS SECTION WILL START A CYCLE TO TEST THE RESET TOTAL COUNTER**

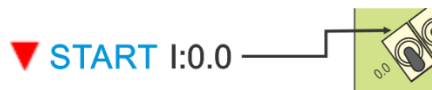
Operator Pressing the START pushbutton:

Push up ▲ the **START** pushbutton.



*The take-up SPOOL turns ON*

Push down ▼ the **START** pushbutton.



---

**As the Motor runs**, toggle up and down **PHOTOELECTRIC** switch until the part count is reached.



*When the count reaches the number of resistors in a kit, the **MOTOR** turns OFF.*

---

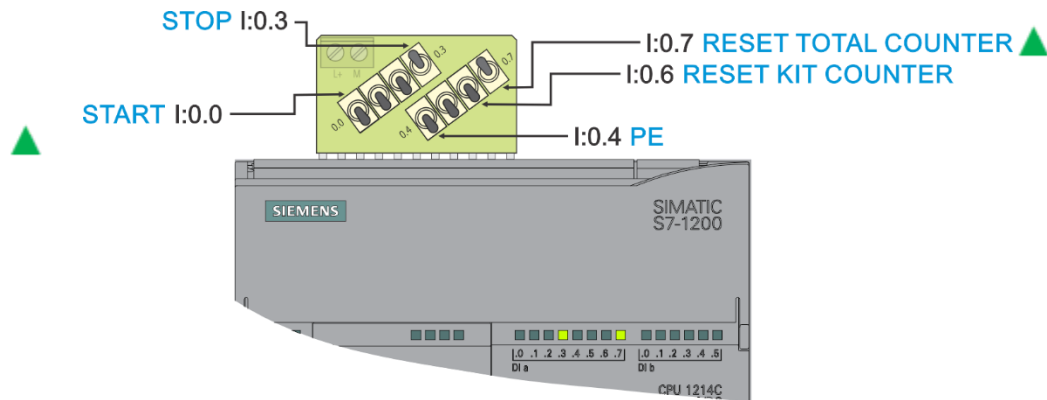
*continued*



---

Operator Pressing the RESET TOTAL pushbutton:

Push up the **RESET TOTAL** switch to close. **TOTAL COUNT** *returns to zero*.



Push down ▼ the **RESET TOTAL** switch.

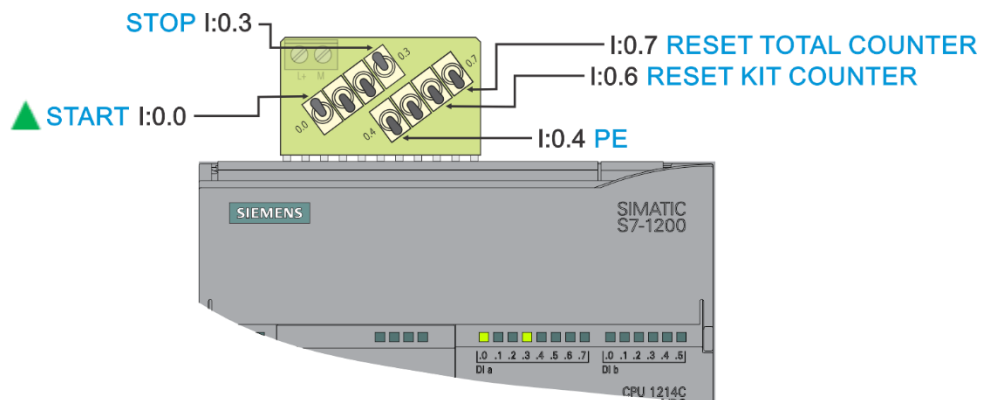


---

THIS SECTION WILL START A CYCLE TO TEST THE **STOP** pushbutton.

Operator Pressing the START pushbutton:

Push up ▲ the **START** pushbutton. *The take-up SPOOL turns ON.*



Push down ▼ the **START** pushbutton. ▼ **START** I:0.0



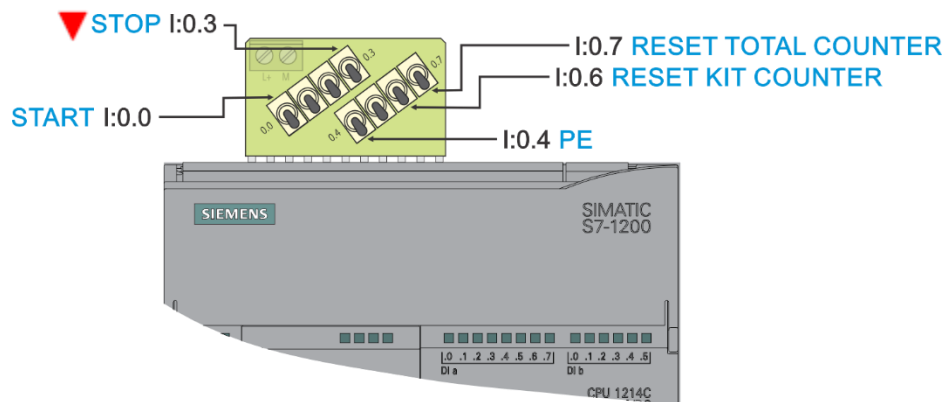
---

*continued*

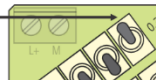
---

Operator Pressing the STOP pushbutton:

Push down ▼ the **STOP** pushbutton. *The take-up SPOOL turns OFF.*



Push up ▲ the **STOP** pushbutton. ▲ STOP I:0.3



---

## THIS SECTION WILL TEST THE HMI PUSHBUTTONS

---

Operator Pressing the HMI START pushbutton:



Press the **HMI START** pushbutton.

*The take-up SPOOL turns ON.*

---

**As the Motor runs**, toggle up and down **PHOTOELECTRIC** switch until the part count is reached.



*When the count reaches the number of resistors in a kit, the **MOTOR** turns OFF.*

---

*continued*

---

Operator Pressing the HMI RESET KIT pushbutton:



Press the **HMI RESET KIT** pushbutton.  
**TOTAL COUNT** *returns to zero.*

---

Operator Pressing the HMI RESET TOTAL pushbutton:



Press the **HMI RESET TOTAL** pushbutton.  
**TOTAL COUNT** *returns to zero.*

---

Operator Pressing the HMI START and then STOP pushbutton:



Press the **HMI START** pushbutton.  
*The take-up SPOOL turns ON.*

---

Operator Pressing the HMI STOP pushbutton:



Press the **HMI STOP** pushbutton.  
*The take-up SPOOL turns OFF.*

## LAB: CONVEYOR TIMER with INDICATORS

**PROCESS FILE:** CONVEYOR INDICATORS

**OBJECTIVE:** Use both direct and indirect information to determine output action.

**SOFTWARE:** SIEMENS TIA Portal

A subroutine Function of the Conveyor Timer is included. You do not need to recreate the logic for the conveyor control. Add the **Tags** and **Ladder Logic subroutine** for the indicators.

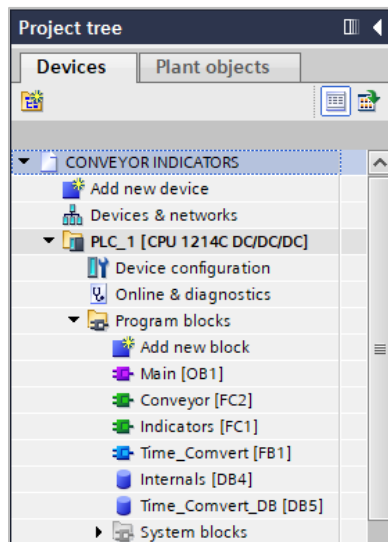
Modify the CONVEYOR TIMER lab with the following indicators for both conveyors: running, in a halting cycle, and stopped.

A RUNNING indicator will be energized as each of the conveyors is energized.

When the STOP pushbutton is pressed the RUNNING indicator for the first conveyor will be turned off. The HALTING indicator for the first conveyor will be turned on while the conveyor is timed delayed to be de-energized.

At the completion of the timing cycle and the first conveyor is off, a STOPPED indicator for the first conveyor will be turned on.

The second conveyor when Started or Stopped will have the sequence with the indicators, STOP → RUN → HALT → STOP.



The program will use two Function programming blocks.

**Function Name:** Block\_Counters  
Block\_Motor

See **Program Assist** on details to create the Function block

When creating the tag, the description MUST BE entered.

### OUTPUTS:

| Address | PLC Tag    | Comments                         |
|---------|------------|----------------------------------|
| Q0.2    | INDC_RUN_1 | Conveyor No. 1 Running Indicator |

|      |             |                                  |
|------|-------------|----------------------------------|
| Q0.3 | INDC_RUN_2  | Conveyor No. 2 Running Indicator |
| Q0.4 | INDC_HALT_1 | Conveyor No. 1 Halting Indicator |
| Q0.5 | INDC_HALT_2 | Conveyor No. 2 Halting Indicator |
| Q0.6 | INDC_STOP_1 | Conveyor No. 1 Stopped Indicator |
| Q0.7 | INDC_STOP_2 | Conveyor No. 2 Stopped Indicator |

---

### FUNCTION: **Indicators**

Create the Function, **Indicators**, for the logic to control the conveyors status indicators.

#### Main OB1:

Drag in the call to the Function Indicators.



Only one (1) status indicator per conveyor will be ON at a time.

|            | START Pushbutton Pressed |  | Both Running | STOP Pushbutton Pressed |  | Both Stopped |
|------------|--------------------------|--|--------------|-------------------------|--|--------------|
| INDC_STOP1 |                          |  |              |                         |  |              |
| INDC_HALT1 |                          |  |              |                         |  |              |
| INDC_RUN1  |                          |  |              |                         |  |              |
| INDC_STOP2 |                          |  |              |                         |  |              |
| INDC_HALT2 |                          |  |              |                         |  |              |
| INDC_RUN2  |                          |  |              |                         |  |              |

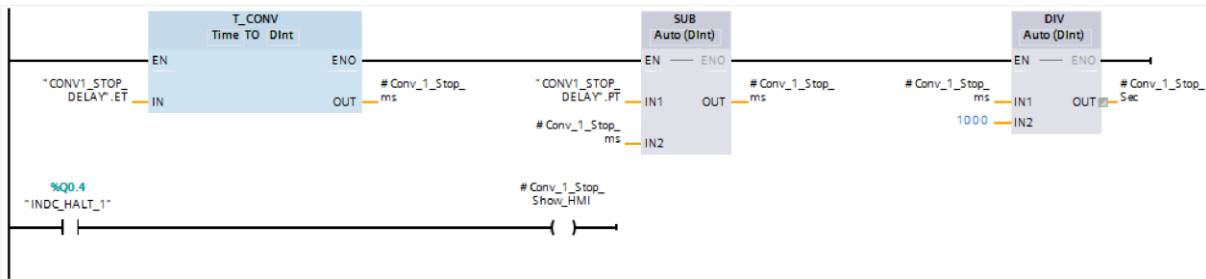
---

### FUNCTION BLOCK: **Time\_Convert**

The Function Block has been created. *Do not delete or modify.*

#### INFORMATION ONLY:

The Function Block is used to convert the timer's value for the HMI display.



First Rung:

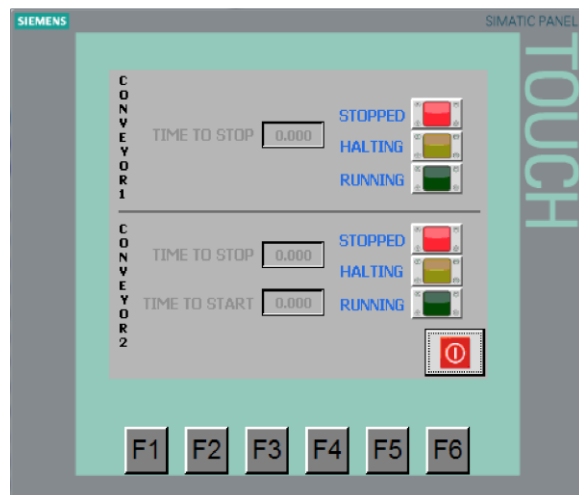
- The T\_CONV instruction converts the timer's formatted Elapsed Time to milliseconds.
- The SUB instruction subtracts the converted elapsed value from the timer's Preset Time for a count-down time display.
- The DIV instruction divides the count-down value, in milliseconds, by 1000 to display as actual time.

Second Rung:

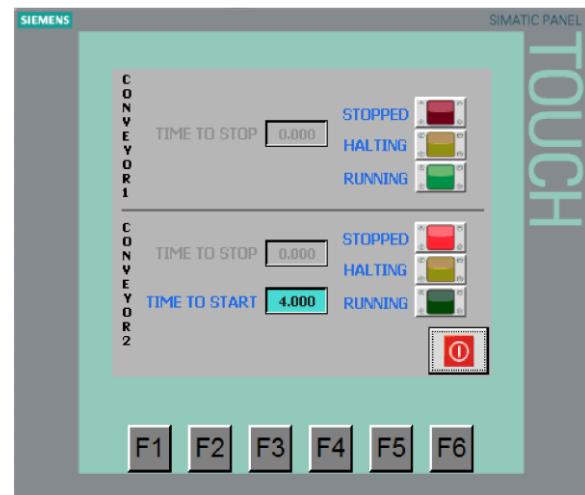
- The indicator status set a status bit to show the timer on the HMI.

## HMI and SWITCHES:

Initial HMI screen.

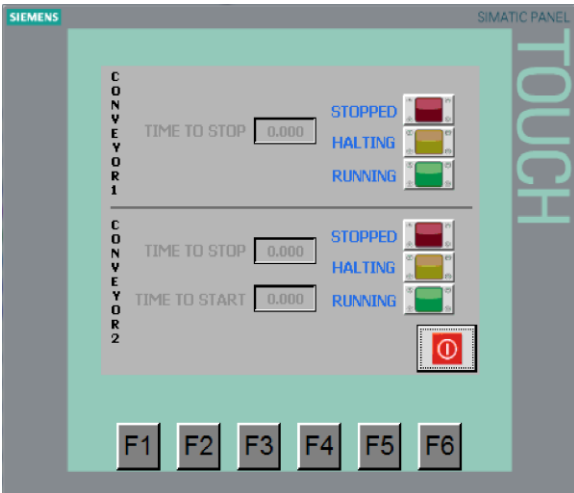


START pressed, CONVEYOR 1 Running, Conveyor 2 Timer elapsing.

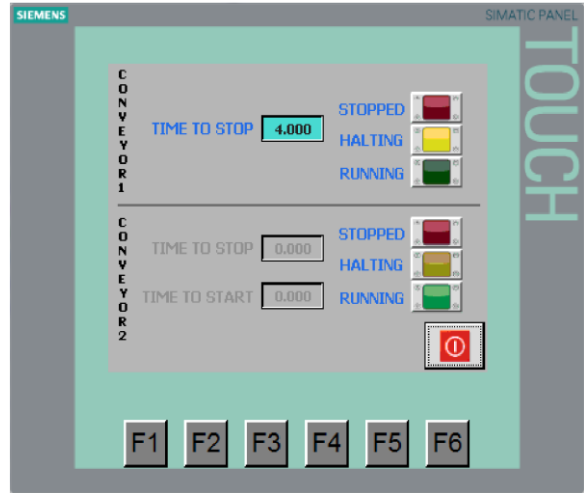


*continued*

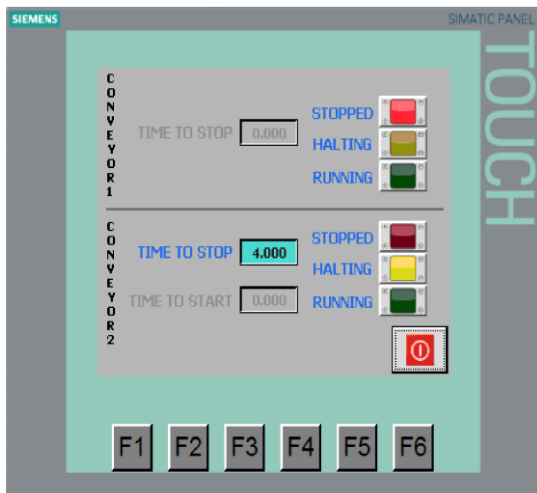
CONVEYOR 2 Running.



STOP pressed, CONVEYOR 1 Halting  
Timer elapsing.

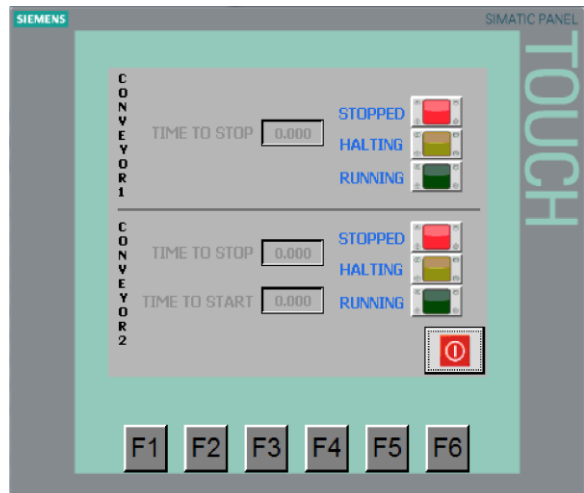


CONVEYOR 1 Stopped, CONVEYOR 2  
Halting Timer elapsing.



*continued*

CONVEYOR 2 Stopped.



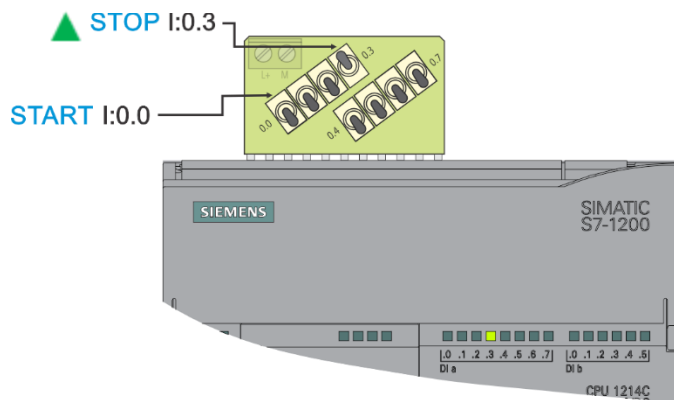
Switch sequences start on the next page.

*continued*

Initial switch settings, push up ▲ the STOP switch to close.



CONVEYOR 1 and CONVEYOR 2 are Red



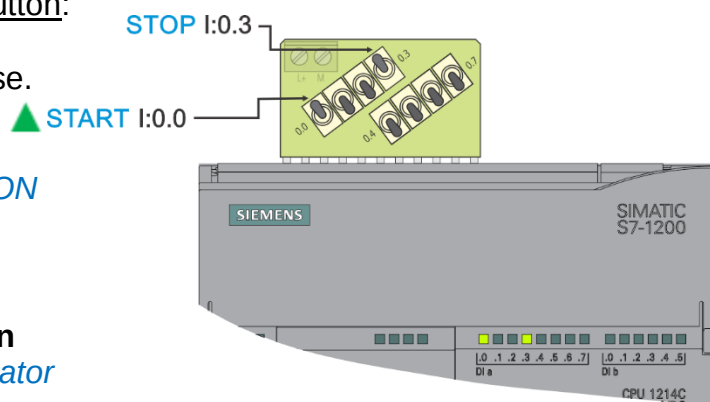
Operator Pressing the START pushbutton:

Push up ▲ the START switch to close.



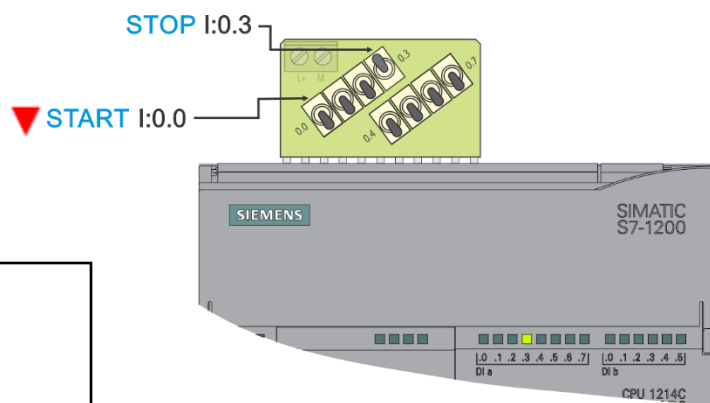
CONVEYOR 1 turns ON immediately and the indicator is Green.

Conveyor 2 Timer On begins elapsing. Indicator is Red.



-- THEN --

Push down ▼ the Start switch to open.



After the time delay, CONVEYOR 2 turns ON.



CONVEYOR 1 and CONVEYOR 2 are Green.



---

Operator Pressing the STOP  
pushbutton:

Push down ▼ the STOP switch to open.

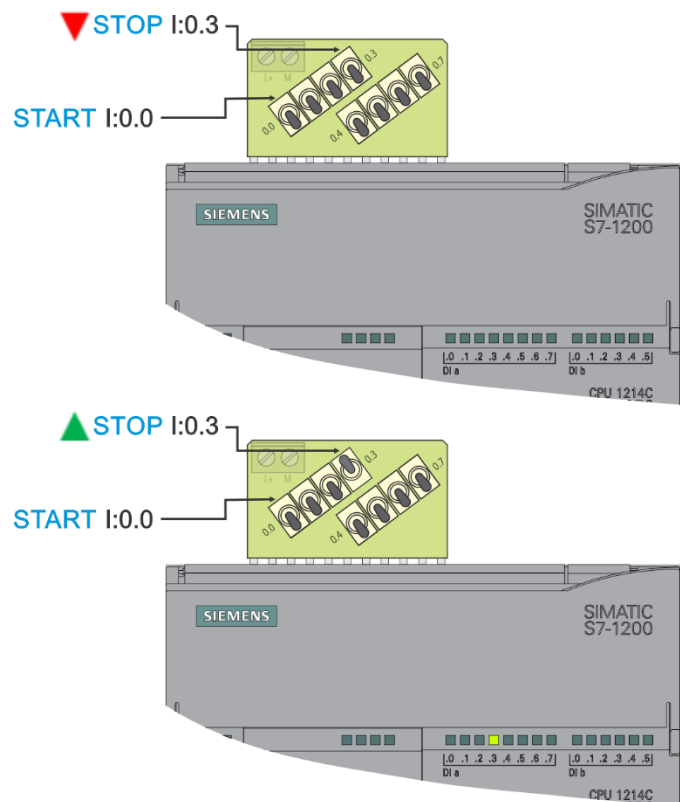


**Conveyor 1 Timer Off**  
*begins elapsing and  
the indicator is Amber.*

**CONVEYOR 2 is**  
**Green.**

- - THEN - -

Push up ▲ the STOP switch to close.



**CONVEYOR 1** *turns OFF after the time delay and the indicator is Red.*

**Conveyor 2 Timer Off** *begins elapsing and the indicator is Amber.*



**CONVEYOR 1** *indicator is Red.*

**CONVEYOR 2** *turns OFF after the time delay and the indicator is Red.*

## LAB: MOVE PART SHIFT

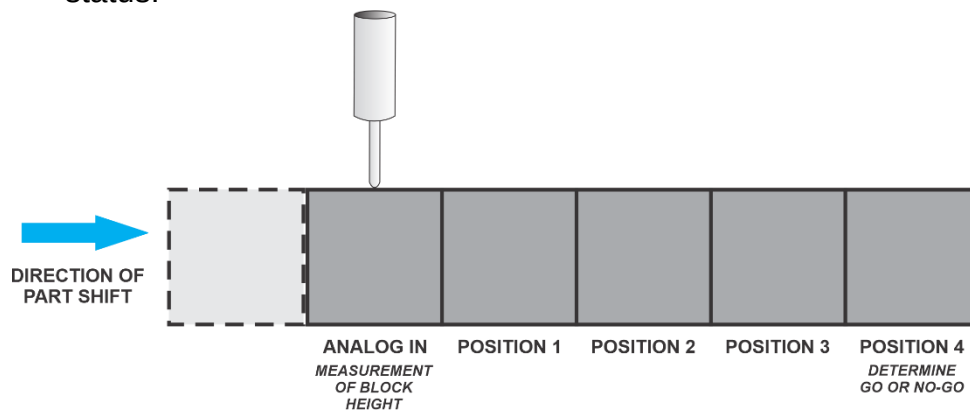
**PROCESS FILE:** MOVE PART SHIFT

**OBJECTIVE:** Create a shift using the MOVE instructions to track a part's data value.

**SOFTWARE:** SIEMENS TIA Portal

The program uses a simplified “First In First Out” type of shift. *The term simplified refers to the logic is fixed to the number of parts and provides no status/error information.*

- The first location is where the Analog sensor reads the part height.
- Positions 1 through 3 are the part being shifted and the data shifted in the program.
- Position 4 is where the height data is read to determine the GO or NO-GO status.



In this lab, only the shift is created.

When creating the tag, the description MUST BE entered.

**INPUT:**

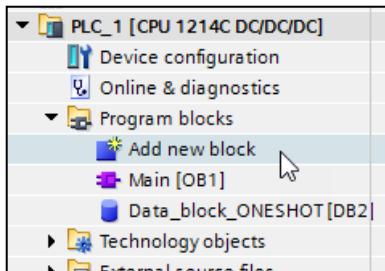
| Address | PLC Tag | Comments                  |
|---------|---------|---------------------------|
| I0.0    | SHIFT   | Momentary N.O. Pushbutton |

*The Data\_Block\_ONESHOT for the one shot CONTROL has already been created.*

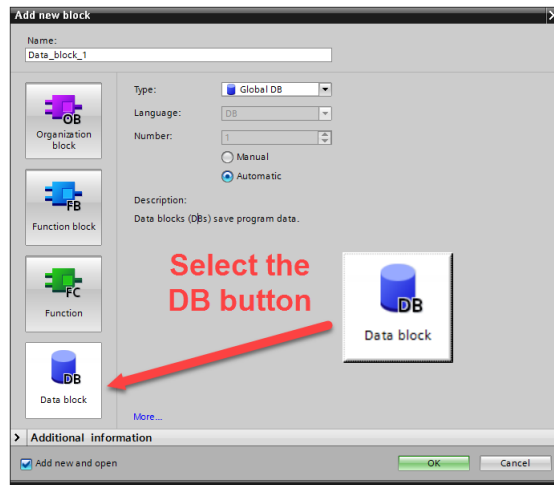
## Data Block

**Create Data Block First**

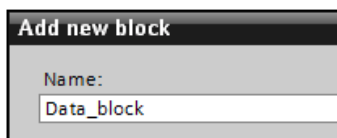
Select the **Add new block**



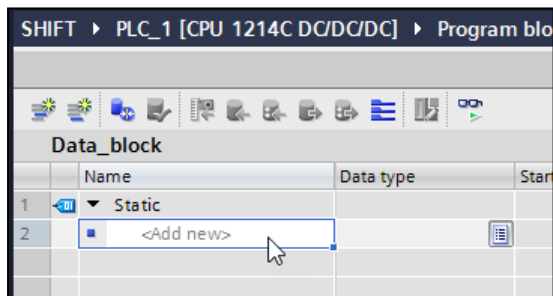
The **Add new block** dialog appears.



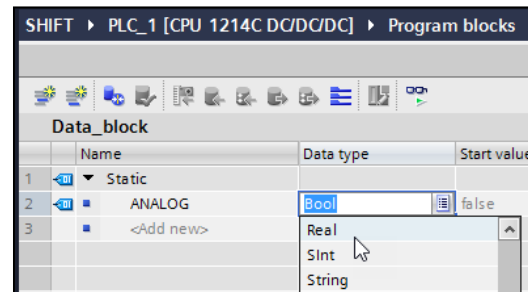
Type in the  
Block name.



Add the tag **Names** to the Data Block



Select the dropdown for **Real** Data Type.



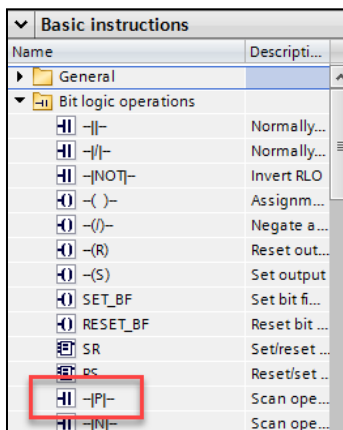
Name of Data Block: **Data\_block**

**DATA BLOCK Data\_block:**

| Static Tag | Data Type | Comments                               |
|------------|-----------|----------------------------------------|
| ANALOG     | REAL      | Read of Part Height                    |
| POSITION1  | REAL      | Shift Position 1                       |
| POSITION2  | REAL      | Shift Position 2                       |
| POSITION3  | REAL      | Shift Position 3                       |
| POSITION4  | REAL      | Shift Position 4, Test for Go or No-Go |

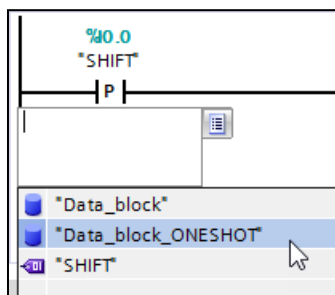
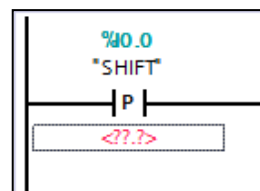
## Positive Signal “One Shot” Instruction

The "Scan operand for positive signal edge" instruction to determine if there is a "0" to "1" change in the signal state of a specified operand. If the instruction detects a change in the result of logic operation (RLO) from "0" to "1", there is a positive, rising edge for one scan only.

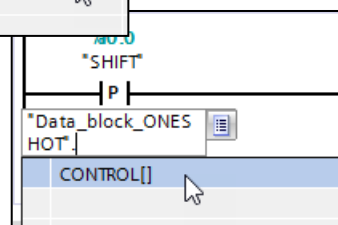


The **Scan operand for positive signal edge** instruction is in **Bit logic operations** folder.

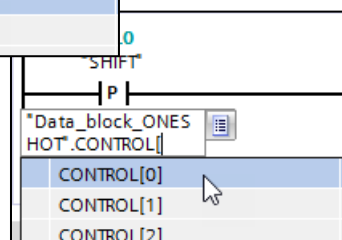
Drag the instruction onto the Network and assign the SHIFT tag.



From the dropdown for the control operand, select **"Data\_block\_ONESHOT"**



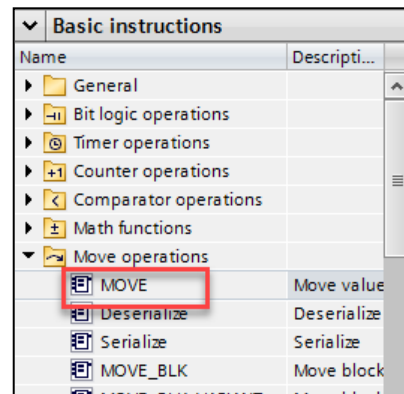
Select **CONTROL[]**



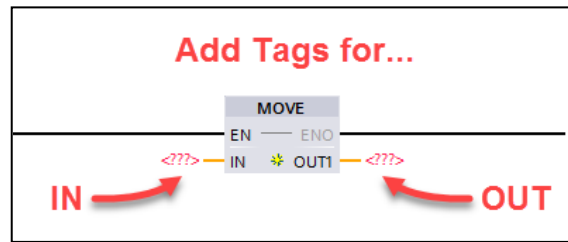
Select **CONTROL[0]**

## MOVE Instruction

The **MOVE** instruction is in **Bit logic operations** folder.



Add the **IN** and **OUT** tags for each MOVE instruction.

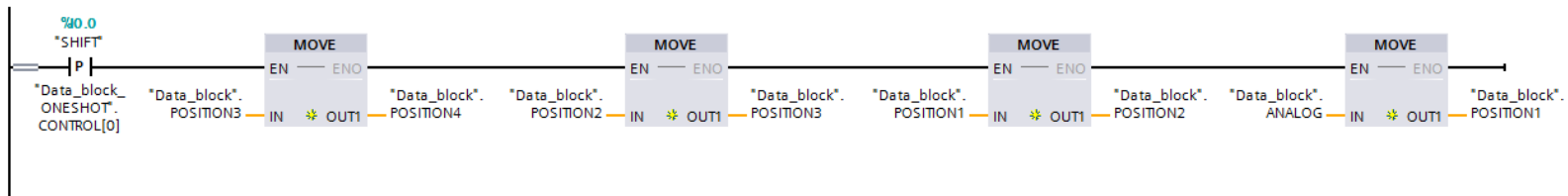


*SEE NEXT PAGE FOR LOGIC PROGRAM*

## PROGRAM

Use the PDF reader's Zoom function to enlarge the page to view the program.

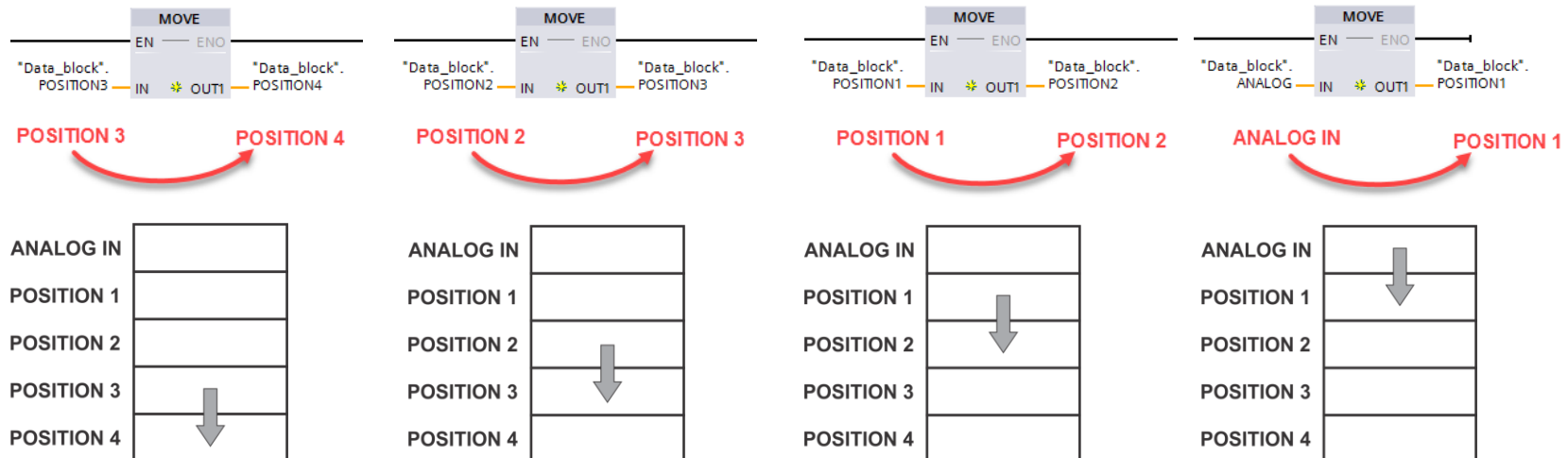
The Network logic uses a “One Shot” to trigger a series of Moves to shift the data down a “stack” holding the height data.



See next page for the instructions on running the program.

## PROGRAM EXPLANATION

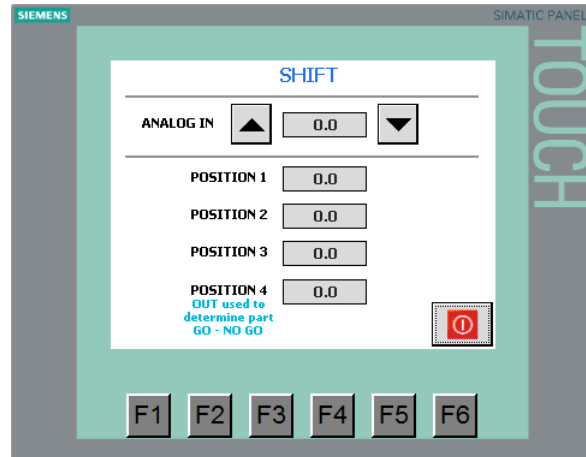
The shift works by using the “left to right” scan of the network by starting at the bottom of the stack and shifting the data into the next position. Then moving up the stack to shift the data to its next position. If the ANALOG IN was shifted first the Data in POSITION 1 would be overwritten before the Data could be moved to POSITION 2.



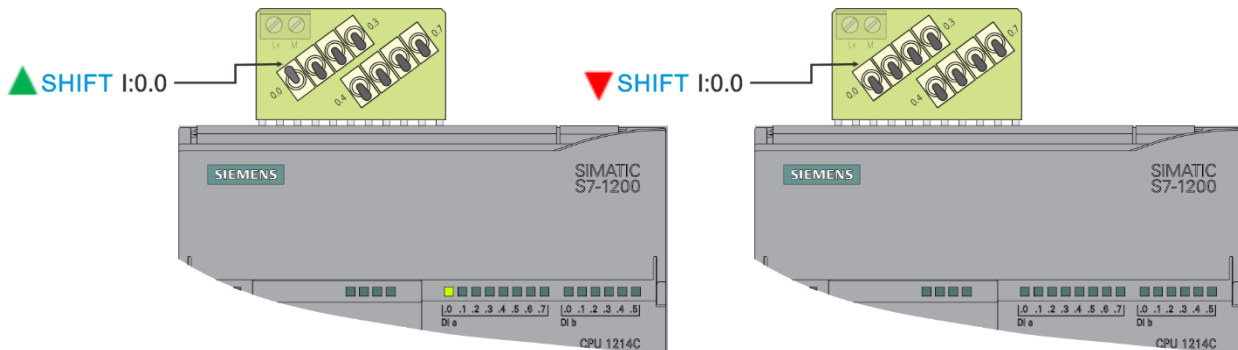
## HMI and SWITCHES:

Repeat the changing of the HMI value and toggling **SHIFT** until a new value is placed into **POSITION 4**.

The **UP - DOWN** arrowhead buttons will change the ANALOG IN value.



Use **SHIFT** I0.0 to trigger the shift by **toggling Up** ▲ then **Down** ▼.



The values on the HMI will shift down on each Up/Down toggle of the **SHIFT** switch.

